Configuration Summary

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FORECAST ANALYSIS — CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-09-06 13:10:17
======================================================================

── Data Preparation ──

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

── Model Training ──

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 3
rollout_epoch_start.................... 2
rollout_epoch_end...................... 40
rollout_loss_decay..................... 0.95
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

── Model Inference ──

  model_path............................. checkpoints_64x128/best_model.pt
  input_data_path........................ data/processed_64x128/sst_final.nc
  run_validation......................... True
  run_test............................... True
  forecast_lead_steps.................... 90
  chain_stride_steps..................... 90
  output_dir............................. results_64x128/
  val_predictions_file................... predictions_val.nc
  test_predictions_file.................. predictions_test.nc

── Results Analysis ──
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val_predictions_path..................... results_64x128/predictions_val.nc
test_predictions_path.................... results_64x128/predictions_test.nc
processed_data_path...................... data/processed_64x128/sst_final.nc
analyse_validation....................... True
analyse_test............................. True
std_params_path.......................... data/processed_64x128/std_params.nc
trend_params_path........................ data/processed_64x128/trend_params.nc
compute_persistence...................... True
compute_climatology...................... True
acc_clim_years........................... (1991, 2020)
acc_clim_smoothing_window................ 5
n_sample_chains.......................... 9
spatial_lead_steps....................... [1, 3, 7, 14, 21, 30, 45, 60, 90]
compute_enso34........................... True
enso34_lat_bounds........................ (-5.0, 5.0)
enso34_lon_bounds........................ (190.0, 240.0)
enso34_clim_years........................ (1991, 2020)
enso34_n_sample_chains................... 9
compute_dhw.............................. True
dhw_baseline_years....................... (1985, 2012)
dhw_output_val........................... dhw_val.nc
dhw_output_test.......................... dhw_test.nc
output_dir............................... results_64x128/
report_file.............................. results_64x128/analysis_report.txt
metrics_csv.............................. results_64x128/metrics_by_lead_step.csv
figures_dir.............................. results_64x128/figures/
analysis_pdf............................. results_64x128/forecast_analysis.pdf
generate_plots........................... True
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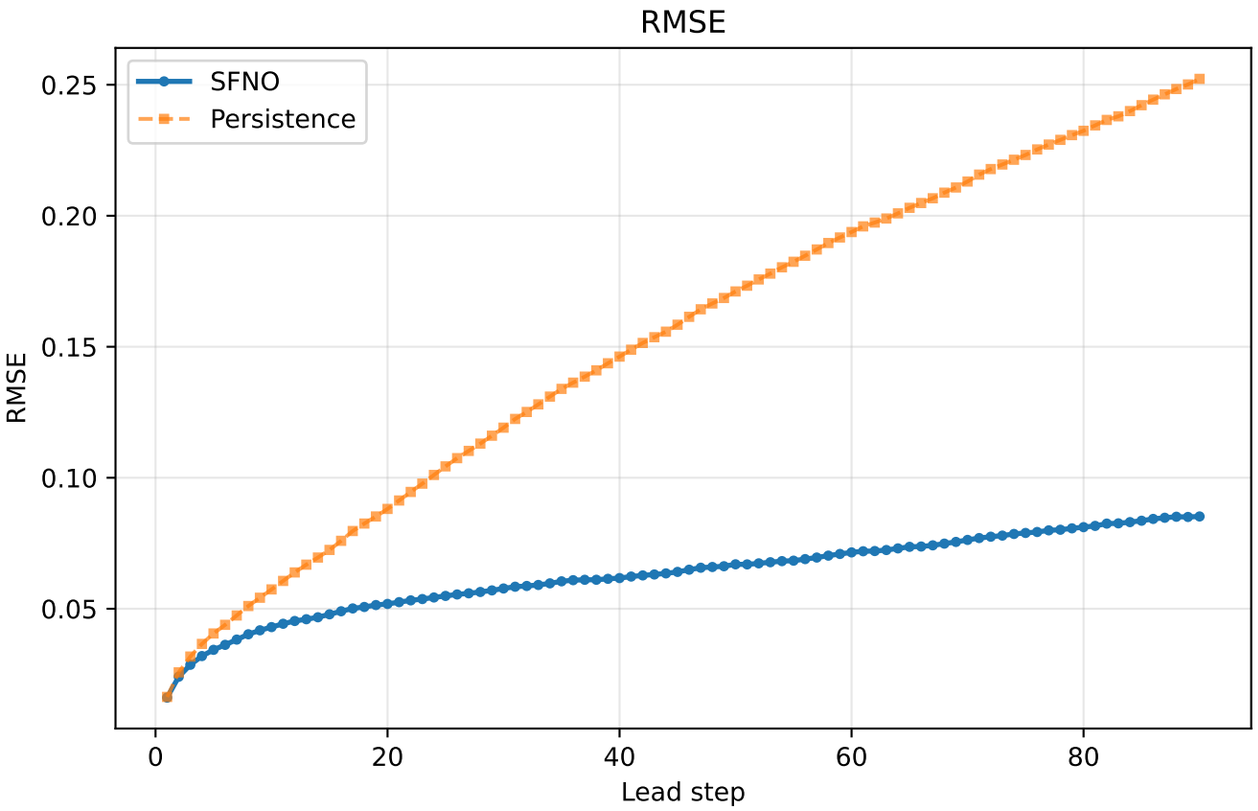
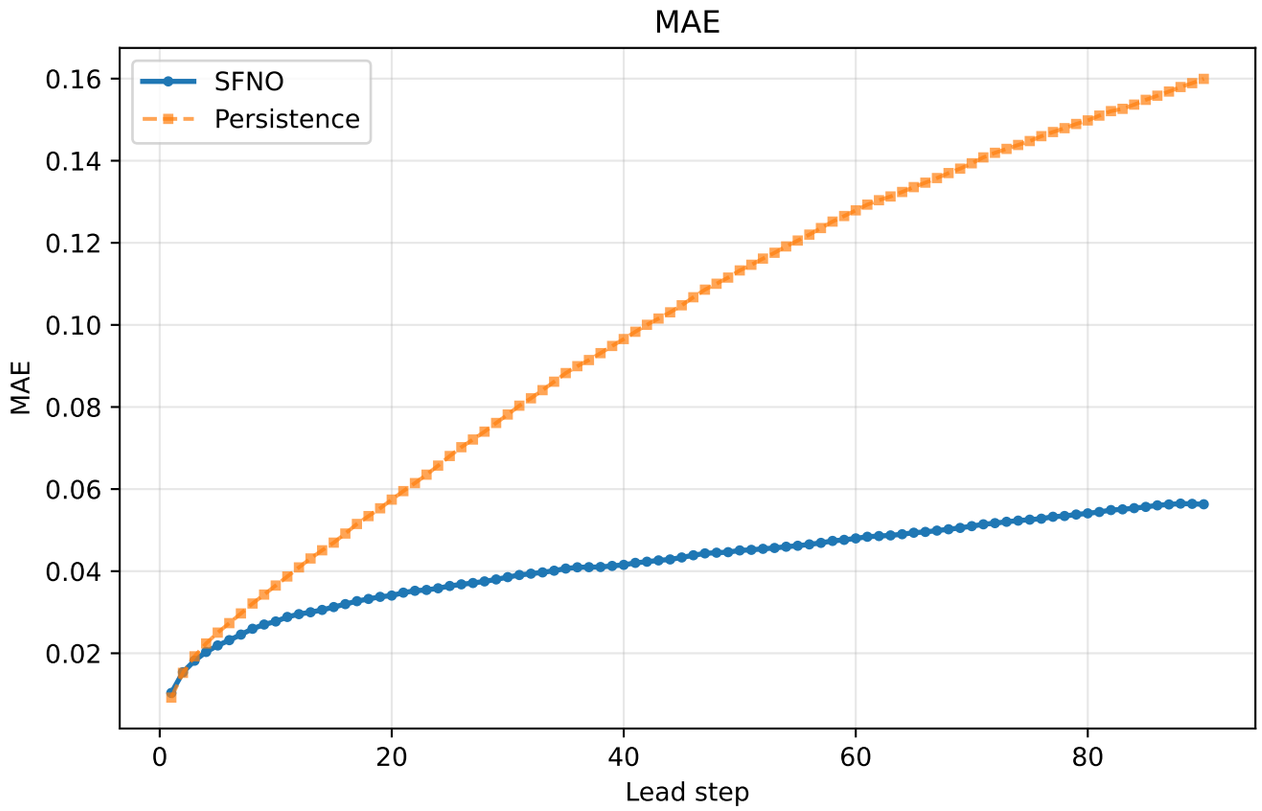
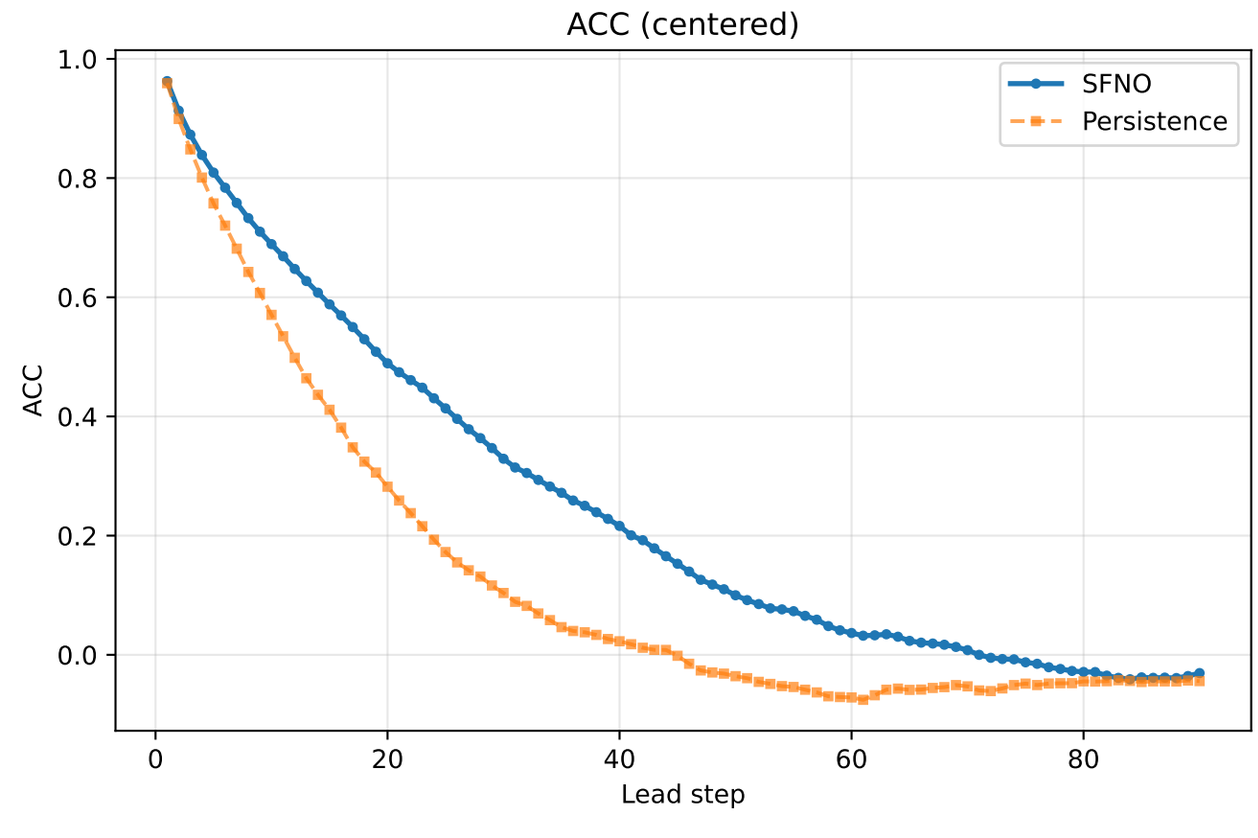
Metrics by Lead Step — Validation

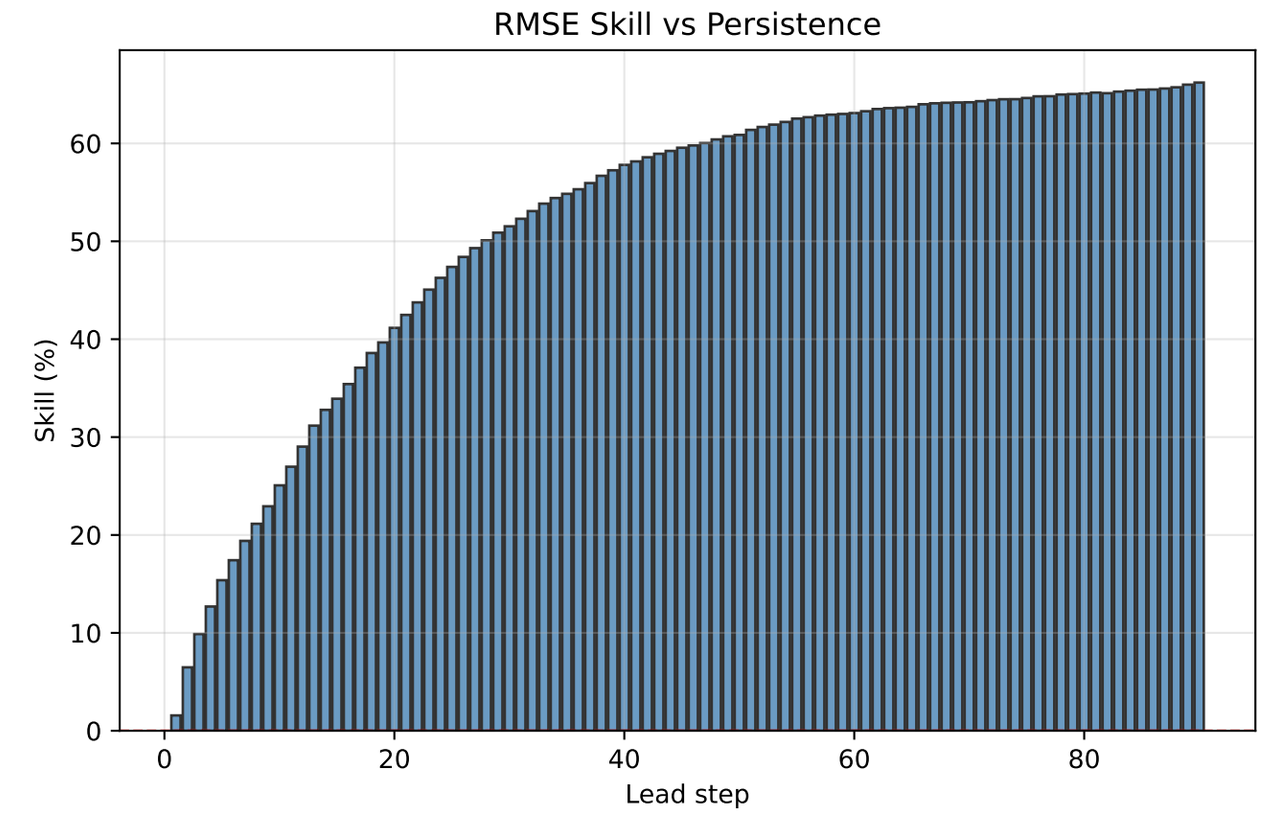
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0161	0.0164	1.6%	0.0103	0.0006	0.9627	0.9592	0.9997
2	0.0241	0.0257	6.5%	0.0154	0.0008	0.9131	0.8991	0.9994
3	0.0286	0.0318	9.9%	0.0182	-0.0000	0.8729	0.8481	0.9992
4	0.0319	0.0366	12.7%	0.0203	-0.0007	0.8388	0.8008	0.9990
5	0.0343	0.0406	15.4%	0.0219	-0.0011	0.8092	0.7574	0.9989
6	0.0362	0.0439	17.4%	0.0232	-0.0014	0.7838	0.7202	0.9987
7	0.0382	0.0474	19.4%	0.0245	-0.0013	0.7583	0.6815	0.9986
8	0.0403	0.0510	21.1%	0.0260	-0.0011	0.7328	0.6425	0.9984
9	0.0418	0.0542	22.9%	0.0270	-0.0012	0.7100	0.6072	0.9983
10	0.0430	0.0574	25.1%	0.0277	-0.0014	0.6891	0.5705	0.9982
11	0.0443	0.0606	27.0%	0.0288	-0.0014	0.6688	0.5345	0.9981
12	0.0453	0.0639	29.0%	0.0295	-0.0015	0.6475	0.4984	0.9980
13	0.0460	0.0668	31.2%	0.0300	-0.0019	0.6272	0.4642	0.9979
14	0.0467	0.0695	32.8%	0.0306	-0.0019	0.6077	0.4363	0.9979
15	0.0479	0.0725	33.9%	0.0313	-0.0017	0.5881	0.4112	0.9978
16	0.0490	0.0759	35.4%	0.0320	-0.0018	0.5695	0.3812	0.9977
17	0.0501	0.0797	37.1%	0.0327	-0.0017	0.5500	0.3482	0.9976
18	0.0507	0.0825	38.6%	0.0332	-0.0014	0.5295	0.3242	0.9975
19	0.0514	0.0852	39.7%	0.0337	-0.0015	0.5085	0.3058	0.9974
20	0.0518	0.0881	41.2%	0.0341	-0.0017	0.4892	0.2821	0.9974
21	0.0525	0.0913	42.5%	0.0348	-0.0017	0.4742	0.2589	0.9973
22	0.0532	0.0946	43.8%	0.0352	-0.0022	0.4609	0.2377	0.9972
23	0.0537	0.0978	45.1%	0.0354	-0.0027	0.4482	0.2157	0.9972
24	0.0543	0.1011	46.3%	0.0358	-0.0030	0.4305	0.1932	0.9971
25	0.0549	0.1043	47.4%	0.0364	-0.0032	0.4135	0.1725	0.9971
26	0.0555	0.1075	48.4%	0.0368	-0.0036	0.3959	0.1551	0.9970
27	0.0559	0.1103	49.3%	0.0371	-0.0035	0.3785	0.1418	0.9970
28	0.0564	0.1130	50.1%	0.0375	-0.0035	0.3636	0.1313	0.9969
29	0.0570	0.1161	50.9%	0.0380	-0.0038	0.3469	0.1164	0.9968
30	0.0577	0.1191	51.5%	0.0385	-0.0041	0.3290	0.1038	0.9968
31	0.0584	0.1225	52.3%	0.0391	-0.0043	0.3143	0.0889	0.9967
32	0.0587	0.1251	53.1%	0.0394	-0.0046	0.3051	0.0821	0.9966
33	0.0591	0.1280	53.8%	0.0397	-0.0049	0.2935	0.0694	0.9966
34	0.0597	0.1310	54.4%	0.0401	-0.0052	0.2824	0.0582	0.9965
35	0.0605	0.1339	54.8%	0.0406	-0.0052	0.2718	0.0464	0.9964
36	0.0609	0.1363	55.3%	0.0409	-0.0056	0.2589	0.0400	0.9964
37	0.0611	0.1386	55.9%	0.0410	-0.0056	0.2501	0.0378	0.9964
38	0.0611	0.1410	56.7%	0.0410	-0.0057	0.2393	0.0334	0.9964
39	0.0614	0.1437	57.3%	0.0413	-0.0057	0.2281	0.0265	0.9963
40	0.0617	0.1462	57.8%	0.0415	-0.0058	0.2163	0.0227	0.9963
41	0.0623	0.1489	58.2%	0.0420	-0.0057	0.2005	0.0178	0.9962
42	0.0627	0.1514	58.6%	0.0423	-0.0059	0.1923	0.0118	0.9962
43	0.0631	0.1536	58.9%	0.0426	-0.0061	0.1787	0.0084	0.9961
44	0.0635	0.1558	59.2%	0.0429	-0.0062	0.1654	0.0082	0.9961
45	0.0641	0.1584	59.6%	0.0433	-0.0062	0.1528	-0.0016	0.9960
46	0.0649	0.1614	59.8%	0.0439	-0.0062	0.1397	-0.0150	0.9959
47	0.0657	0.1643	60.0%	0.0443	-0.0064	0.1259	-0.0262	0.9958
48	0.0659	0.1665	60.4%	0.0445	-0.0064	0.1179	-0.0296	0.9958
49	0.0662	0.1686	60.7%	0.0446	-0.0068	0.1100	-0.0316	0.9957
50	0.0670	0.1711	60.9%	0.0451	-0.0076	0.0999	-0.0359	0.9956
51	0.0669	0.1733	61.4%	0.0452	-0.0078	0.0918	-0.0392	0.9956
52	0.0673	0.1757	61.7%	0.0454	-0.0082	0.0852	-0.0454	0.9956
53	0.0678	0.1779	61.9%	0.0457	-0.0084	0.0780	-0.0488	0.9955
54	0.0682	0.1803	62.2%	0.0460	-0.0090	0.0762	-0.0525	0.9955
55	0.0684	0.1824	62.5%	0.0462	-0.0090	0.0732	-0.0540	0.9954
56	0.0690	0.1847	62.7%	0.0465	-0.0095	0.0654	-0.0585	0.9954
57	0.0695	0.1871	62.8%	0.0469	-0.0100	0.0589	-0.0631	0.9953
58	0.0703	0.1896	62.9%	0.0474	-0.0101	0.0482	-0.0696	0.9952
59	0.0709	0.1917	63.0%	0.0476	-0.0099	0.0411	-0.0708	0.9951
60	0.0715	0.1938	63.1%	0.0480	-0.0102	0.0366	-0.0716	0.9950
61	0.0719	0.1959	63.3%	0.0484	-0.0102	0.0321	-0.0755	0.9950
62	0.0720	0.1974	63.5%	0.0486	-0.0102	0.0327	-0.0678	0.9950
63	0.0724	0.1989	63.6%	0.0487	-0.0104	0.0345	-0.0586	0.9949
64	0.0730	0.2009	63.6%	0.0490	-0.0107	0.0303	-0.0567	0.9948
65	0.0736	0.2030	63.7%	0.0494	-0.0106	0.0234	-0.0588	0.9947
66	0.0738	0.2049	64.0%	0.0496	-0.0107	0.0204	-0.0583	0.9947
67	0.0742	0.2067	64.1%	0.0499	-0.0108	0.0190	-0.0556	0.9946
68	0.0749	0.2088	64.1%	0.0502	-0.0109	0.0169	-0.0542	0.9945
69	0.0755	0.2108	64.2%	0.0505	-0.0108	0.0132	-0.0508	0.9945
70	0.0763	0.2131	64.2%	0.0510	-0.0109	0.0078	-0.0529	0.9943
71	0.0770	0.2157	64.3%	0.0514	-0.0109	-0.0001	-0.0598	0.9942
72	0.0775	0.2178	64.4%	0.0517	-0.0109	-0.0050	-0.0609	0.9942
73	0.0779	0.2196	64.5%	0.0520	-0.0109	-0.0071	-0.0564	0.9941
74	0.0786	0.2214	64.5%	0.0523	-0.0112	-0.0076	-0.0508	0.9940
75	0.0789	0.2232	64.6%	0.0525	-0.0112	-0.0126	-0.0483	0.9939
76	0.0793	0.2253	64.8%	0.0528	-0.0113	-0.0150	-0.0507	0.9939
77	0.0799	0.2272	64.8%	0.0533	-0.0115	-0.0208	-0.0482	0.9938
78	0.0802	0.2290	65.0%	0.0534	-0.0113	-0.0237	-0.0477	0.9937
79	0.0807	0.2308	65.0%	0.0538	-0.0117	-0.0271	-0.0476	0.9936
80	0.0811	0.2324	65.1%	0.0541	-0.0118	-0.0288	-0.0445	0.9936
81	0.0816	0.2345	65.2%	0.0544	-0.0117	-0.0289	-0.0448	0.9935
82	0.0825	0.2366	65.1%	0.0549	-0.0120	-0.0349	-0.0443	0.9934
83	0.0826	0.2380	65.3%	0.0551	-0.0116	-0.0387	-0.0426	0.9934
84	0.0831	0.2399	65.4%	0.0554	-0.0119	-0.0410	-0.0439	0.9933
85	0.0836	0.2422	65.5%	0.0557	-0.0122	-0.0378	-0.0456	0.9932
86	0.0843	0.2443	65.5%	0.0560	-0.0125	-0.0386	-0.0445	0.9931
87	0.0847	0.2464	65.6%	0.0563	-0.0126	-0.0383	-0.0446	0.9930
88	0.0851	0.2484	65.7%	0.0565	-0.0129	-0.0390	-0.0447	0.9929
89	0.0850	0.2501	66.0%	0.0564	-0.0129	-0.0356	-0.0431	0.9930
90	0.0852	0.2522	66.2%	0.0563	-0.0131	-0.0307	-0.0441	0.9929

Forecast Skill vs Lead Step — Validation

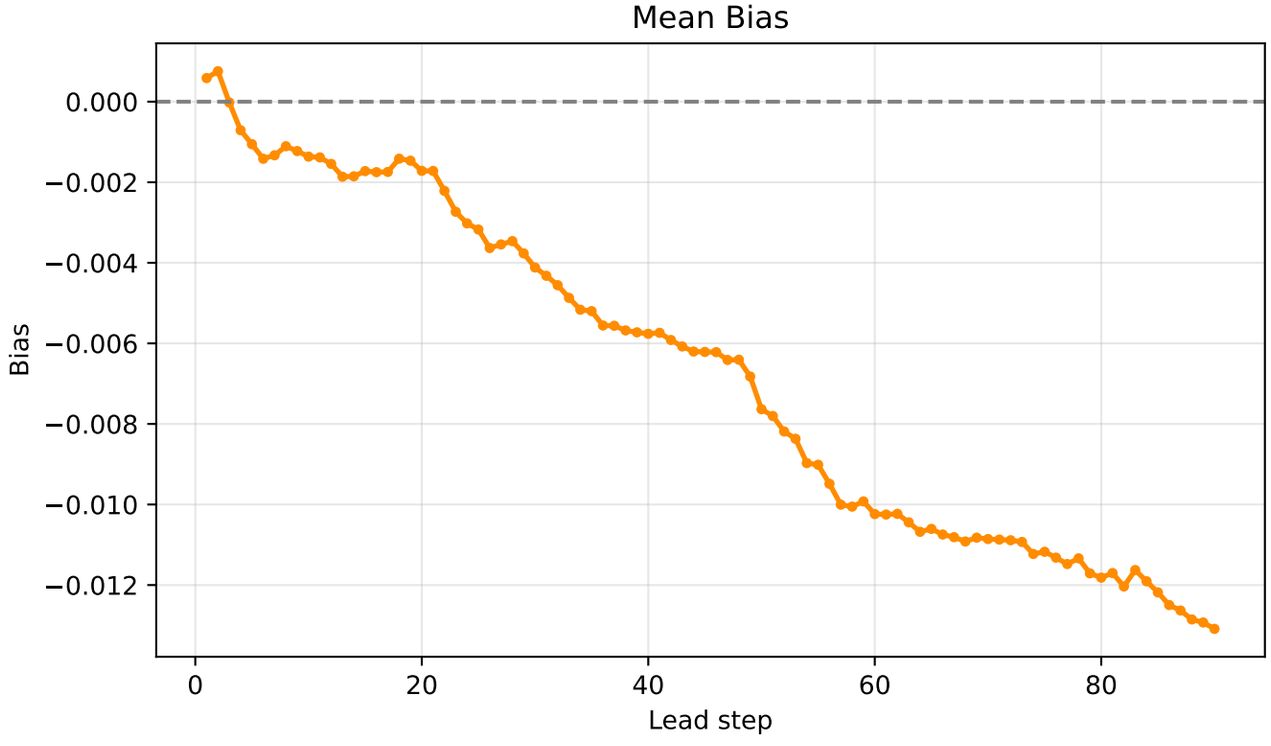


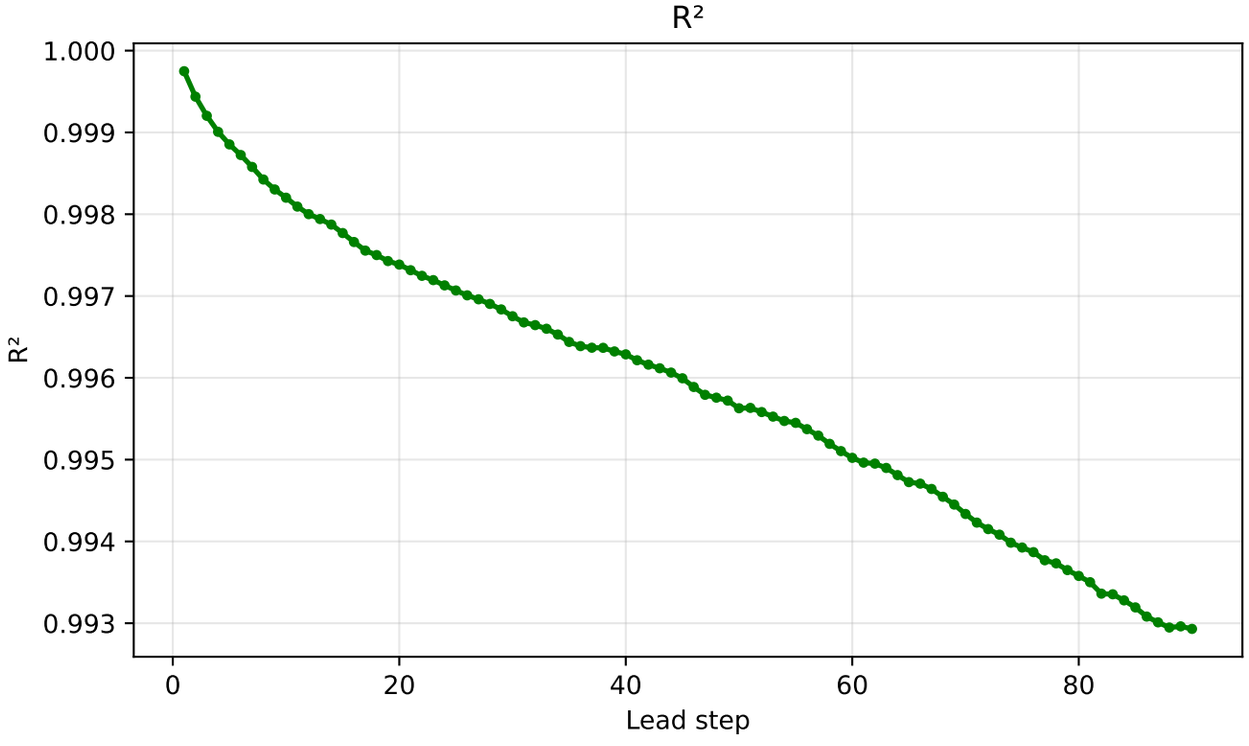






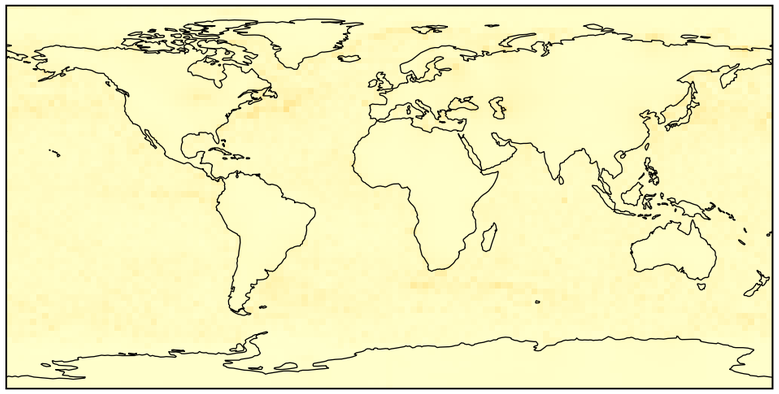
Bias and R² — Validation



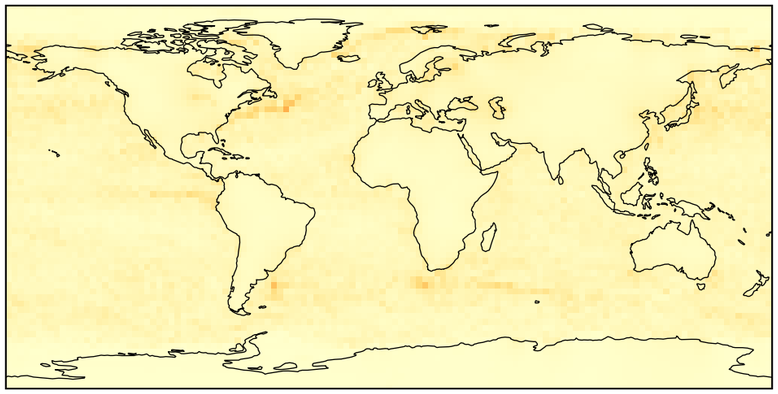


Spatial RMSE (all grid) — Validation

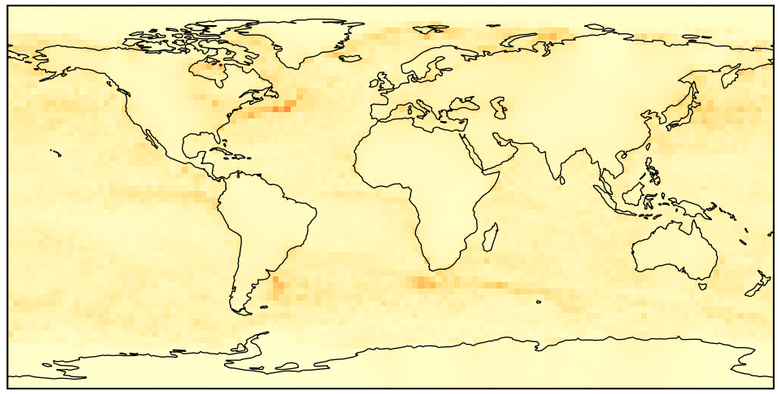
Step 1



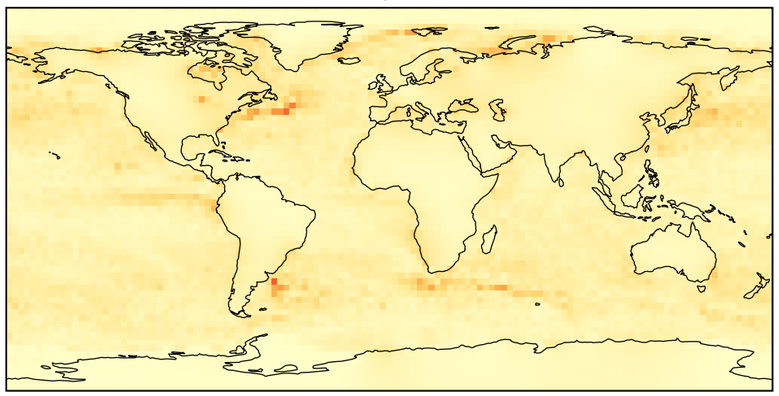
Step 3



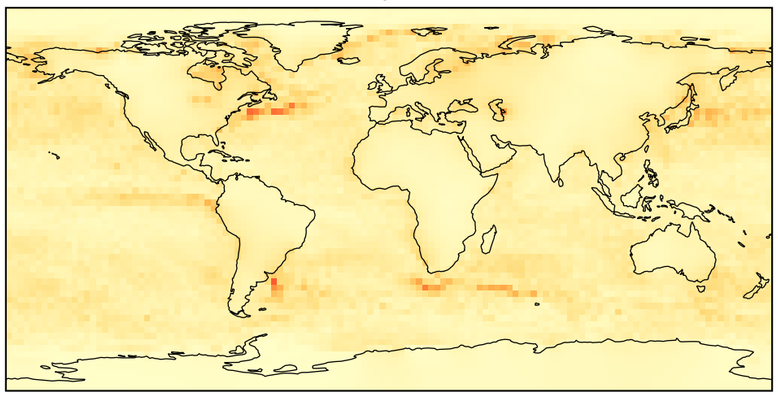
Step 7



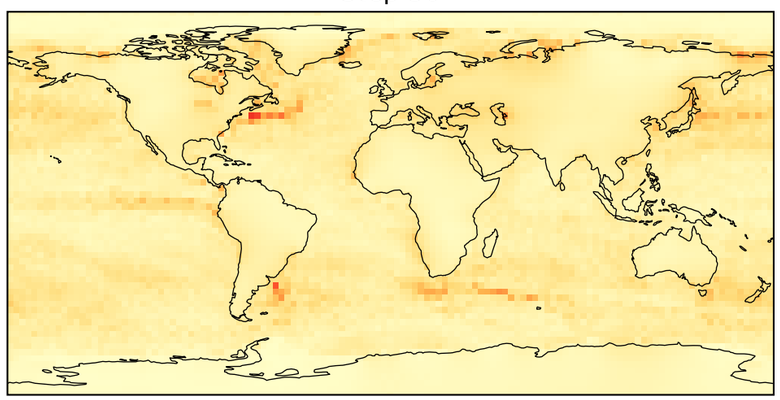
Step 14



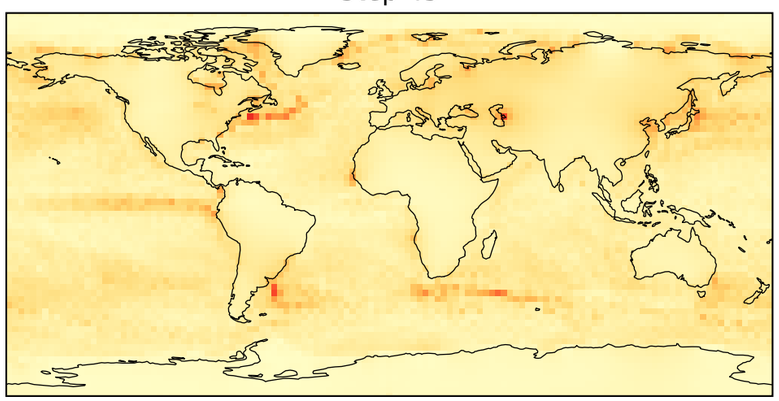
Step 21



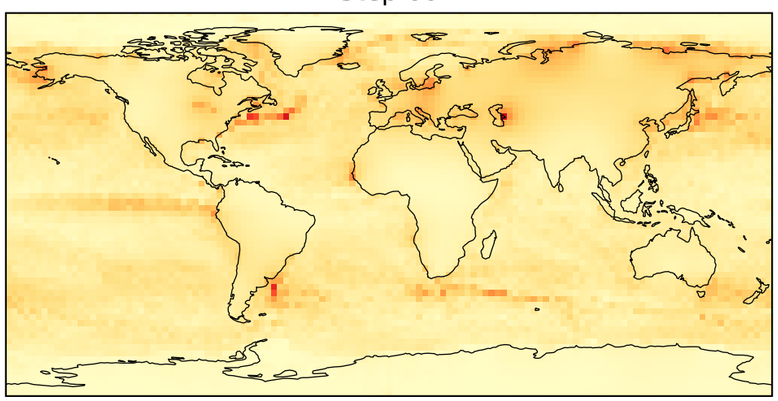
Step 30



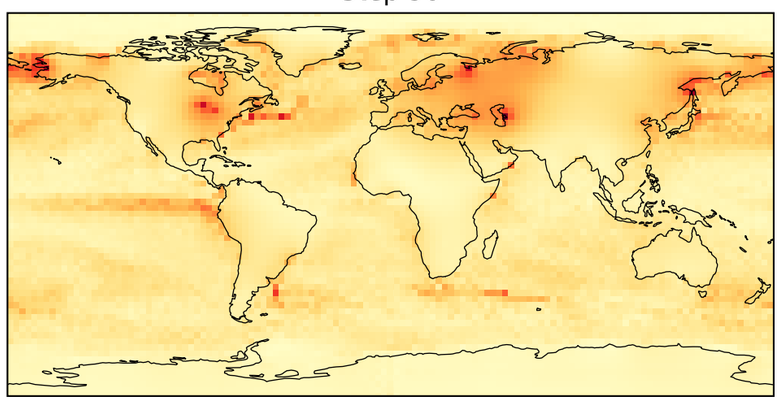
Step 45



Step 60

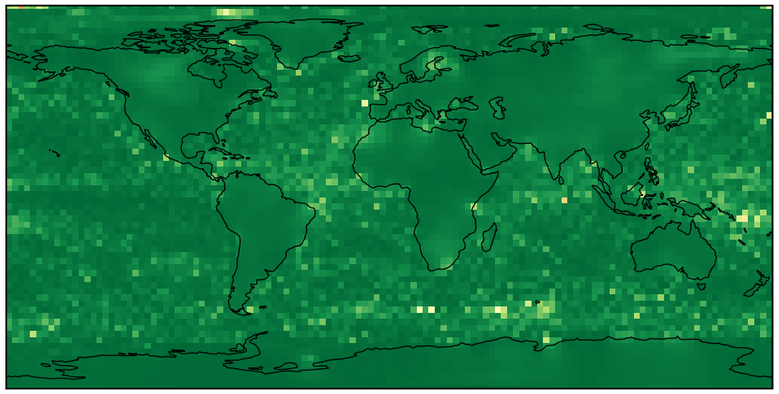


Step 90

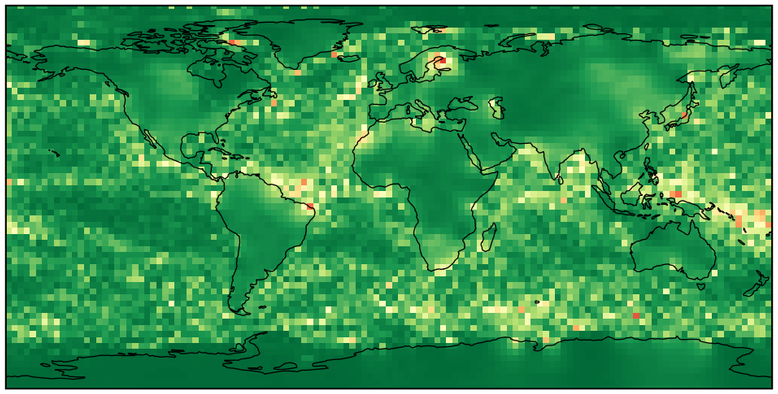


Spatial ACC (all grid) — Validation

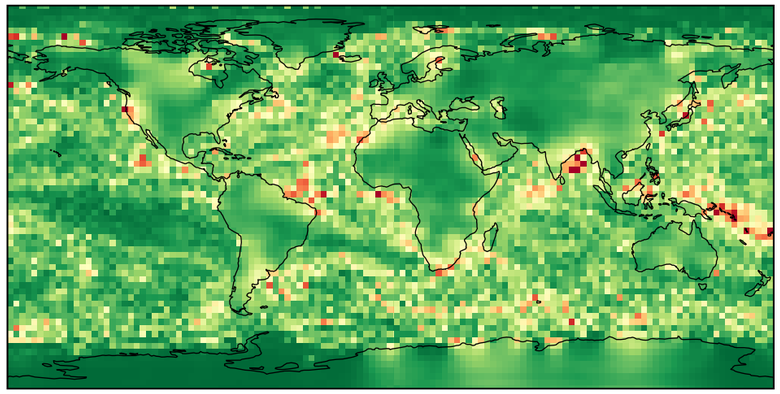
Step 1



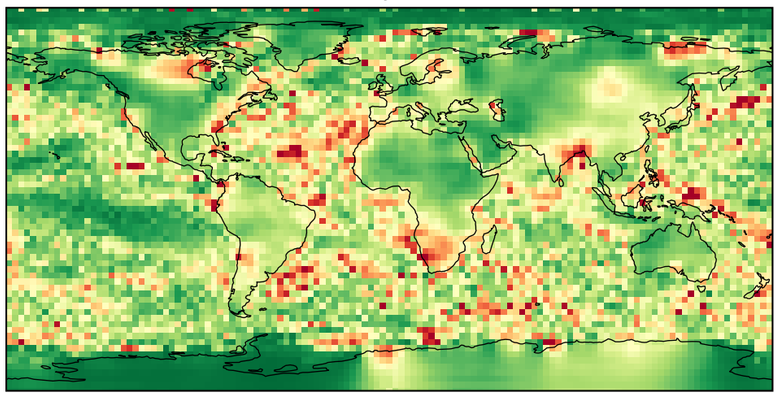
Step 3



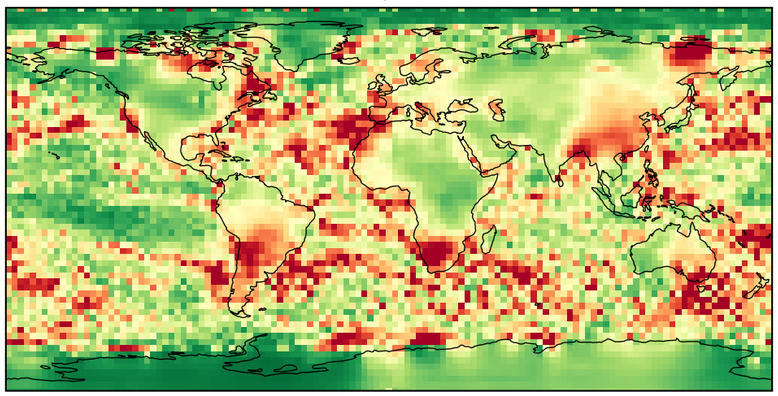
Step 7



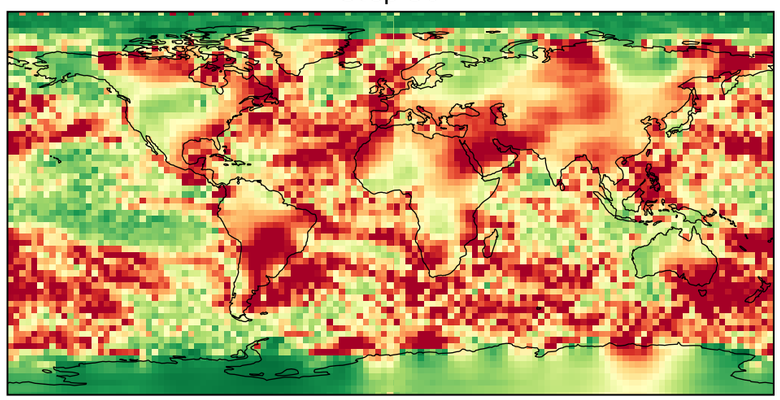
Step 14



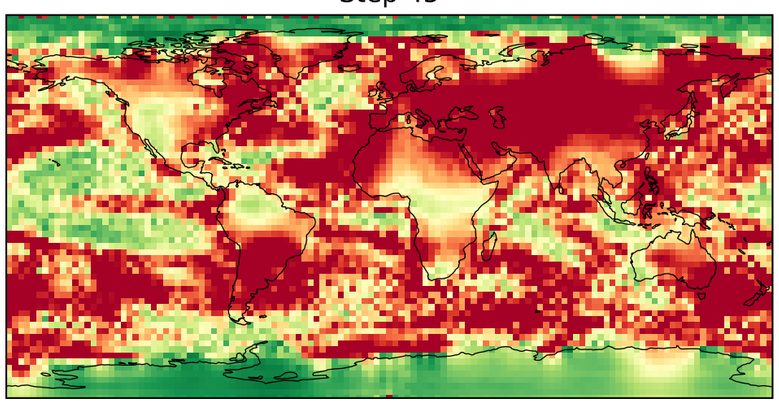
Step 21



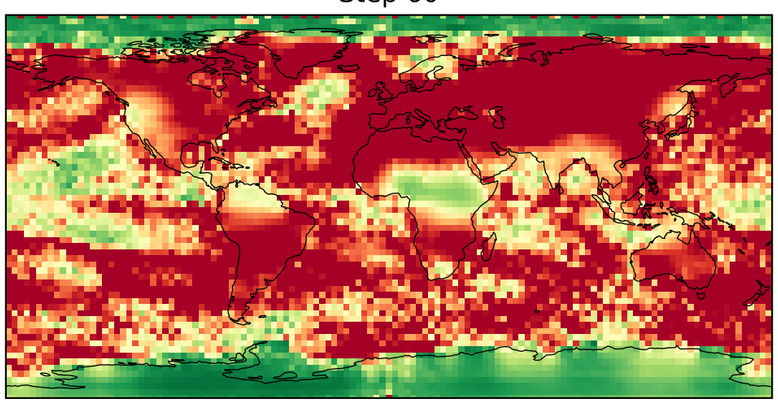
Step 30



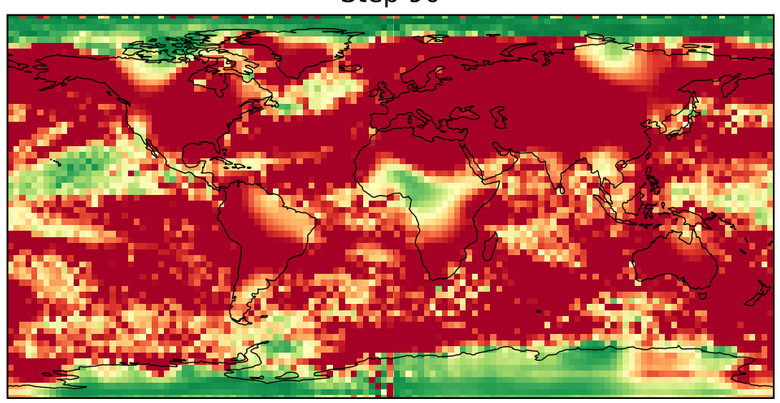
Step 45

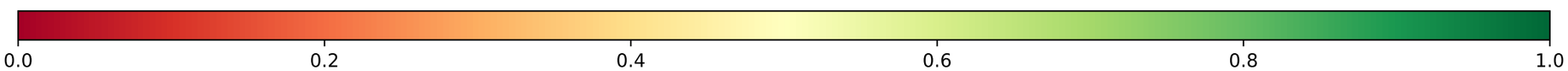


Step 60



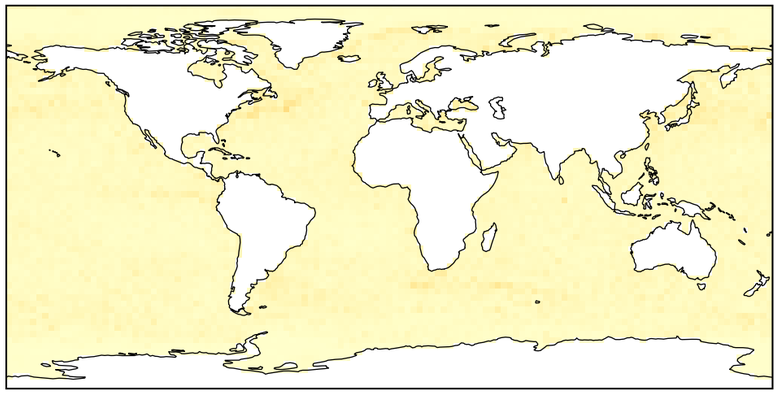
Step 90



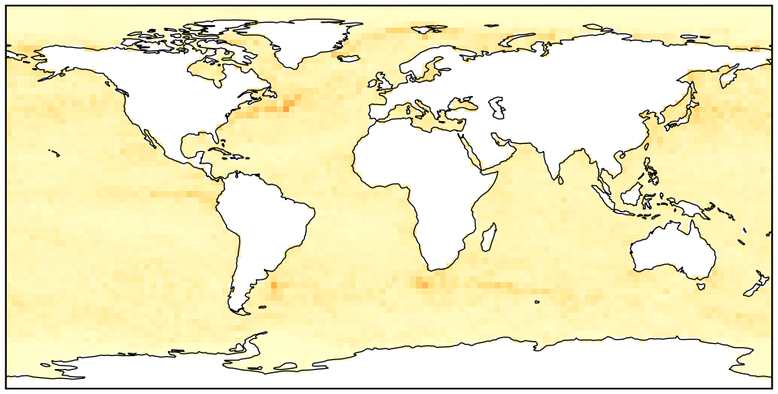


Spatial RMSE (ocean only) — Validation

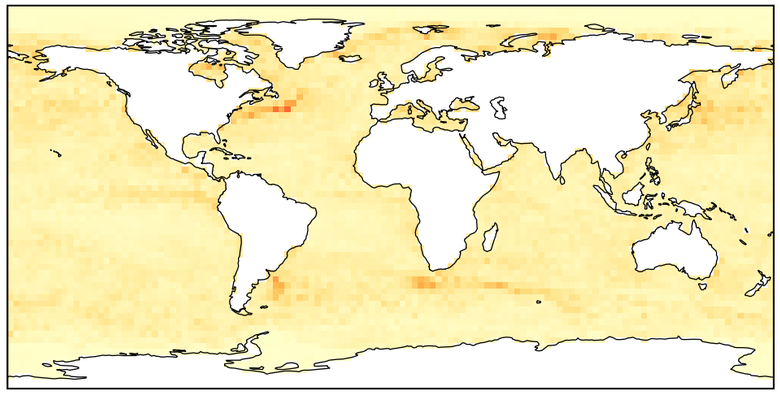
Step 1



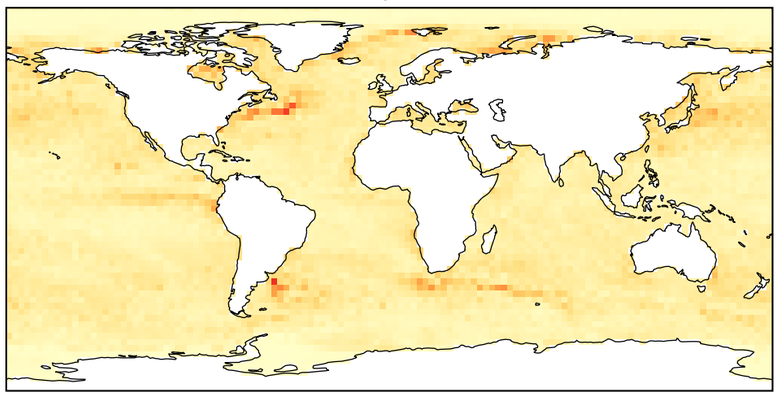
Step 3



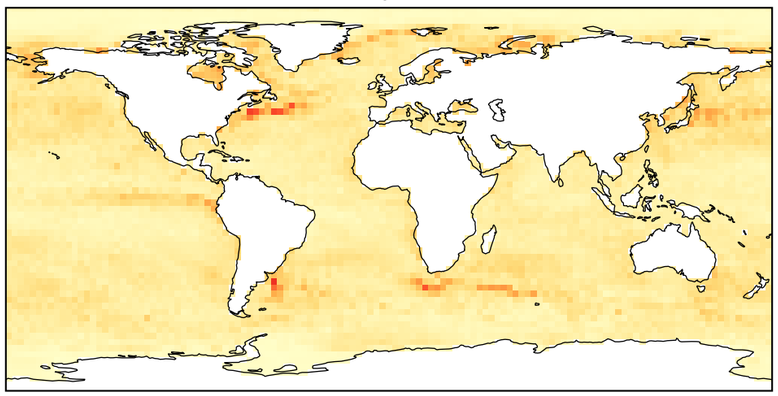
Step 7



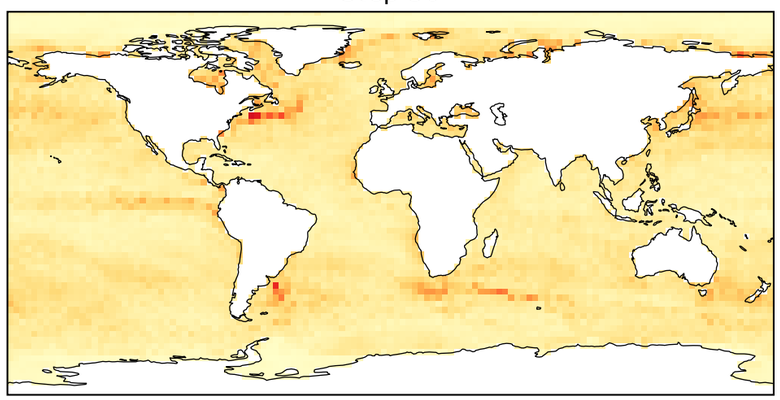
Step 14



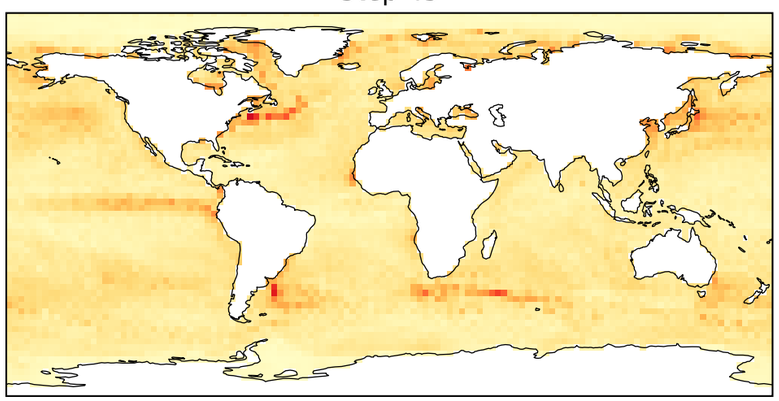
Step 21



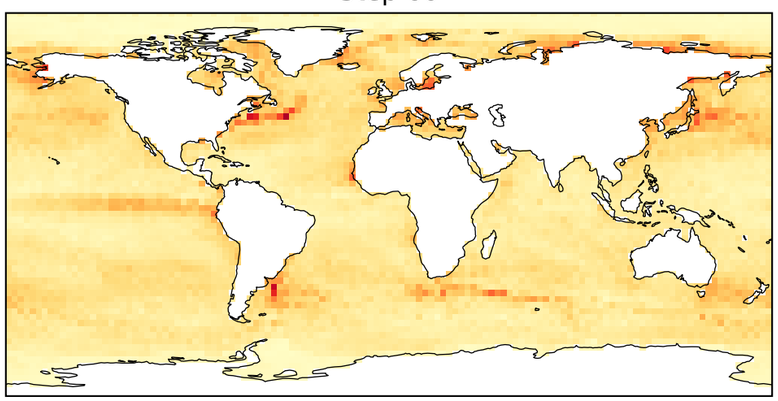
Step 30



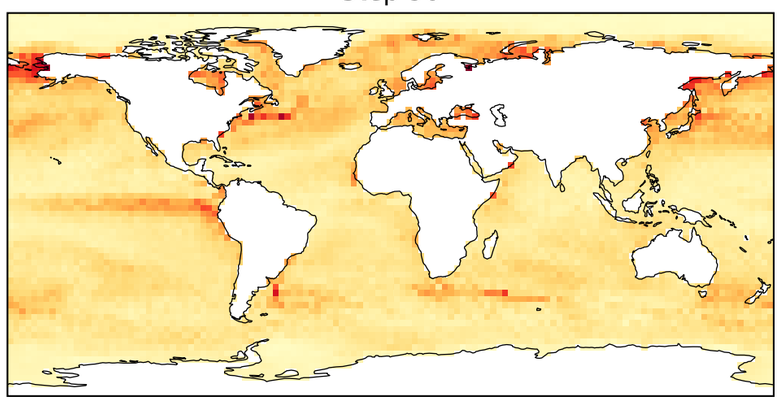
Step 45

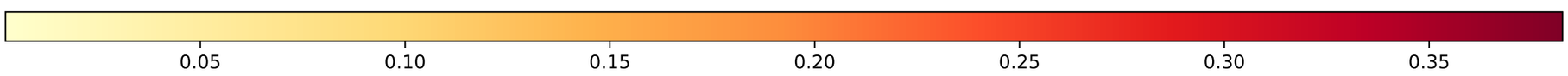


Step 60



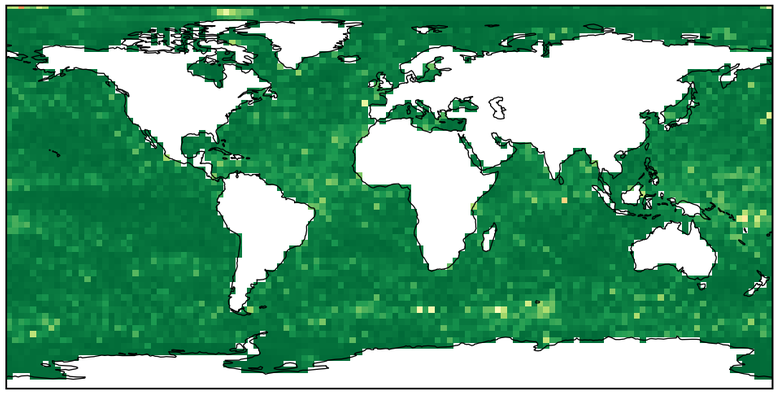
Step 90



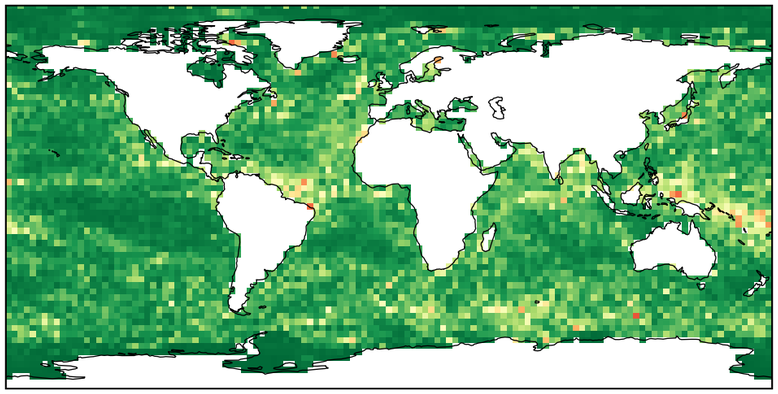


Spatial ACC (ocean only) — Validation

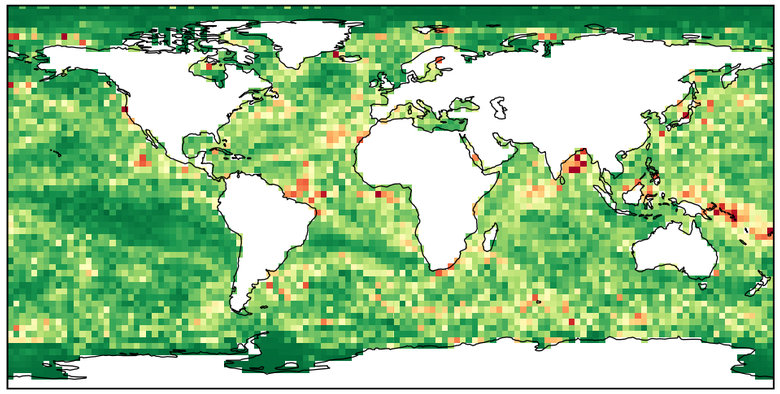
Step 1



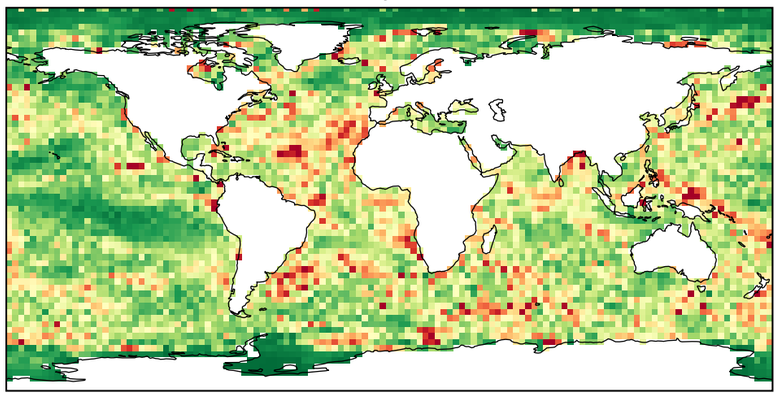
Step 3



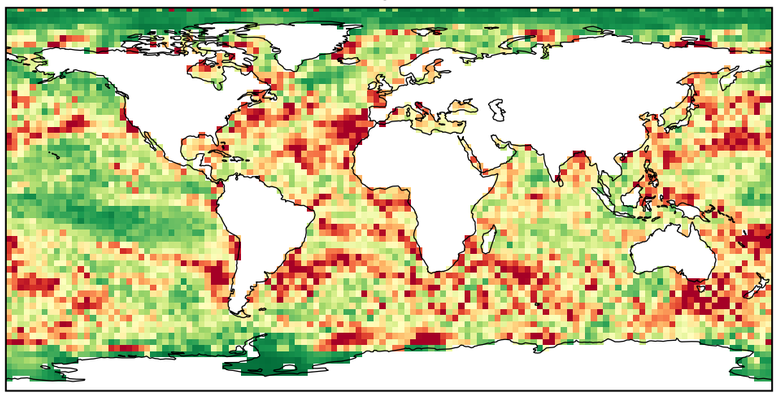
Step 7



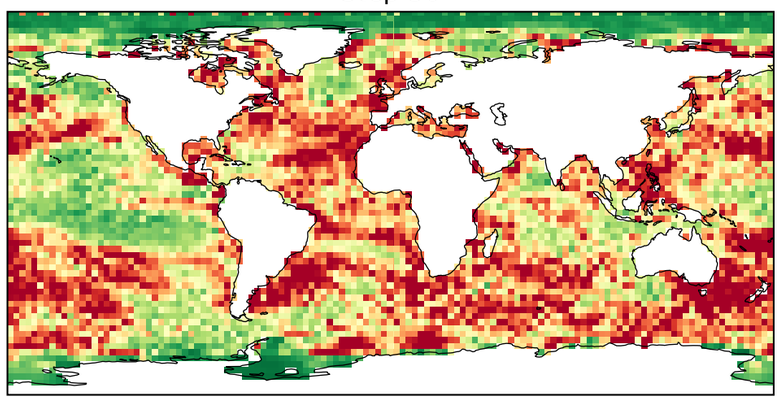
Step 14



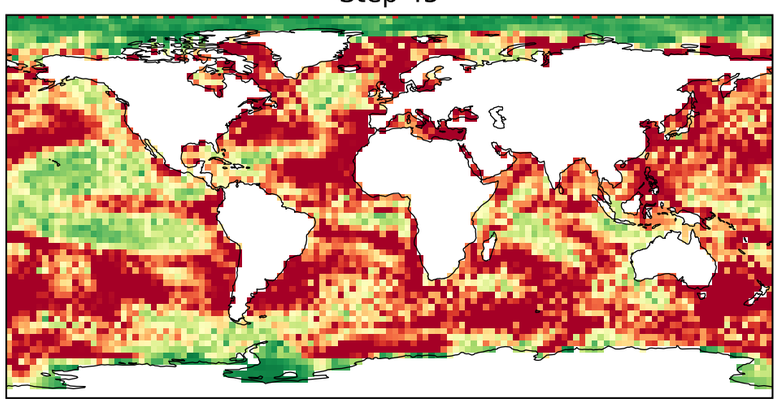
Step 21



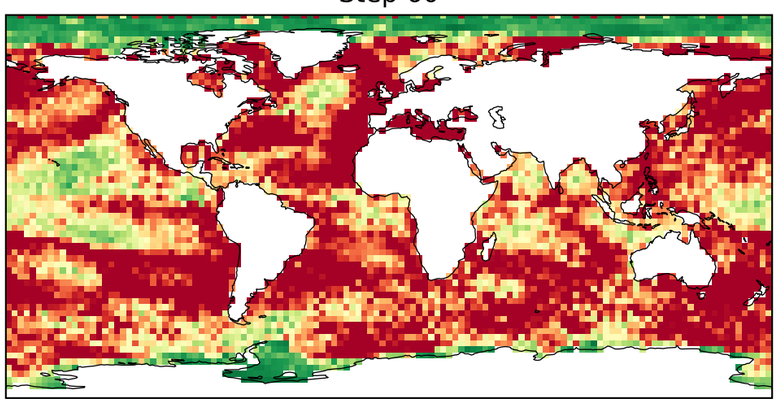
Step 30



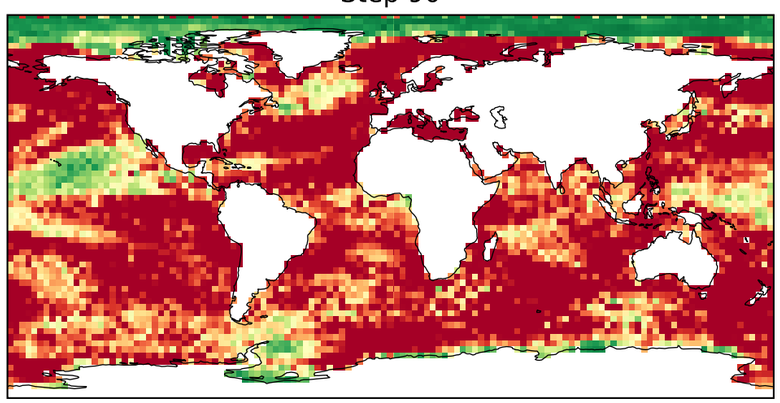
Step 45

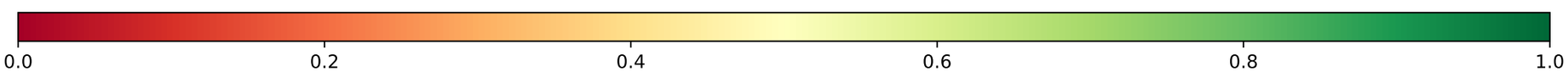


Step 60



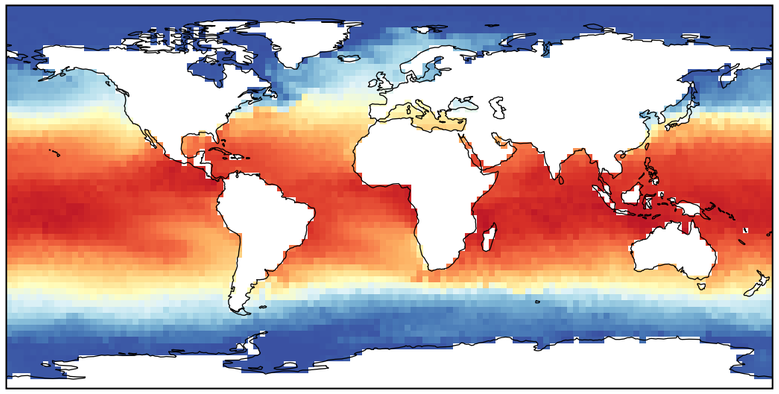
Step 90



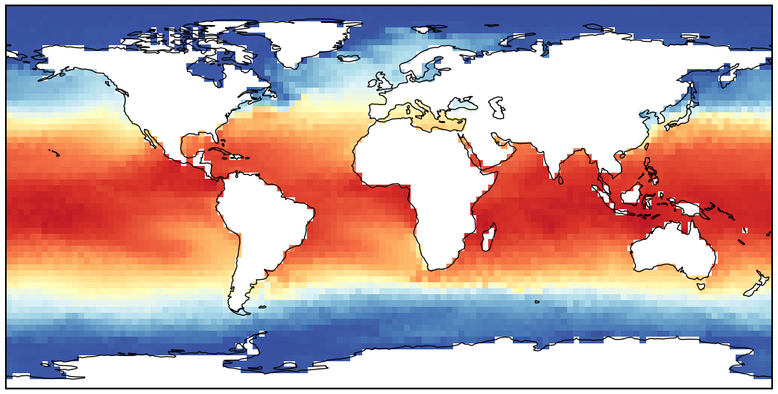


Chain init: 2016-01-11 — Validation (°C)

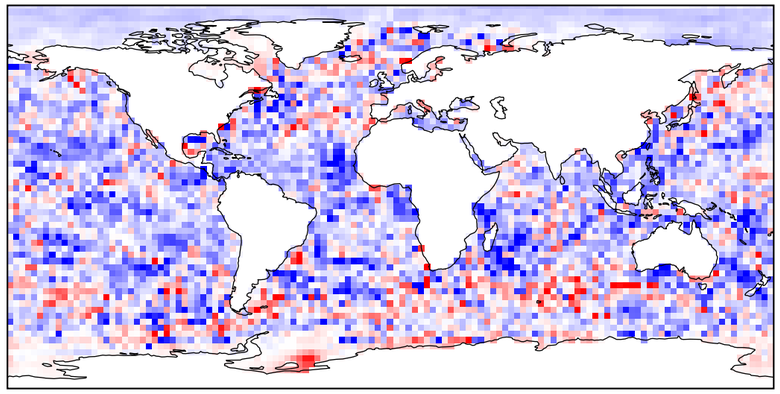
Actual — Step 1 (°C)



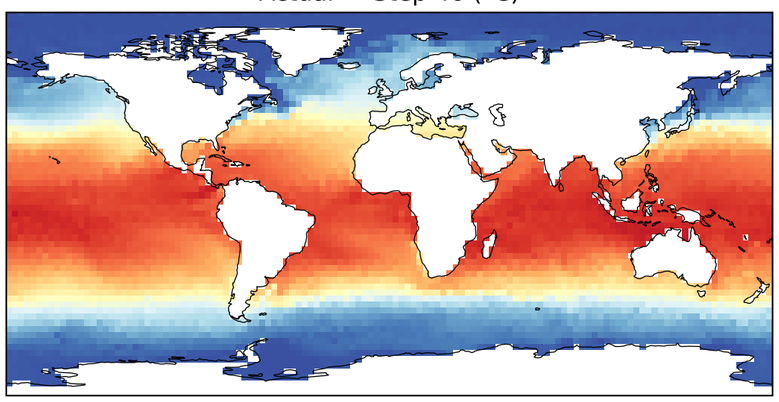
SFNO — Step 1 (°C)



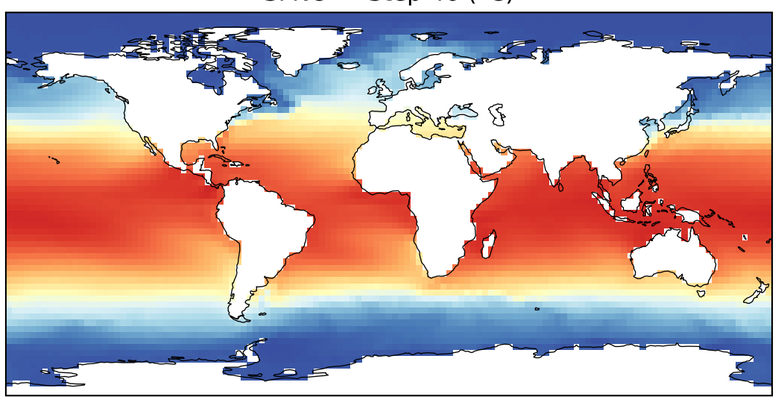
Error — Step 1 (°C)



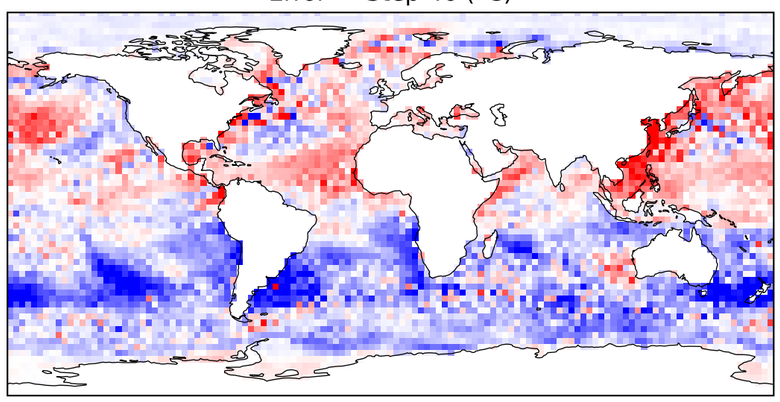
Actual — Step 46 (°C)



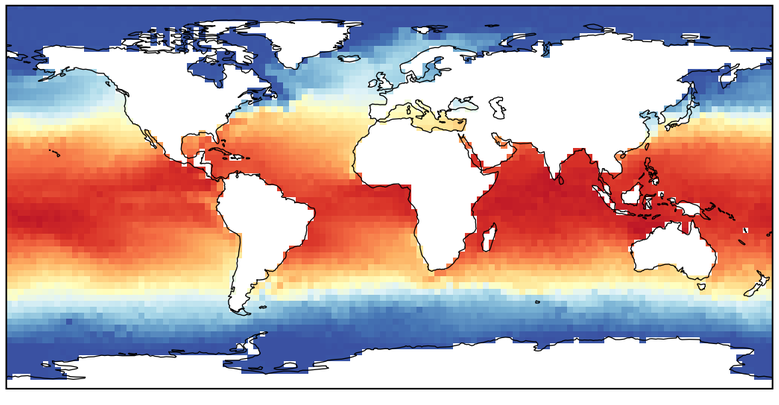
SFNO — Step 46 (°C)



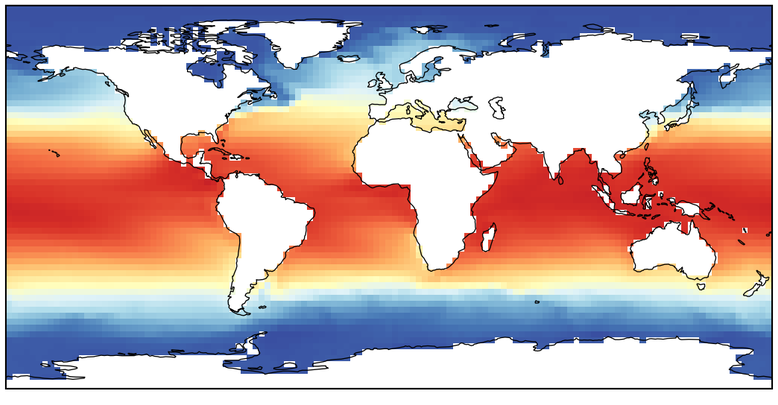
Error — Step 46 (°C)



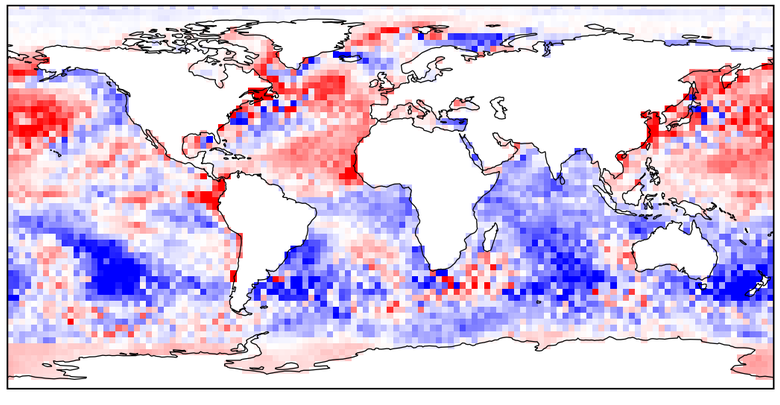
Actual — Step 90 (°C)

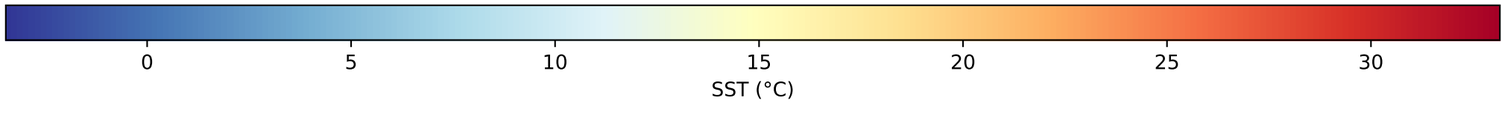


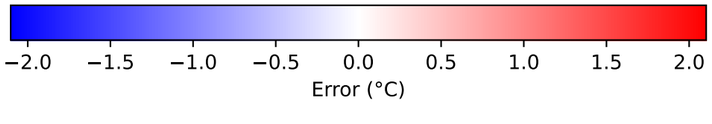
SFNO — Step 90 (°C)



Error — Step 90 (°C)

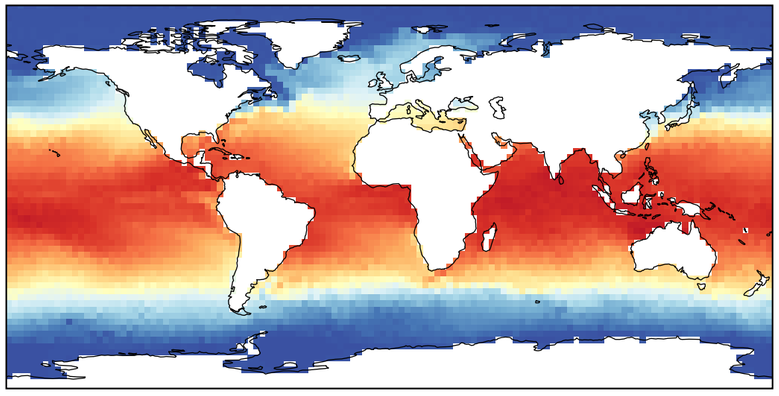




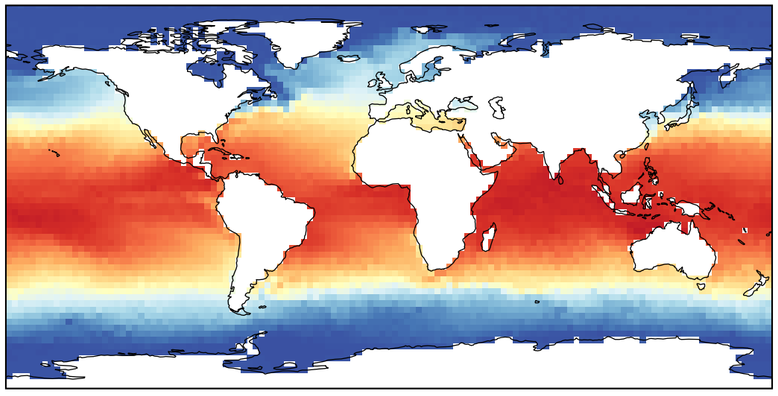


Chain init: 2016-04-10 — Validation (°C)

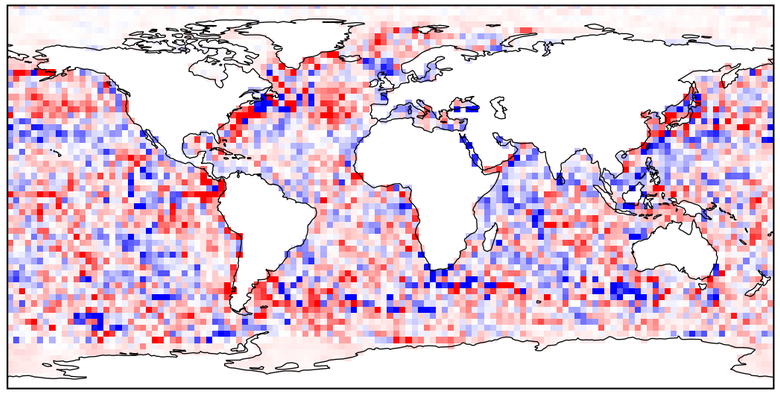
Actual — Step 1 (°C)



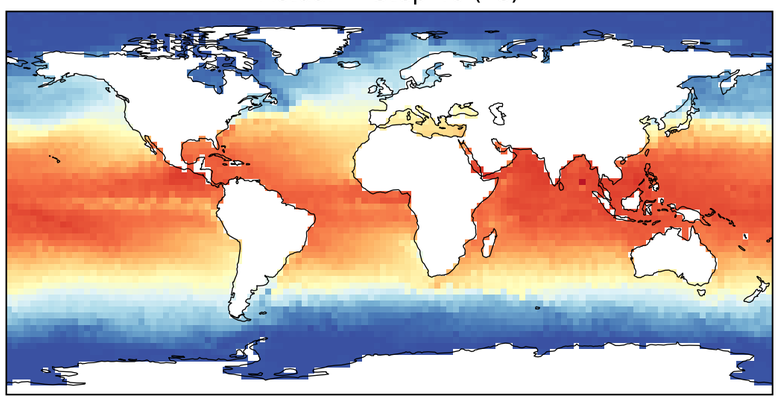
SFNO — Step 1 (°C)



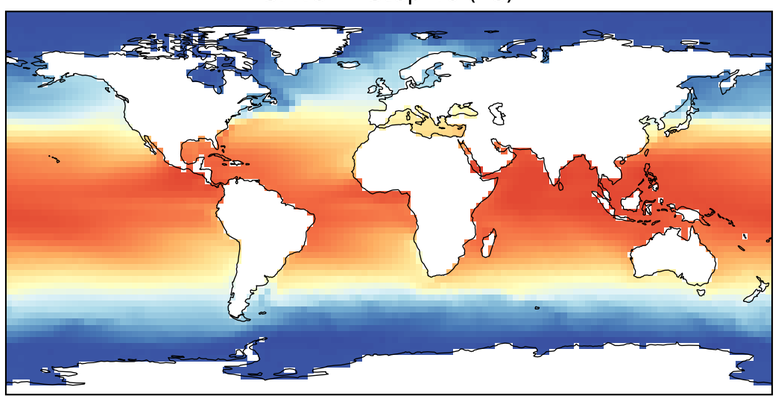
Error — Step 1 (°C)



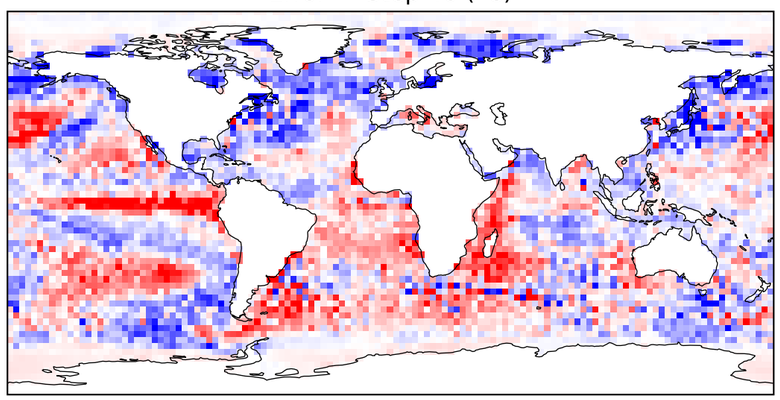
Actual — Step 46 (°C)



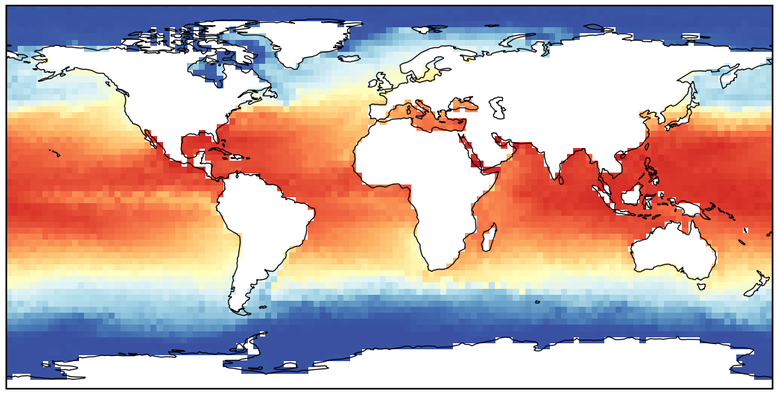
SFNO — Step 46 (°C)



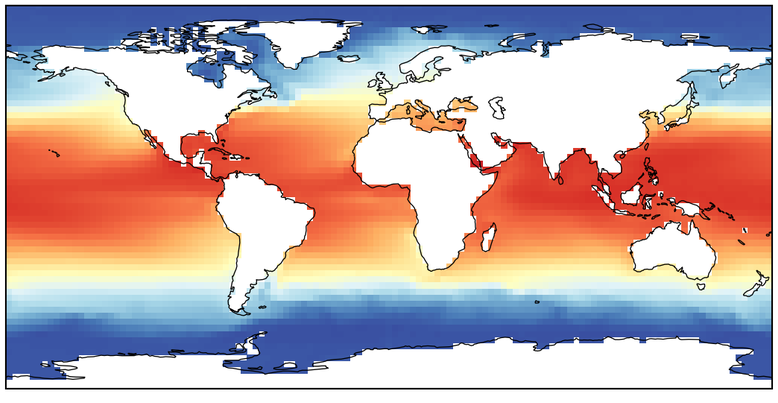
Error — Step 46 (°C)



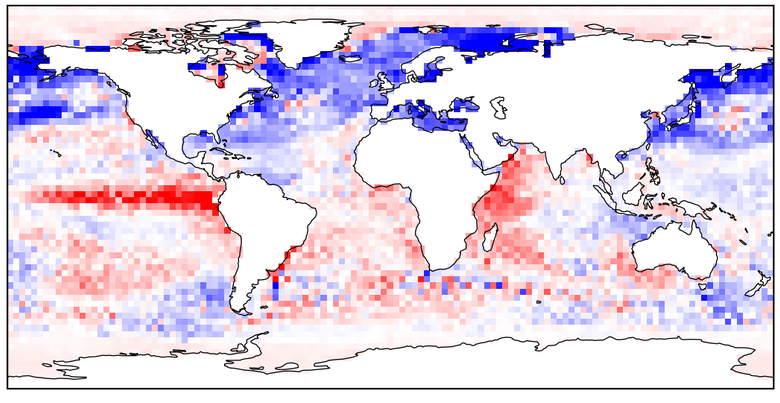
Actual — Step 90 (°C)

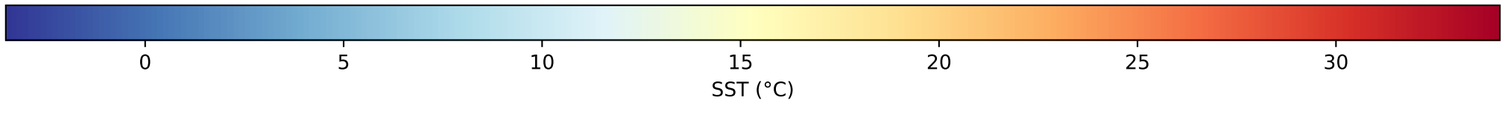


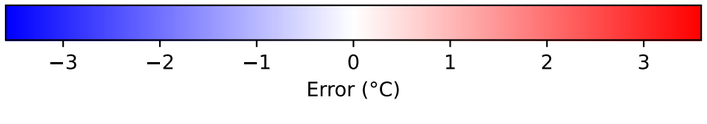
SFNO — Step 90 (°C)



Error — Step 90 (°C)

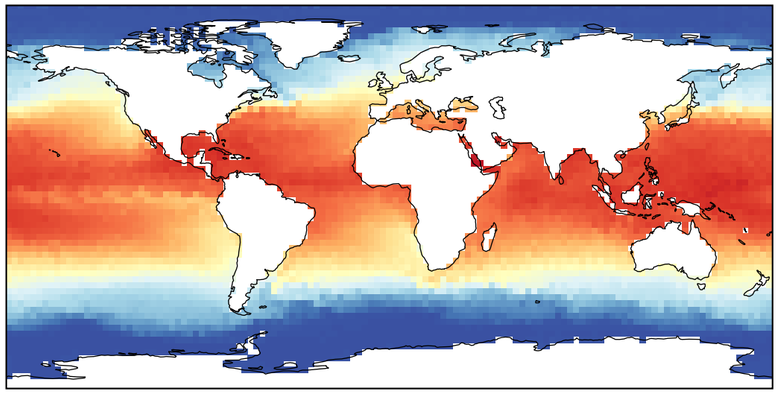




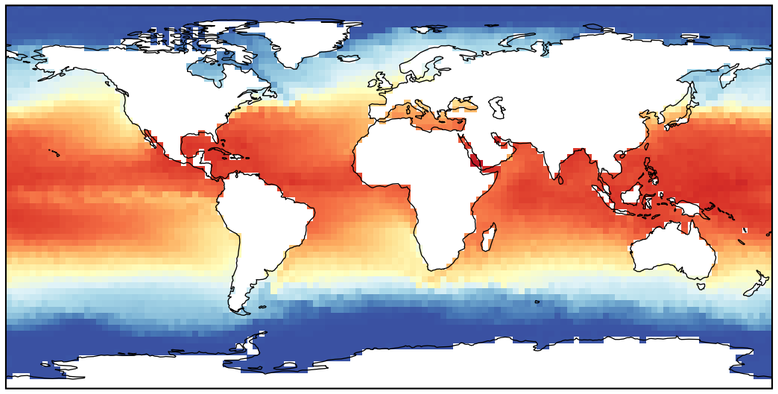


Chain init: 2016-10-07 — Validation (°C)

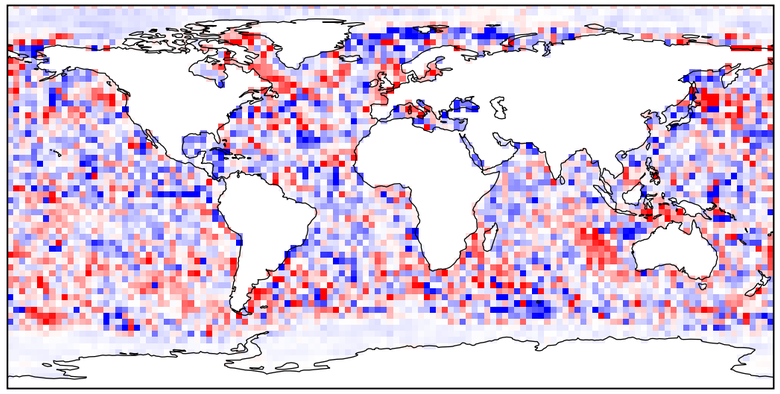
Actual — Step 1 (°C)



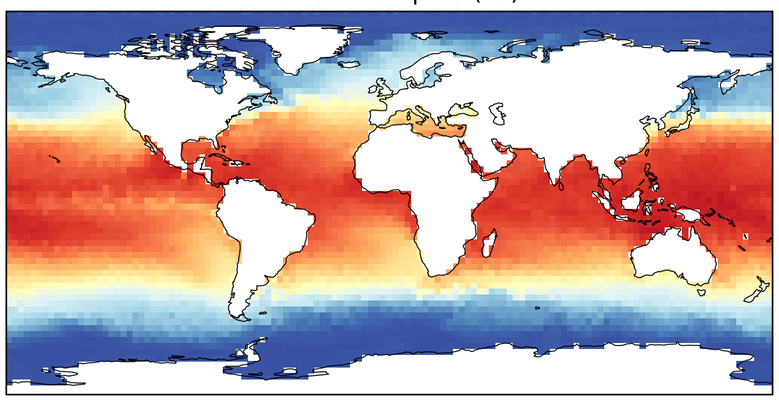
SFNO — Step 1 (°C)



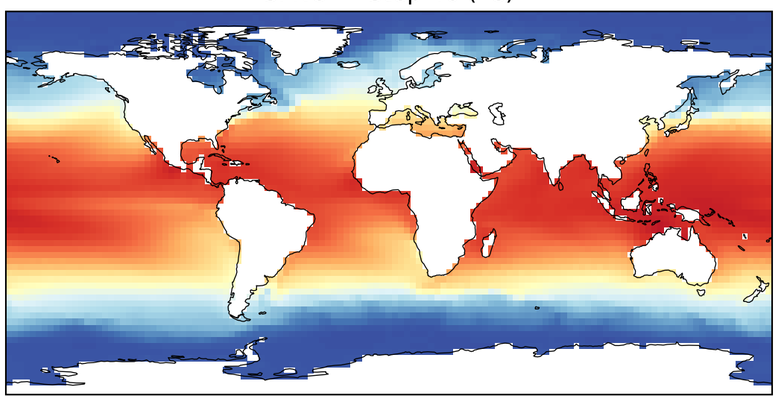
Error — Step 1 (°C)



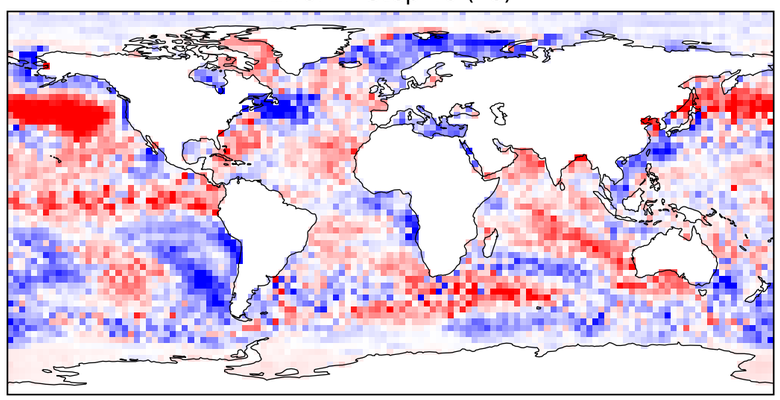
Actual — Step 46 (°C)



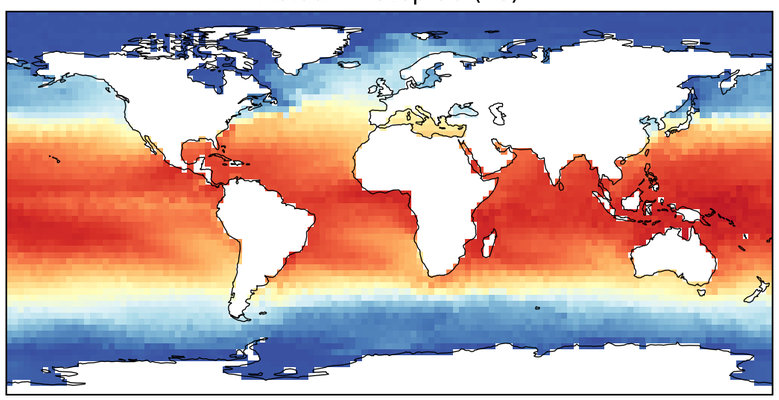
SFNO — Step 46 (°C)



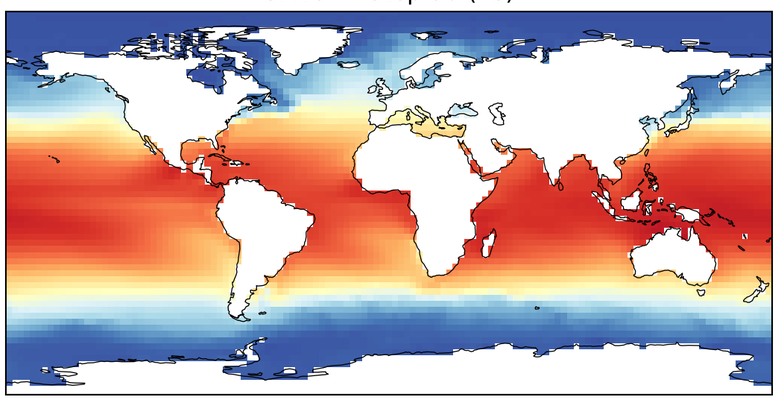
Error — Step 46 (°C)



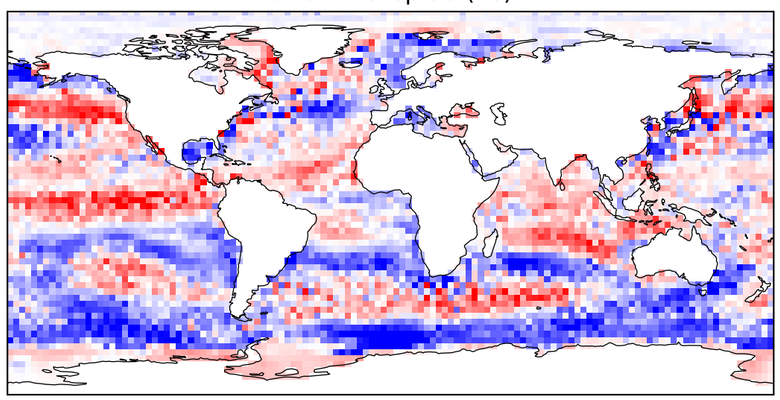
Actual — Step 90 (°C)

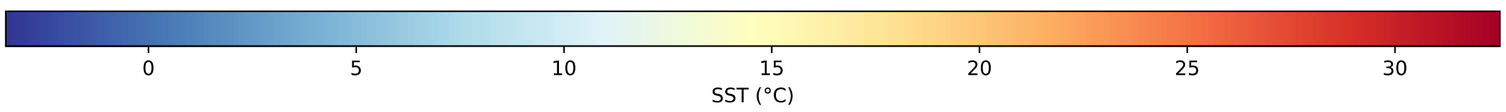


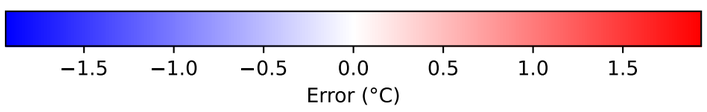
SFNO — Step 90 (°C)



Error — Step 90 (°C)

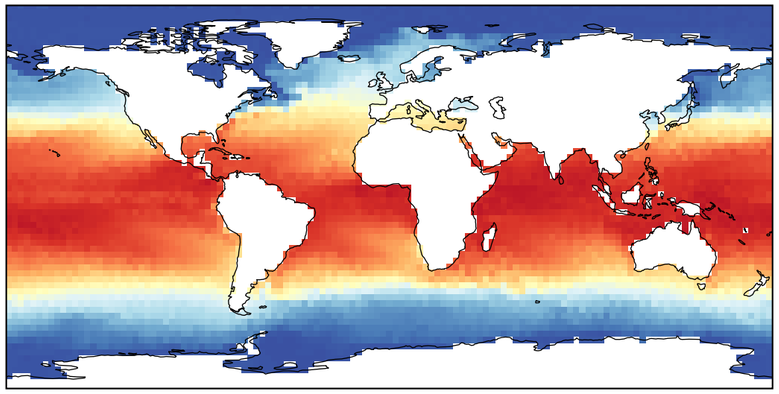




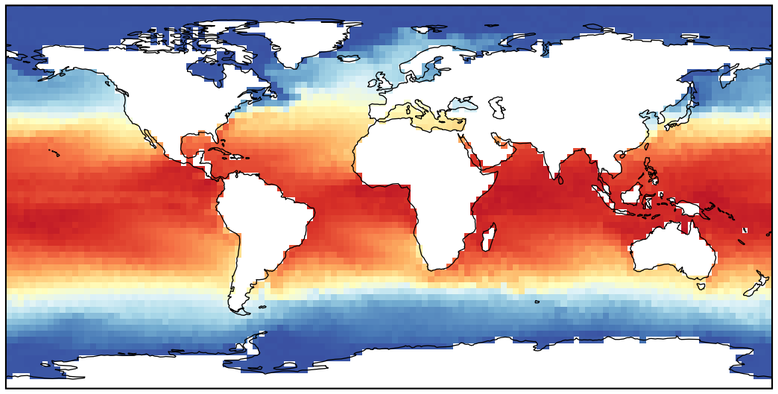


Chain init: 2017-04-05 — Validation (°C)

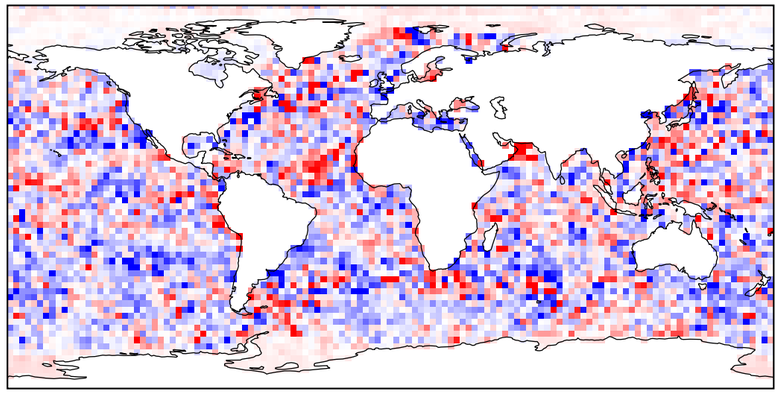
Actual — Step 1 (°C)



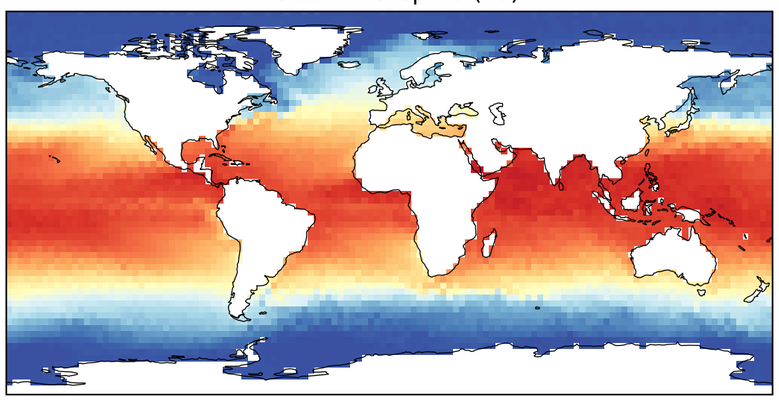
SFNO — Step 1 (°C)



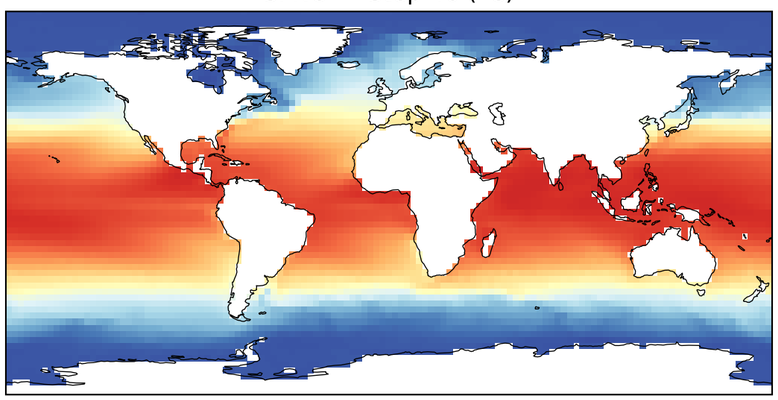
Error — Step 1 (°C)



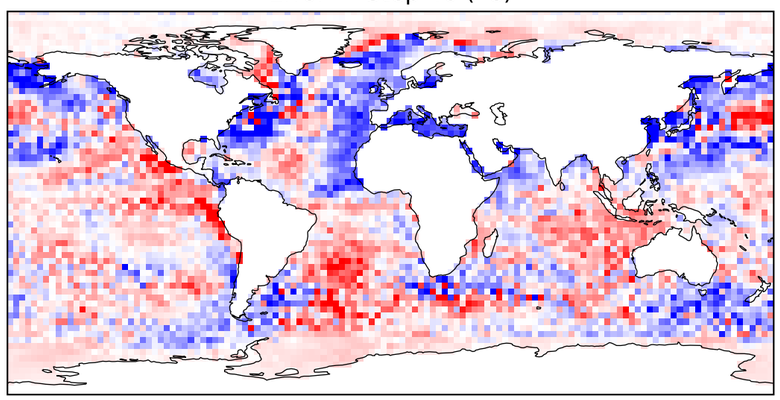
Actual — Step 46 (°C)



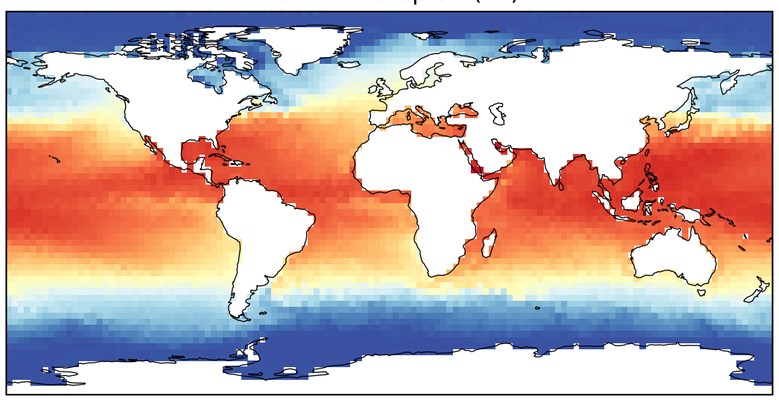
SFNO — Step 46 (°C)



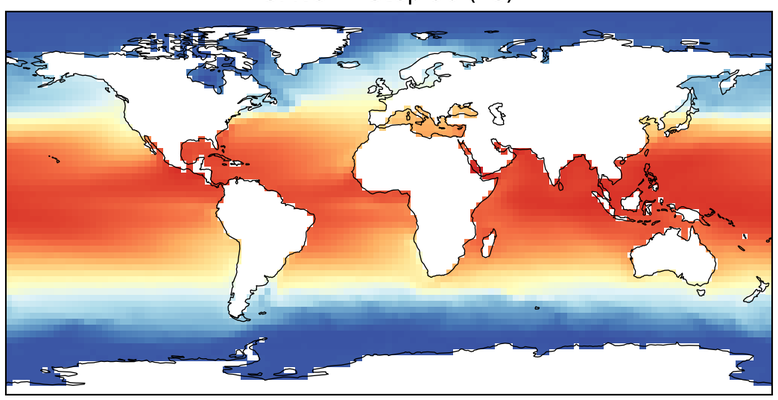
Error — Step 46 (°C)



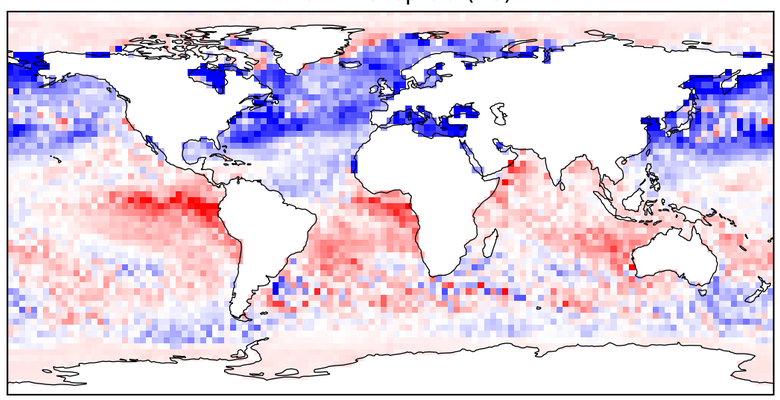
Actual — Step 90 (°C)

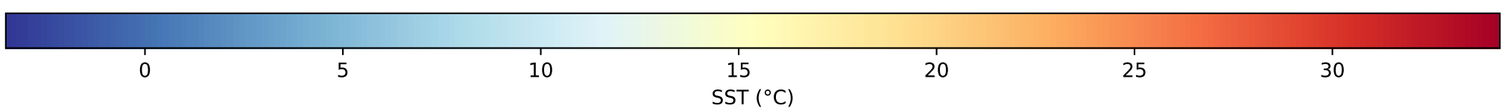


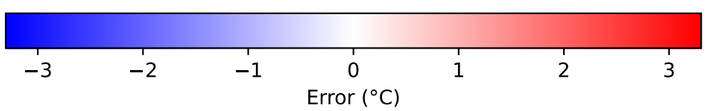
SFNO — Step 90 (°C)



Error — Step 90 (°C)

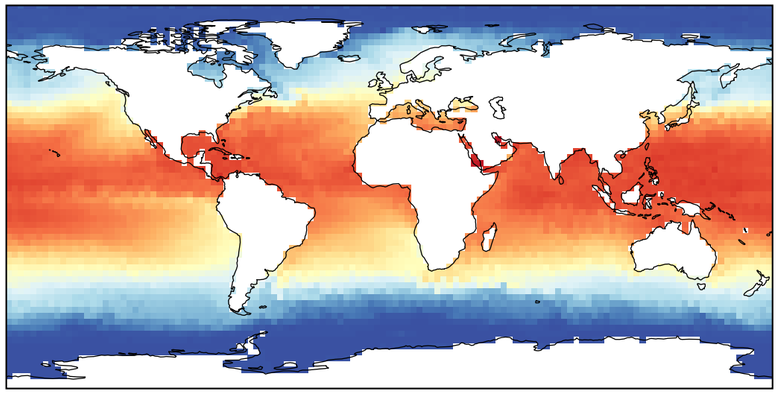




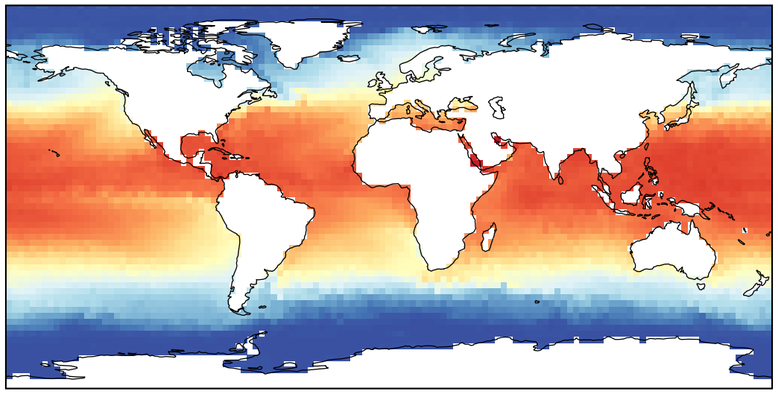


Chain init: 2017-10-02 — Validation (°C)

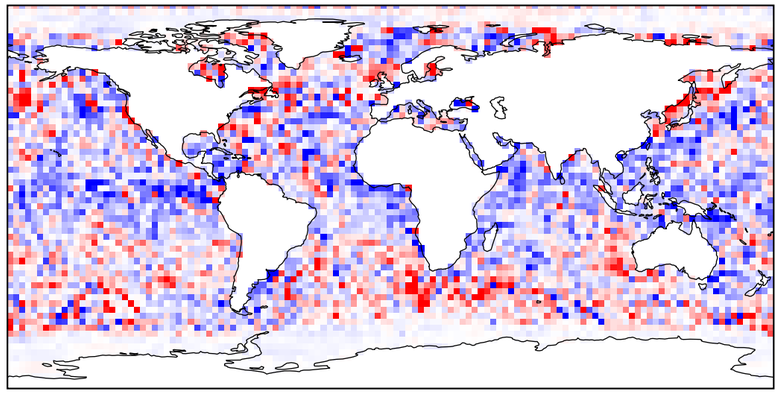
Actual — Step 1 (°C)



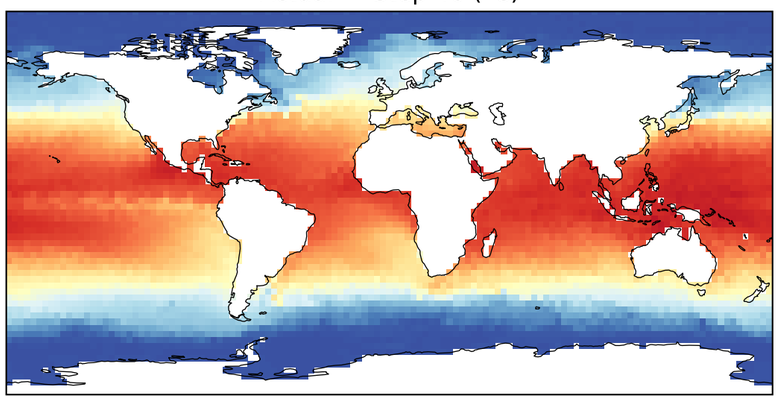
SFNO — Step 1 (°C)



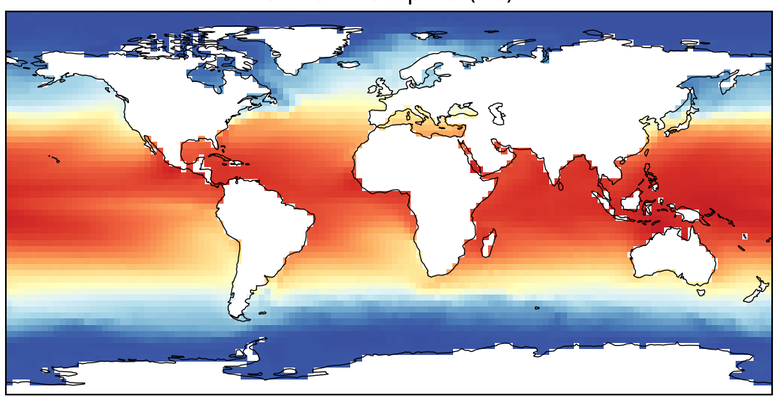
Error — Step 1 (°C)



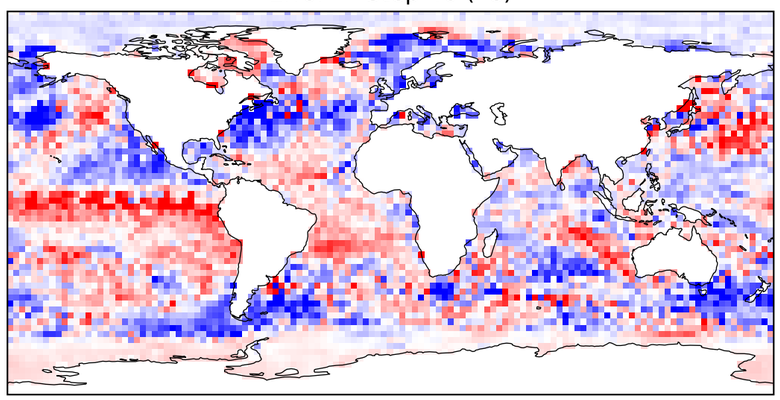
Actual — Step 46 (°C)



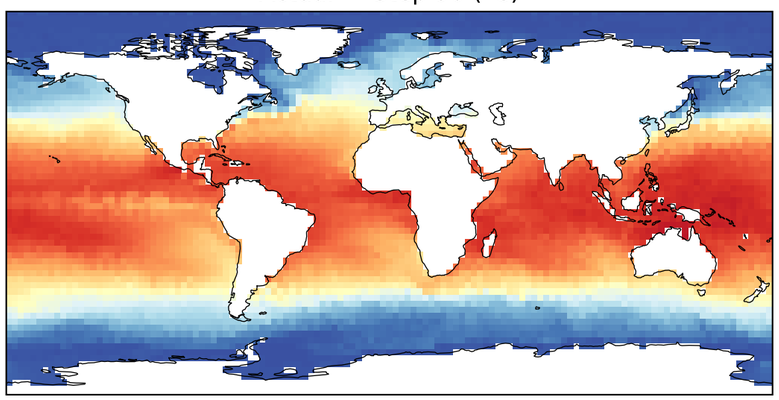
SFNO — Step 46 (°C)



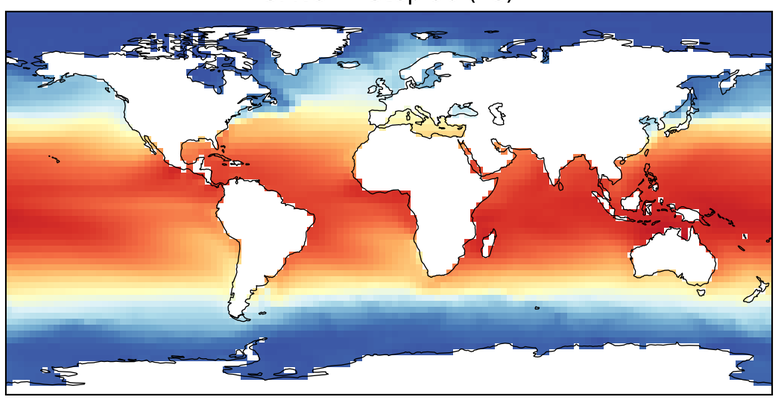
Error — Step 46 (°C)



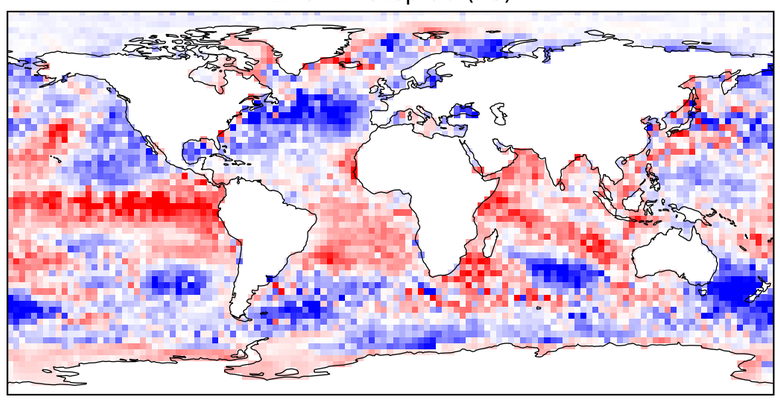
Actual — Step 90 (°C)

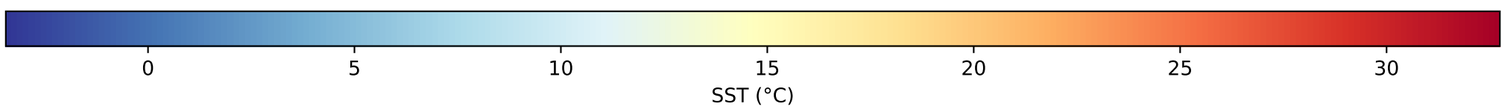


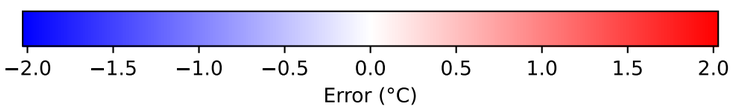
SFNO — Step 90 (°C)



Error — Step 90 (°C)

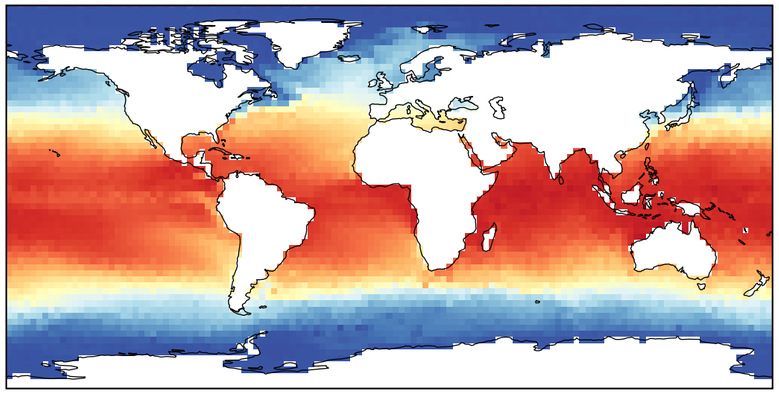




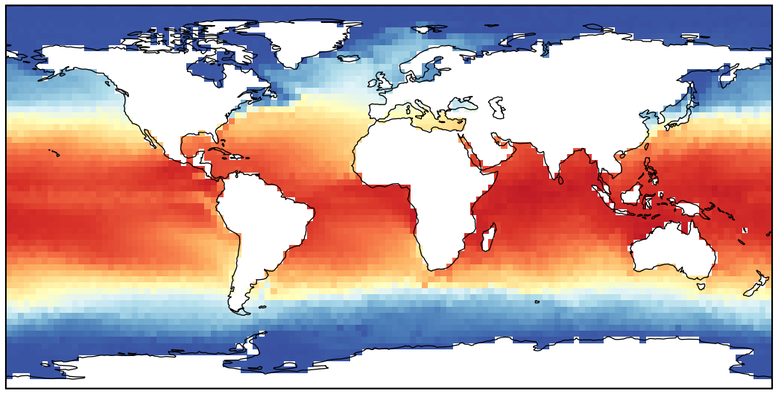


Chain init: 2018-03-31 — Validation (°C)

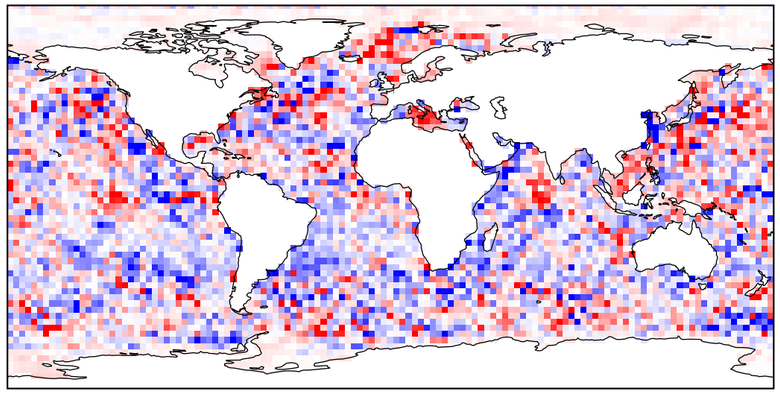
Actual — Step 1 (°C)



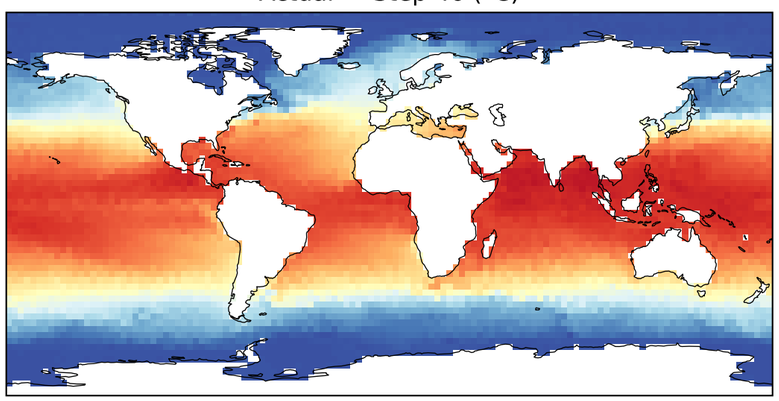
SFNO — Step 1 (°C)



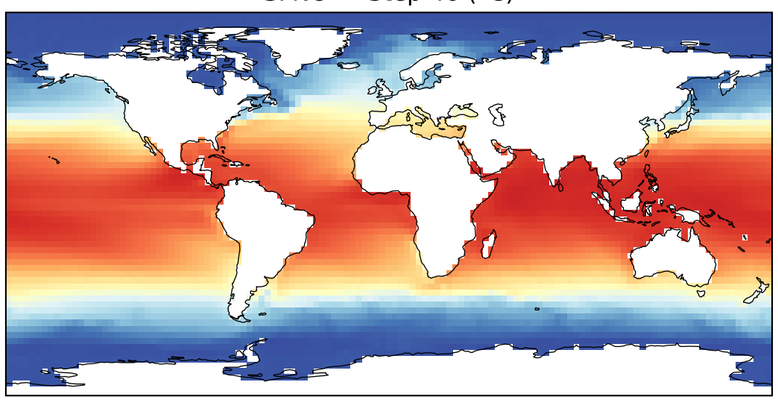
Error — Step 1 (°C)



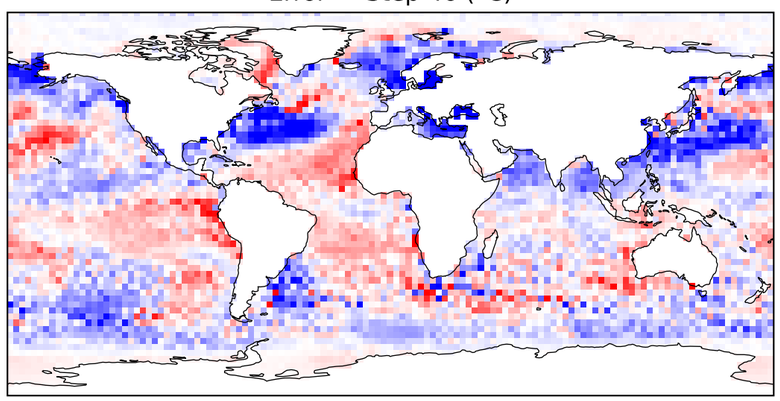
Actual — Step 46 (°C)



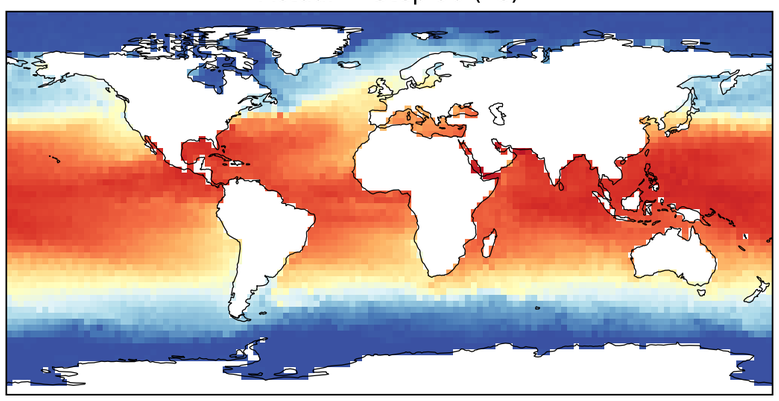
SFNO — Step 46 (°C)



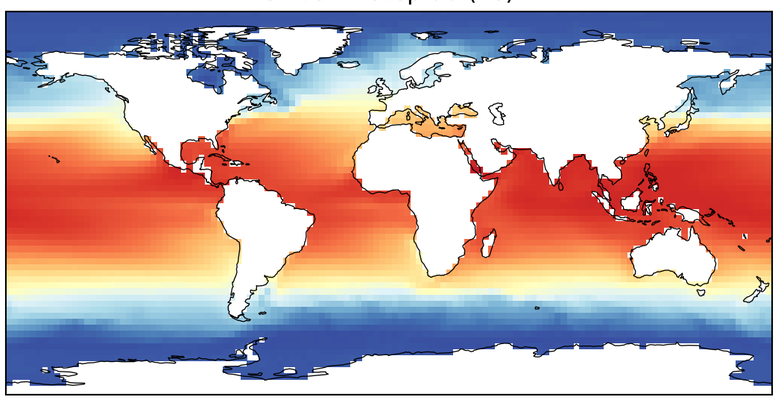
Error — Step 46 (°C)



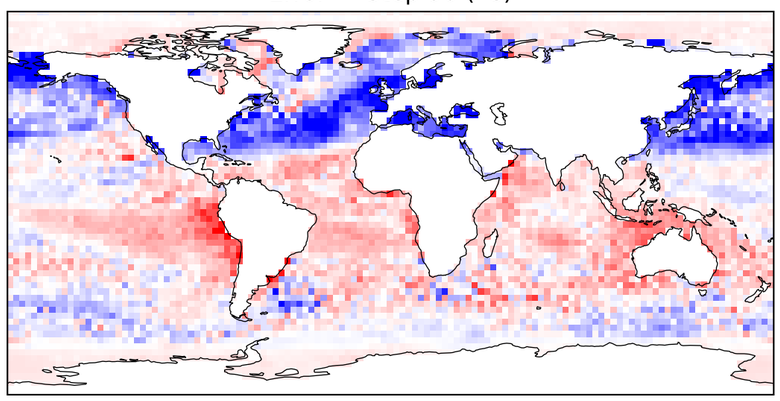
Actual — Step 90 (°C)

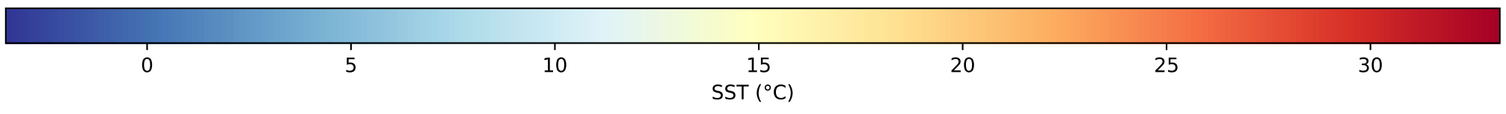


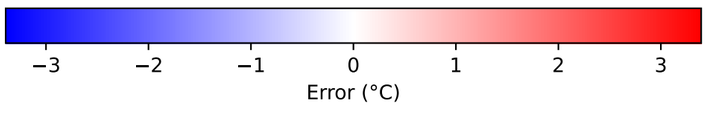
SFNO — Step 90 (°C)



Error — Step 90 (°C)

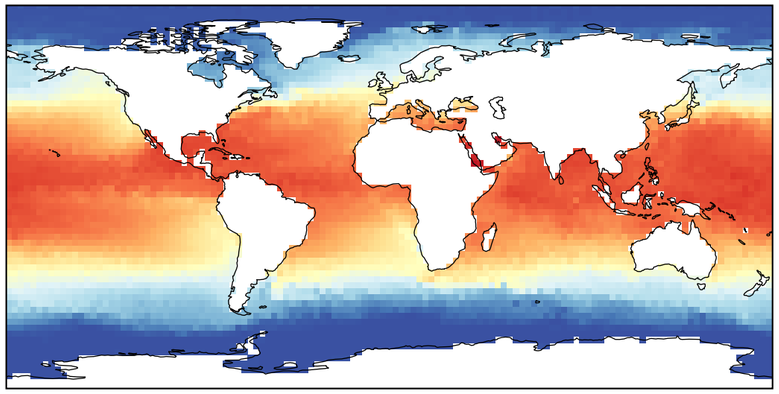




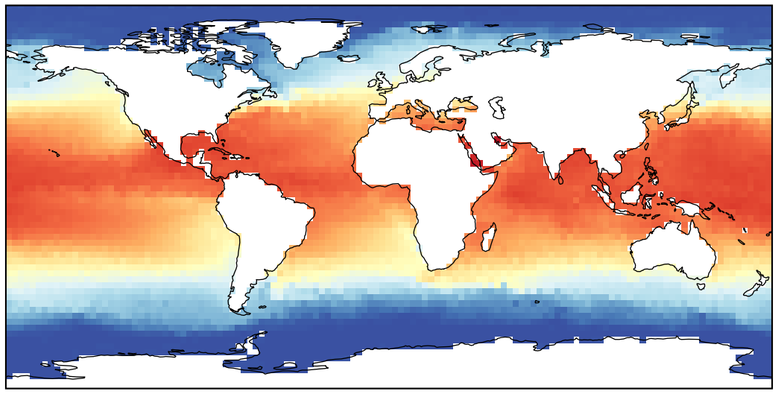


Chain init: 2018-09-27 — Validation (°C)

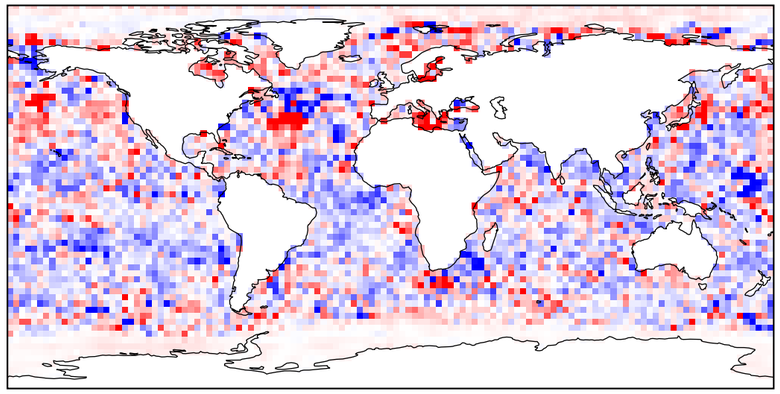
Actual — Step 1 (°C)



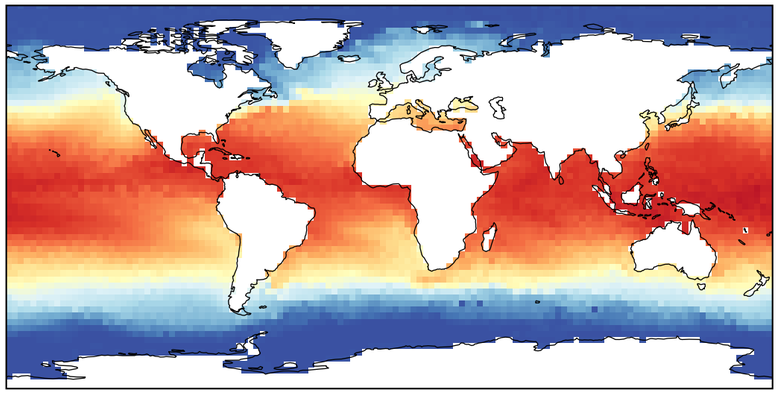
SFNO — Step 1 (°C)



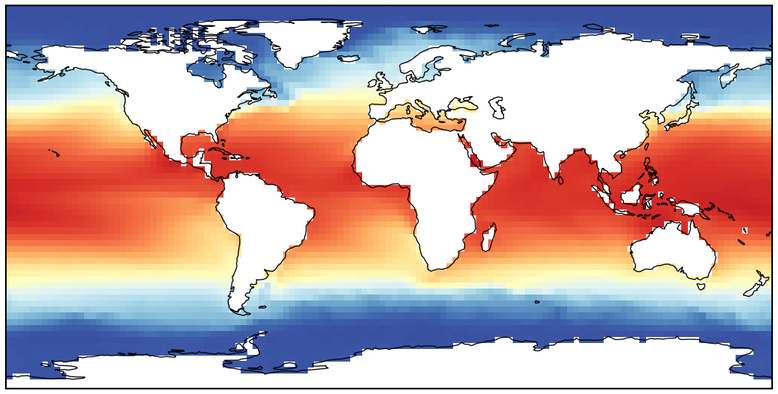
Error — Step 1 (°C)



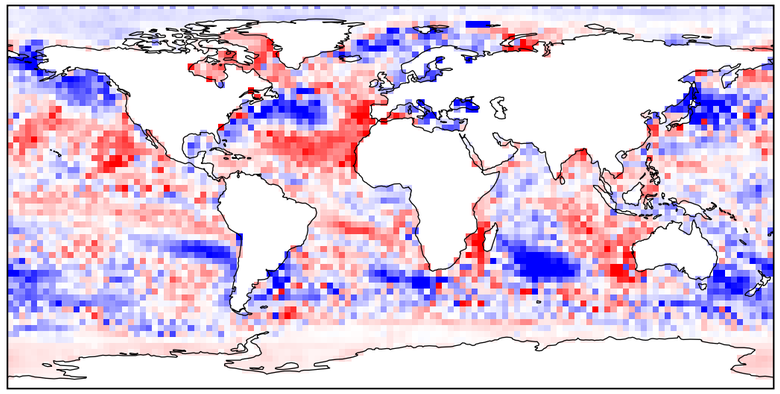
Actual — Step 46 (°C)



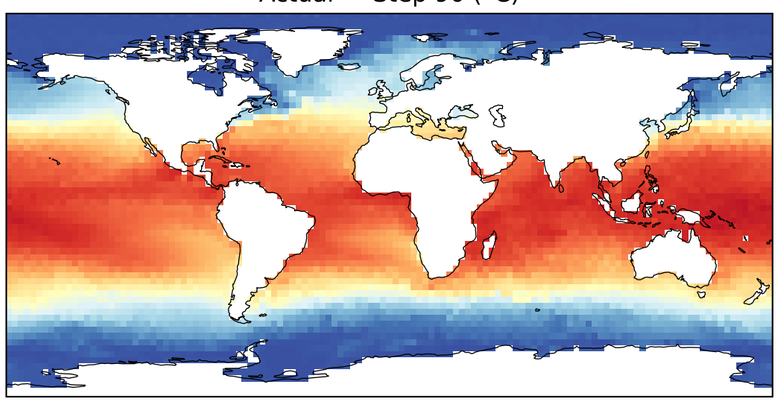
SFNO — Step 46 (°C)



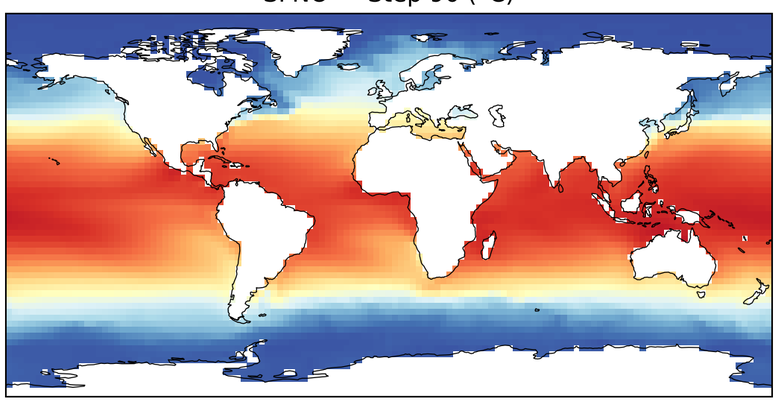
Error — Step 46 (°C)



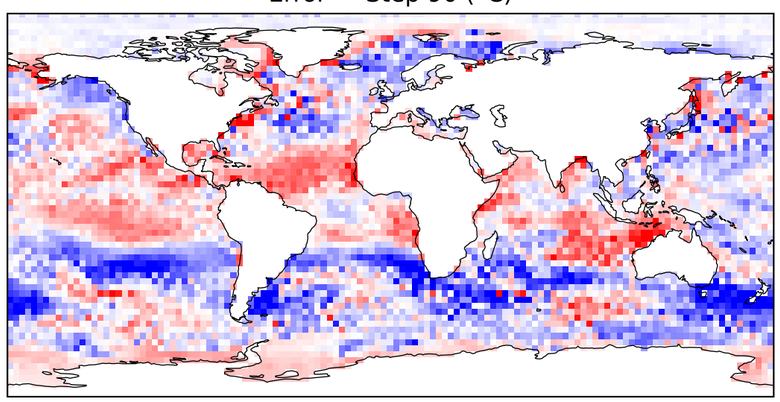
Actual — Step 90 (°C)

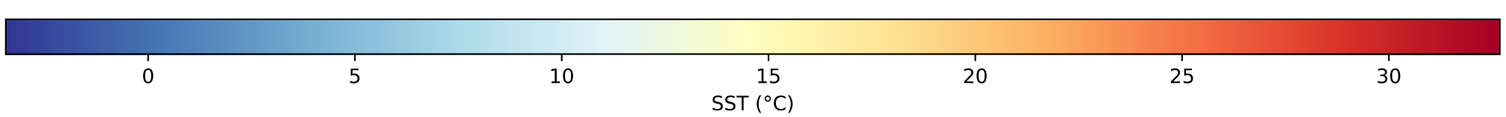


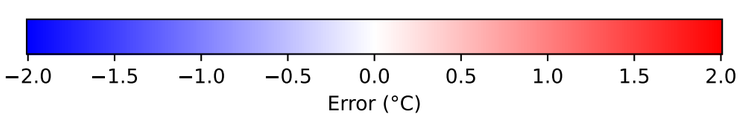
SFNO — Step 90 (°C)



Error — Step 90 (°C)

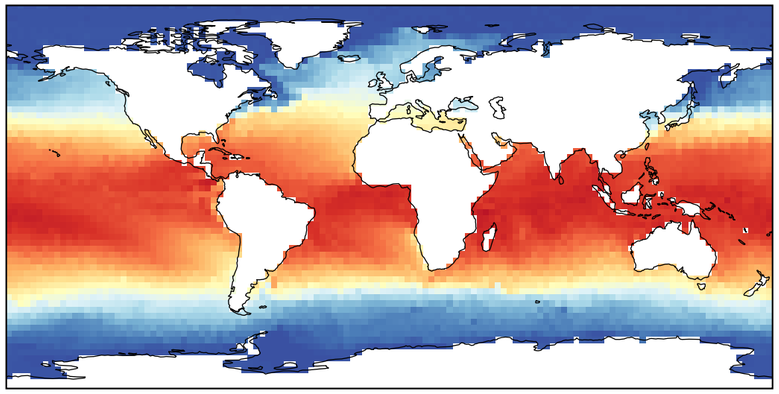




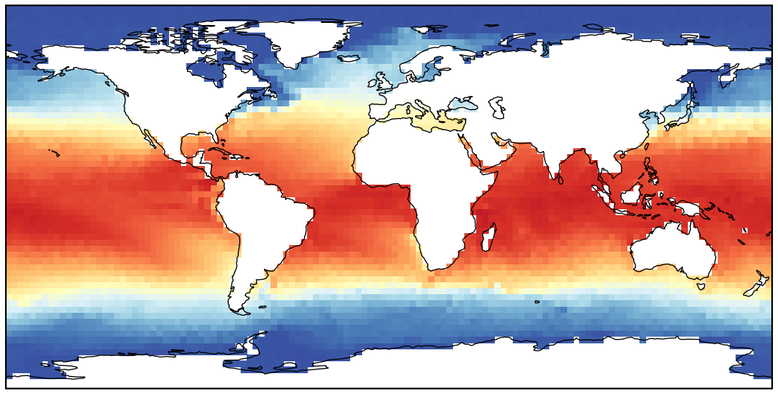


Chain init: 2019-03-26 — Validation (°C)

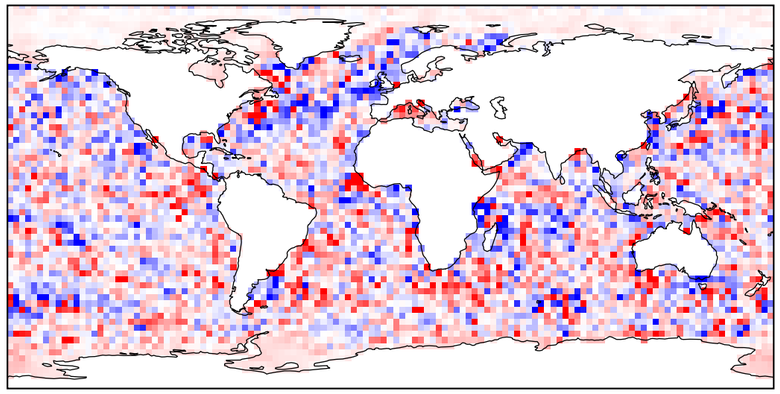
Actual — Step 1 (°C)



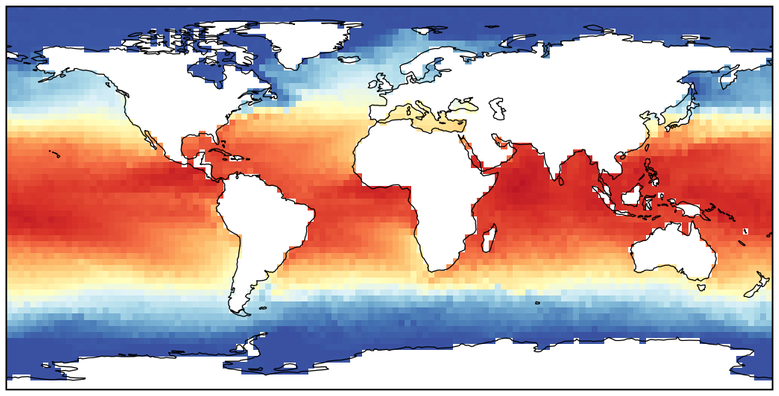
SFNO — Step 1 (°C)



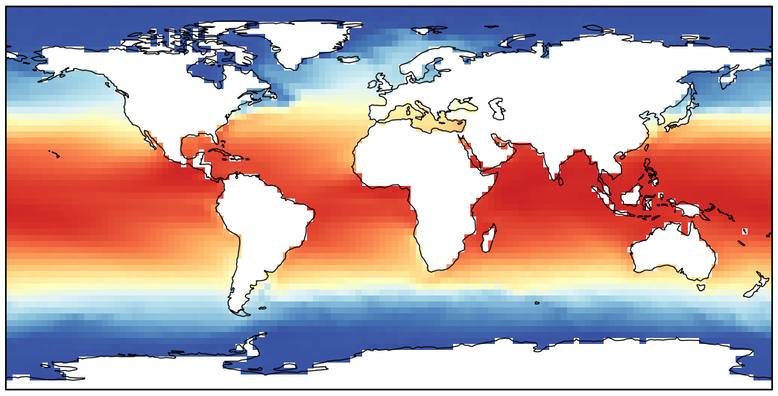
Error — Step 1 (°C)



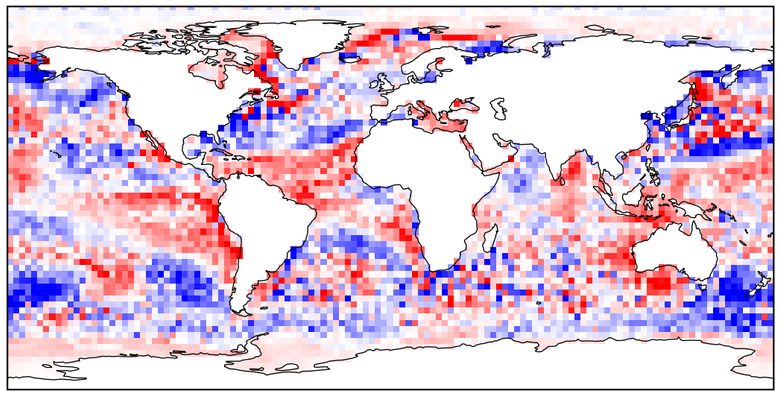
Actual — Step 46 (°C)



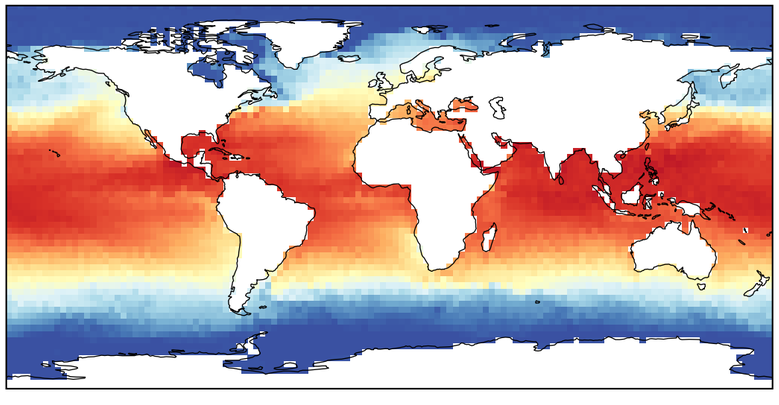
SFNO — Step 46 (°C)



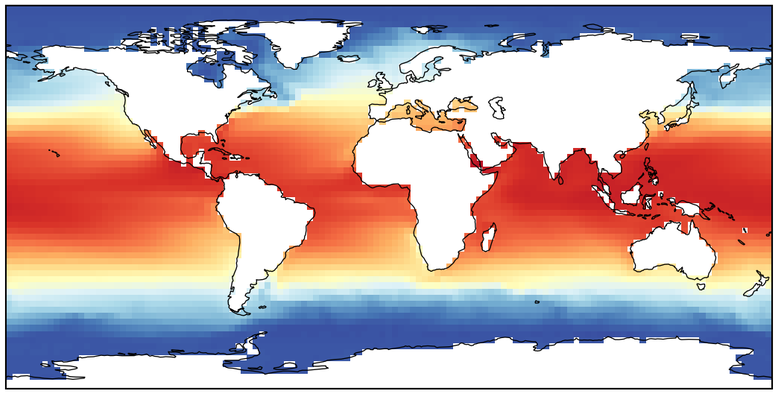
Error — Step 46 (°C)



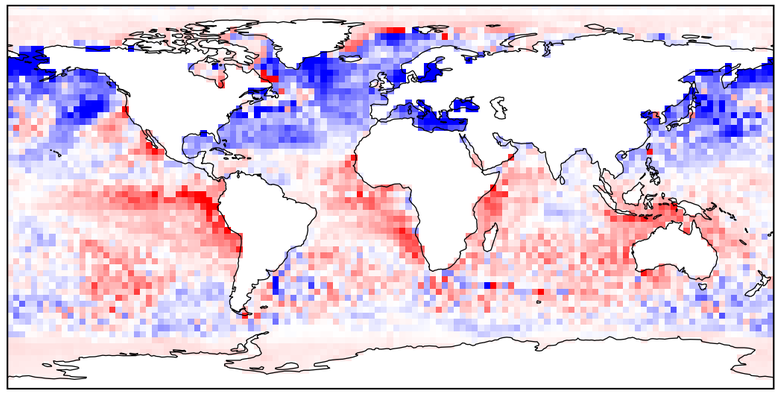
Actual — Step 90 (°C)

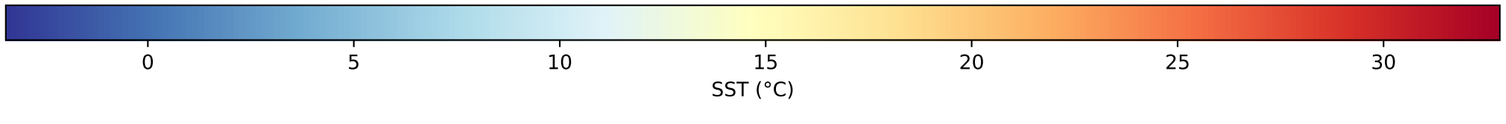


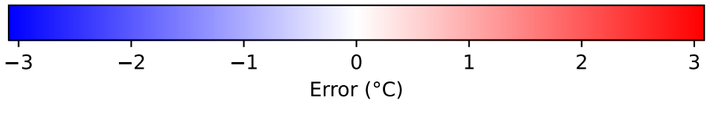
SFNO — Step 90 (°C)



Error — Step 90 (°C)

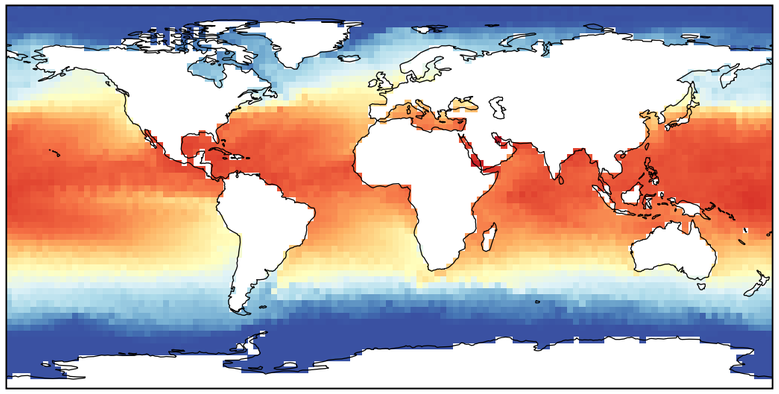




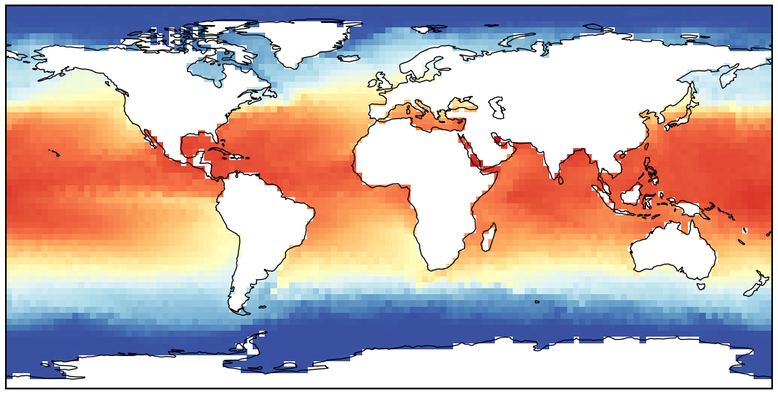


Chain init: 2019-09-22 — Validation (°C)

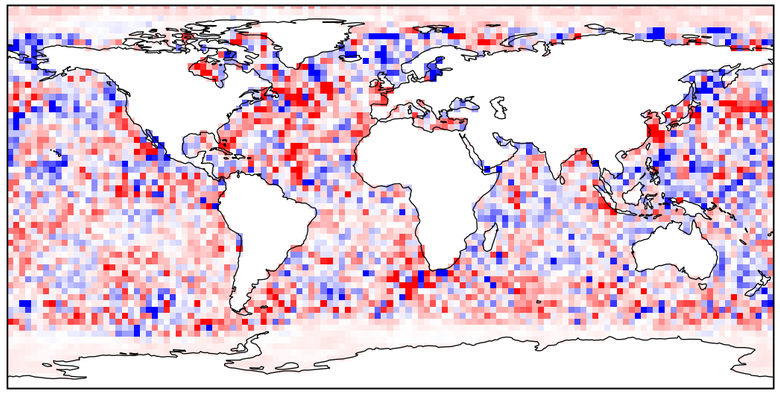
Actual — Step 1 (°C)



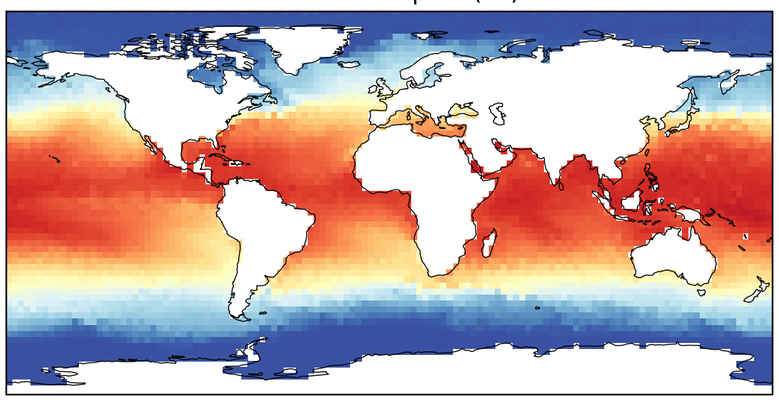
SFNO — Step 1 (°C)



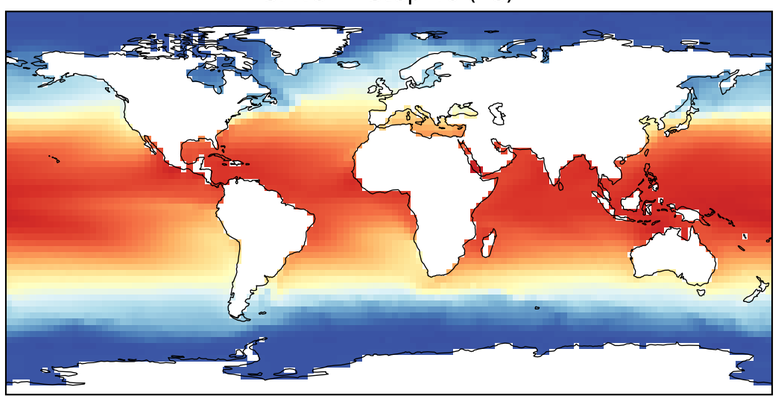
Error — Step 1 (°C)



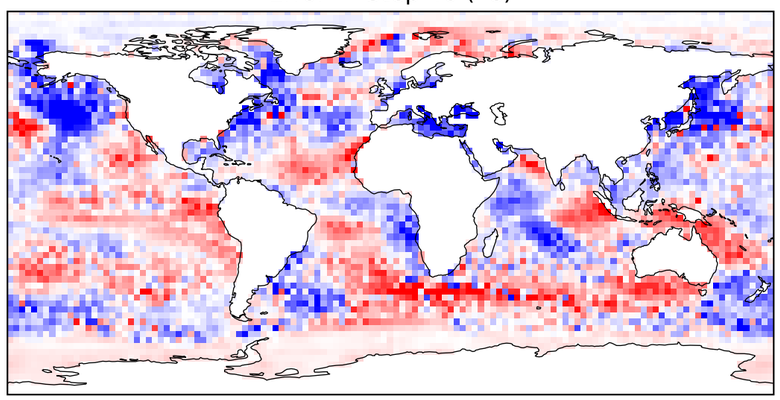
Actual — Step 46 (°C)



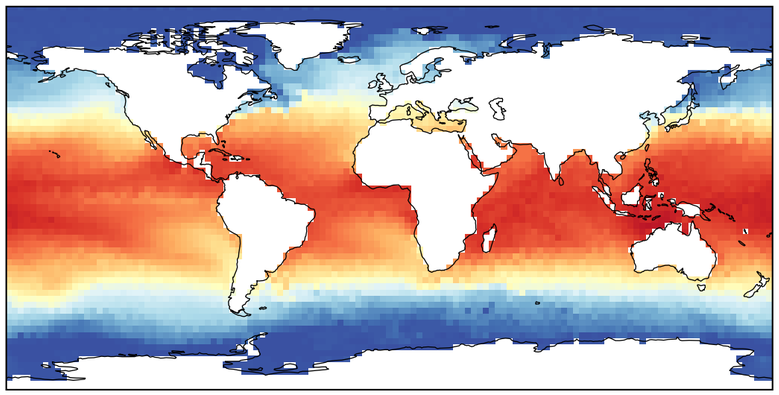
SFNO — Step 46 (°C)



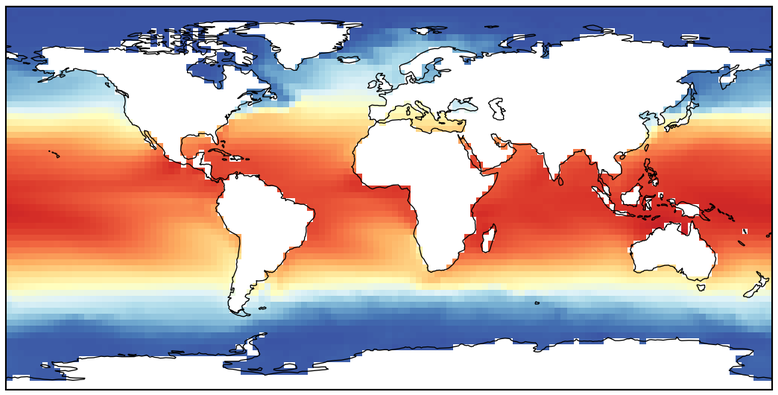
Error — Step 46 (°C)



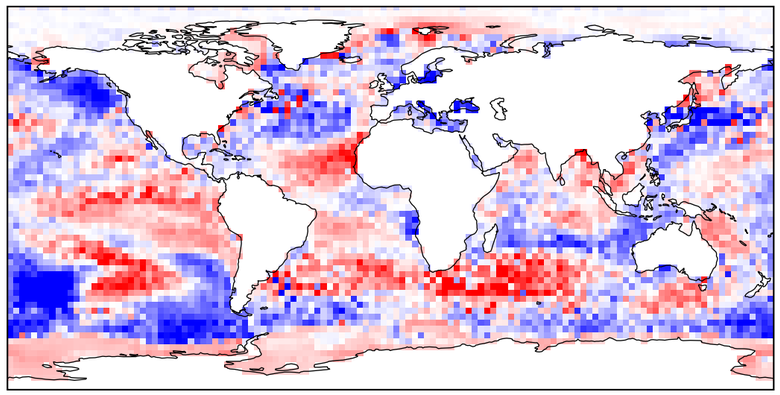
Actual — Step 90 (°C)

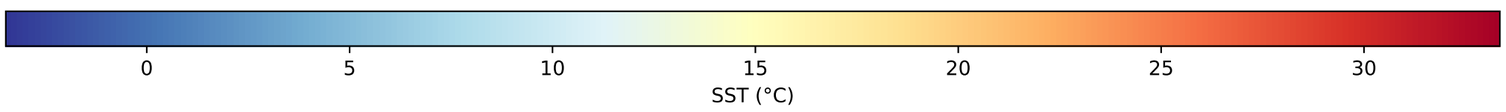


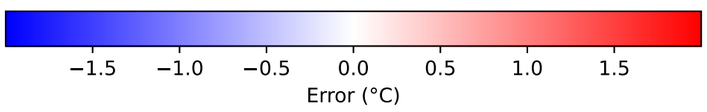
SFNO — Step 90 (°C)



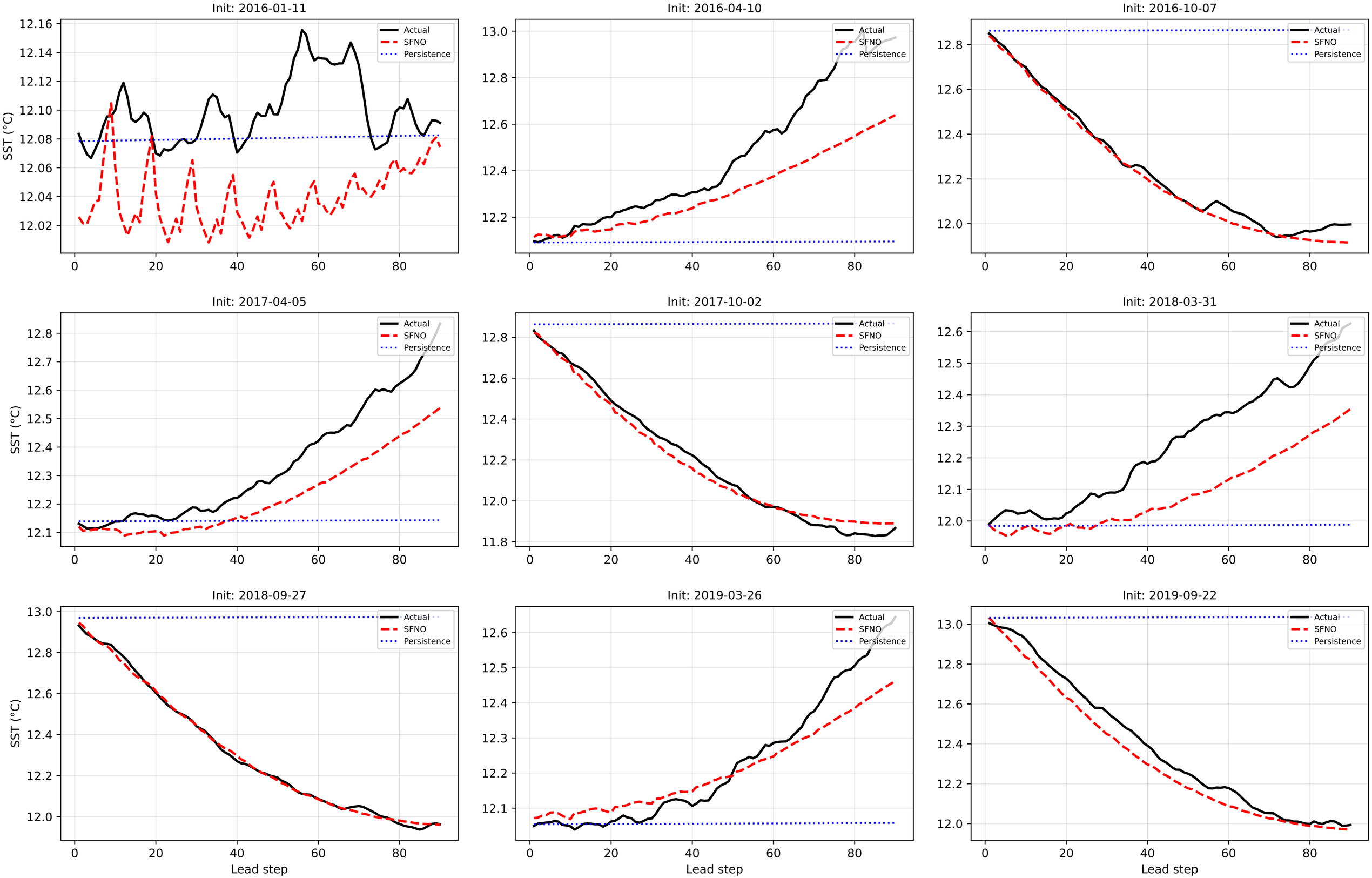
Error — Step 90 (°C)





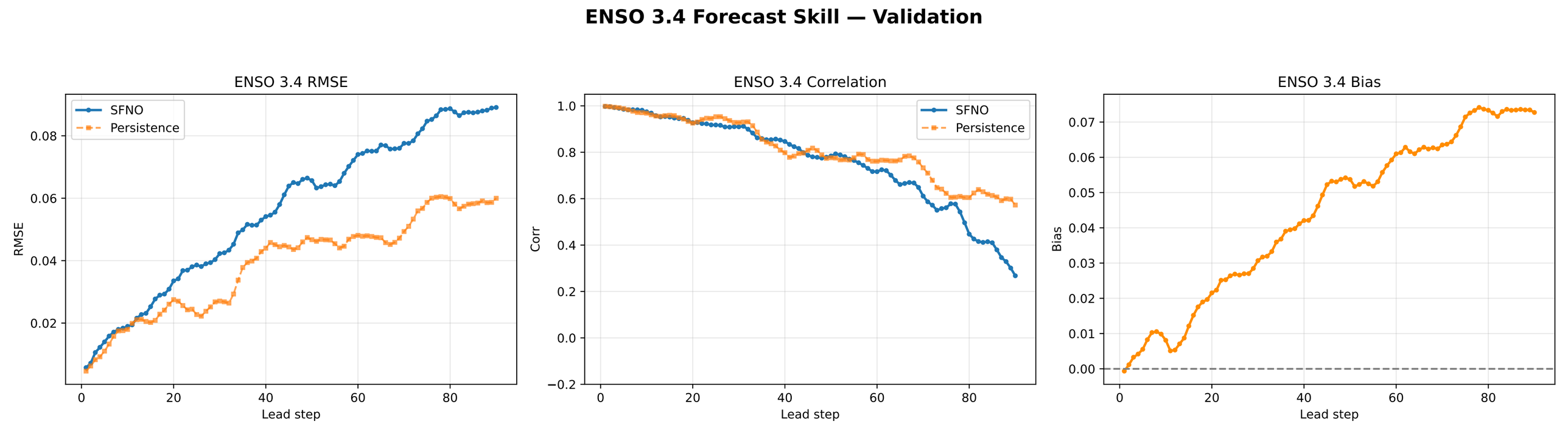


Global-Mean SST — Validation (°C)

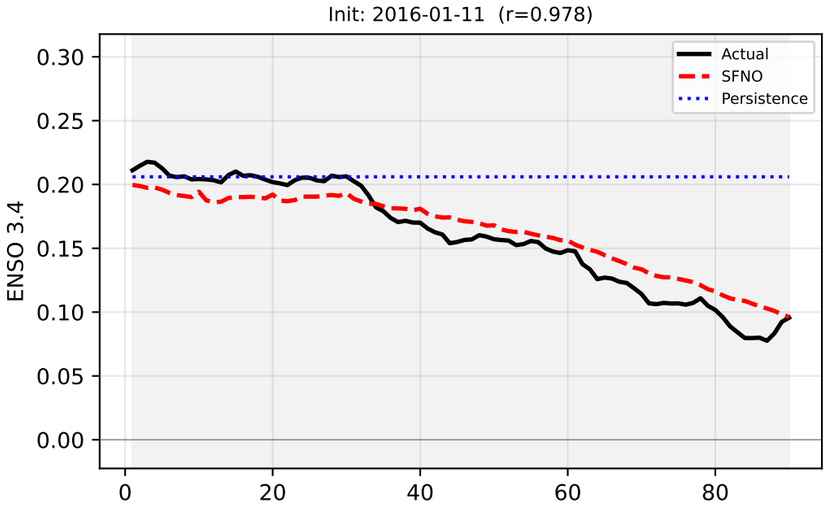


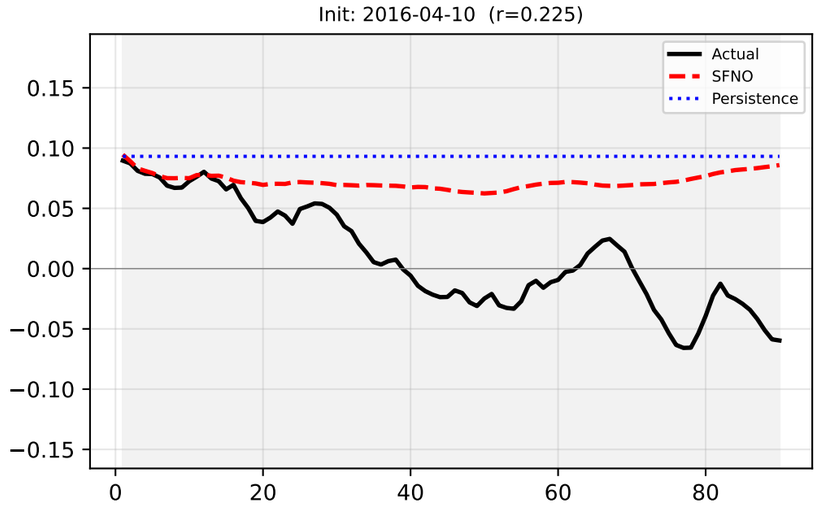
ENSO 3.4 Metrics — Validation

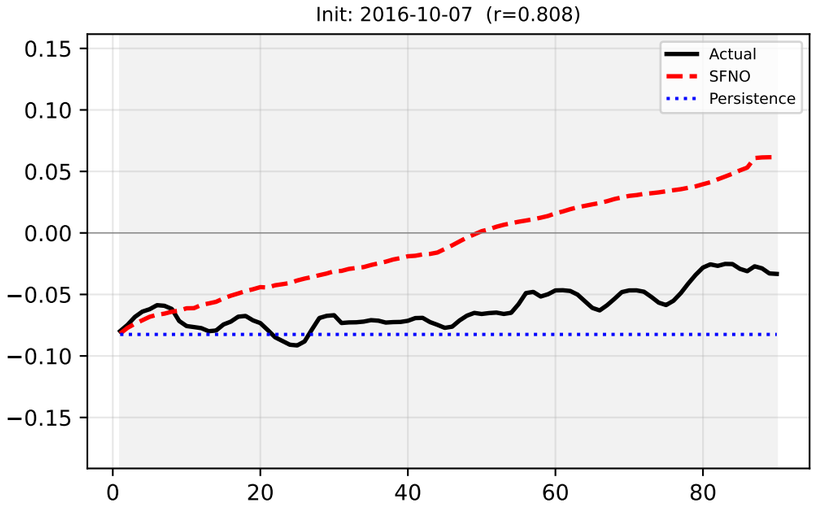
Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0057	0.0046	0.9973	0.9981	-0.0006
2	0.0071	0.0063	0.9964	0.9964	0.0012
3	0.0106	0.0083	0.9927	0.9938	0.0033
4	0.0122	0.0092	0.9897	0.9920	0.0042
5	0.0140	0.0110	0.9853	0.9881	0.0055
6	0.0158	0.0133	0.9830	0.9833	0.0083
7	0.0171	0.0158	0.9833	0.9771	0.0103
8	0.0180	0.0175	0.9829	0.9714	0.0106
9	0.0184	0.0176	0.9811	0.9705	0.0098
10	0.0190	0.0180	0.9745	0.9684	0.0081
11	0.0194	0.0199	0.9688	0.9615	0.0051
12	0.0216	0.0211	0.9577	0.9562	0.0053
13	0.0227	0.0212	0.9523	0.9548	0.0071
14	0.0231	0.0205	0.9540	0.9576	0.0087
15	0.0253	0.0202	0.9518	0.9592	0.0121
16	0.0277	0.0209	0.9474	0.9576	0.0152
17	0.0290	0.0228	0.9449	0.9494	0.0176
18	0.0293	0.0241	0.9461	0.9431	0.0190
19	0.0309	0.0261	0.9379	0.9328	0.0197
20	0.0335	0.0275	0.9277	0.9257	0.0215
21	0.0342	0.0270	0.9281	0.9303	0.0223
22	0.0368	0.0256	0.9237	0.9410	0.0251
23	0.0370	0.0243	0.9225	0.9466	0.0253
24	0.0381	0.0245	0.9180	0.9458	0.0264
25	0.0386	0.0227	0.9176	0.9531	0.0269
26	0.0381	0.0222	0.9162	0.9533	0.0266
27	0.0390	0.0238	0.9091	0.9448	0.0270
28	0.0394	0.0252	0.9083	0.9366	0.0270
29	0.0404	0.0268	0.9103	0.9284	0.0285
30	0.0423	0.0271	0.9098	0.9280	0.0307
31	0.0425	0.0268	0.9120	0.9302	0.0317
32	0.0434	0.0264	0.8992	0.9312	0.0319
33	0.0452	0.0293	0.8827	0.9144	0.0333
34	0.0489	0.0338	0.8620	0.8869	0.0360
35	0.0499	0.0378	0.8589	0.8557	0.0368
36	0.0516	0.0394	0.8541	0.8439	0.0390
37	0.0513	0.0399	0.8542	0.8370	0.0394
38	0.0514	0.0408	0.8568	0.8272	0.0398
39	0.0530	0.0429	0.8526	0.8091	0.0411
40	0.0541	0.0440	0.8465	0.7981	0.0421
41	0.0546	0.0458	0.8339	0.7782	0.0421
42	0.0555	0.0451	0.8242	0.7835	0.0434
43	0.0580	0.0445	0.8164	0.7945	0.0461
44	0.0611	0.0449	0.7986	0.7965	0.0493
45	0.0639	0.0444	0.7871	0.8085	0.0522
46	0.0650	0.0436	0.7803	0.8178	0.0532
47	0.0647	0.0441	0.7783	0.8079	0.0530
48	0.0661	0.0460	0.7748	0.7888	0.0538
49	0.0665	0.0474	0.7786	0.7738	0.0542
50	0.0657	0.0467	0.7840	0.7763	0.0537
51	0.0633	0.0462	0.7931	0.7751	0.0517
52	0.0637	0.0468	0.7887	0.7672	0.0523
53	0.0643	0.0467	0.7811	0.7680	0.0531
54	0.0645	0.0466	0.7703	0.7665	0.0525
55	0.0640	0.0454	0.7645	0.7763	0.0518
56	0.0654	0.0441	0.7549	0.7919	0.0531
57	0.0680	0.0446	0.7437	0.7902	0.0557
58	0.0702	0.0469	0.7306	0.7685	0.0577
59	0.0721	0.0477	0.7168	0.7616	0.0592
60	0.0740	0.0481	0.7165	0.7614	0.0610
61	0.0744	0.0478	0.7247	0.7659	0.0613
62	0.0752	0.0480	0.7210	0.7649	0.0628
63	0.0751	0.0477	0.7011	0.7628	0.0616
64	0.0751	0.0474	0.6784	0.7625	0.0610
65	0.0771	0.0473	0.6611	0.7666	0.0622
66	0.0768	0.0458	0.6655	0.7823	0.0629
67	0.0757	0.0452	0.6698	0.7845	0.0623
68	0.0758	0.0459	0.6679	0.7758	0.0627
69	0.0760	0.0472	0.6483	0.7581	0.0624
70	0.0776	0.0493	0.6110	0.7330	0.0635
71	0.0776	0.0510	0.5866	0.7100	0.0637
72	0.0784	0.0533	0.5719	0.6791	0.0644
73	0.0807	0.0559	0.5502	0.6473	0.0662
74	0.0823	0.0567	0.5574	0.6412	0.0686
75	0.0847	0.0587	0.5609	0.6229	0.0714
76	0.0852	0.0600	0.5782	0.6055	0.0726
77	0.0864	0.0603	0.5766	0.6065	0.0733
78	0.0884	0.0605	0.5429	0.6097	0.0741
79	0.0885	0.0603	0.4961	0.6050	0.0737
80	0.0887	0.0599	0.4471	0.6048	0.0733
81	0.0877	0.0581	0.4263	0.6243	0.0726
82	0.0865	0.0566	0.4154	0.6391	0.0716
83	0.0874	0.0574	0.4117	0.6298	0.0730
84	0.0875	0.0581	0.4146	0.6190	0.0736
85	0.0874	0.0583	0.4100	0.6136	0.0733
86	0.0876	0.0585	0.3794	0.6072	0.0734
87	0.0880	0.0591	0.3461	0.5917	0.0736
88	0.0882	0.0586	0.3288	0.5994	0.0734
89	0.0889	0.0587	0.3010	0.5977	0.0734
90	0.0891	0.0600	0.2677	0.5725	0.0727

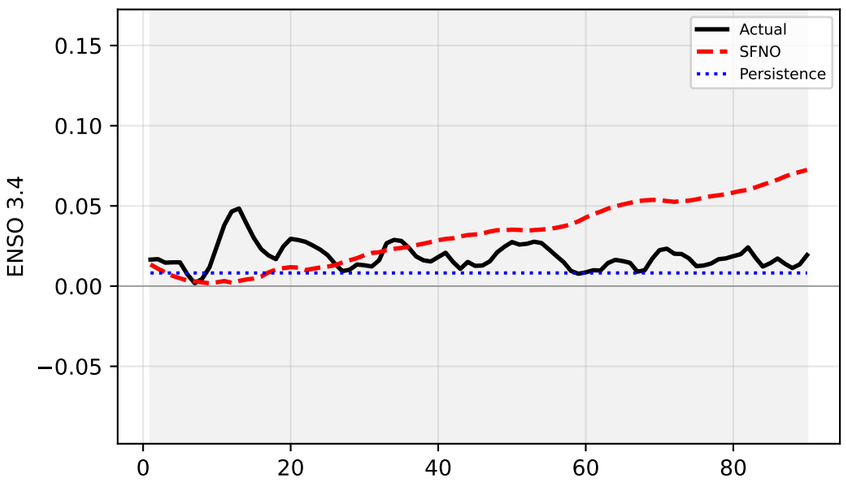

ENSO 3.4 Forecast Skill — Validation
ENSO 3.4 RMSE
SFNO
Persistence
RMSE
0.02
0.04
0.06
0.08
0
20
40
60
80
Lead step
ENSO 3.4 Correlation
SFNO
Persistence
Corr
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
Lead step
ENSO 3.4 Bias
Bias
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0
20
40
60
80
Lead step

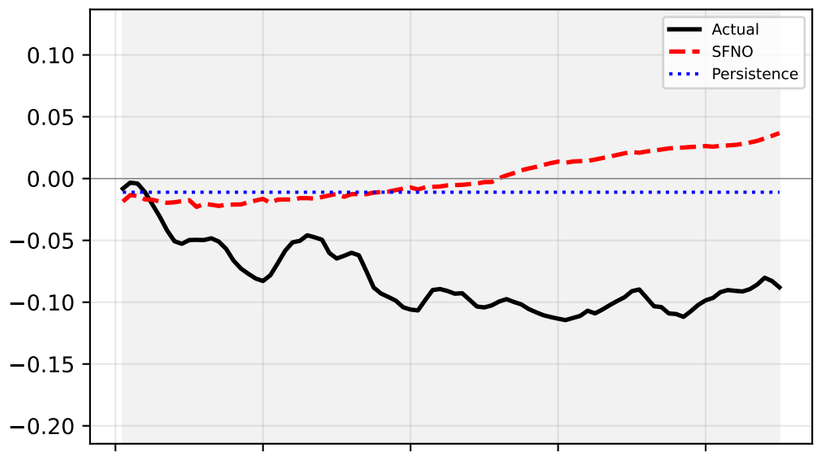
ENSO 3.4 Index — Validation

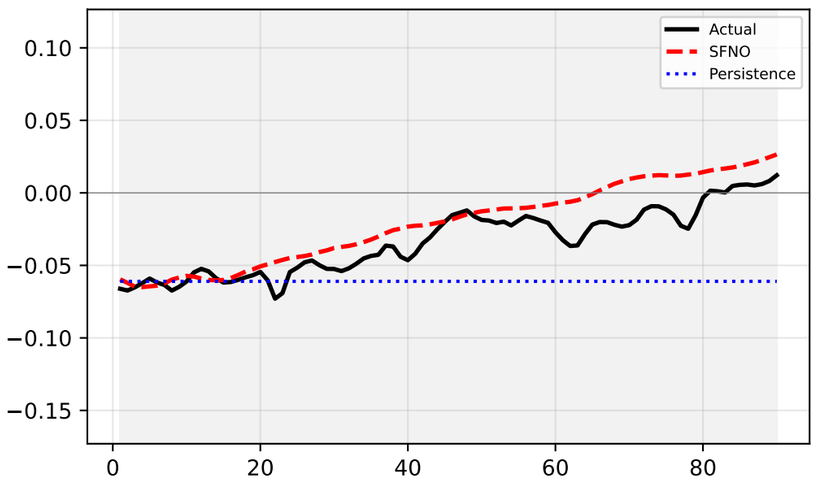


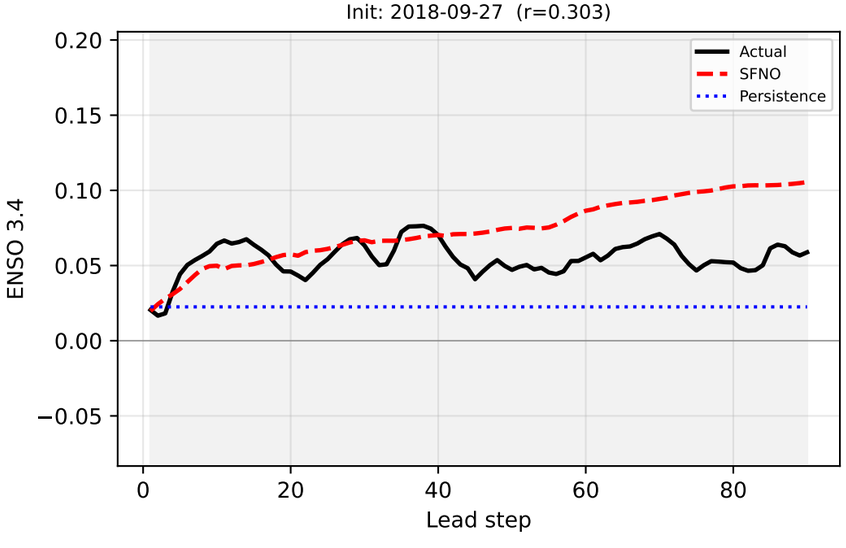


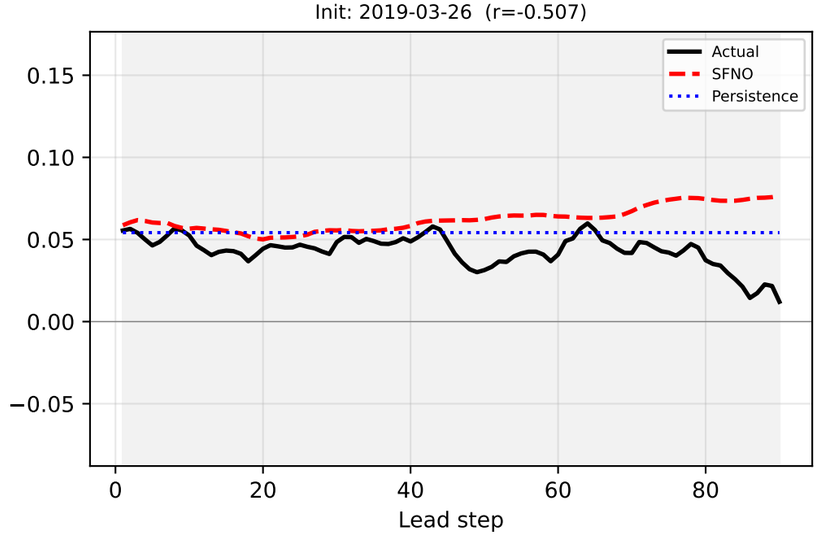


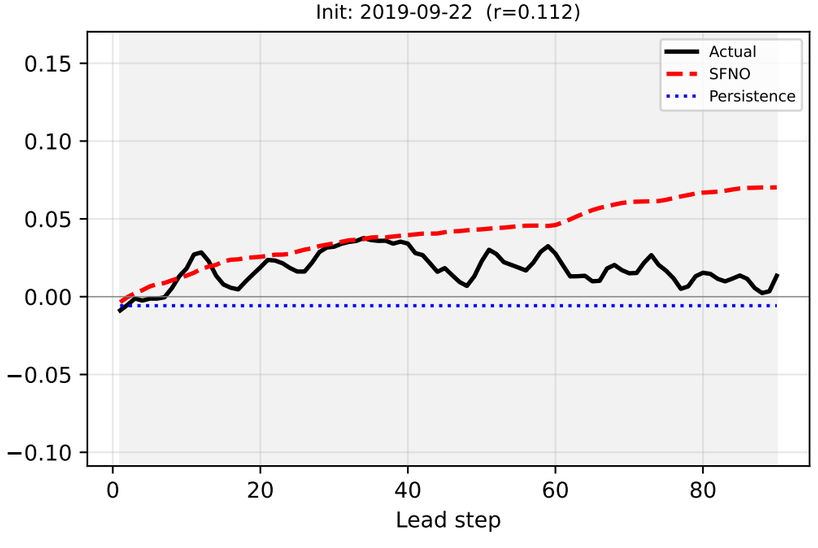




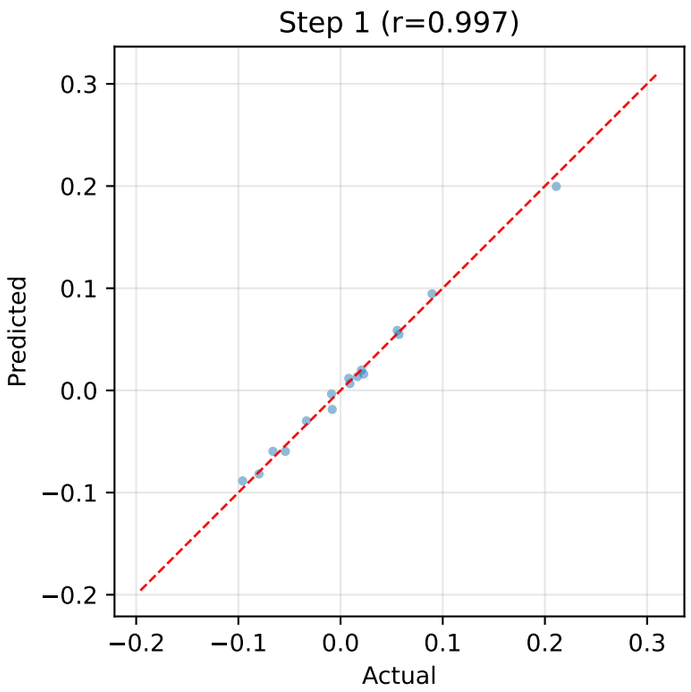


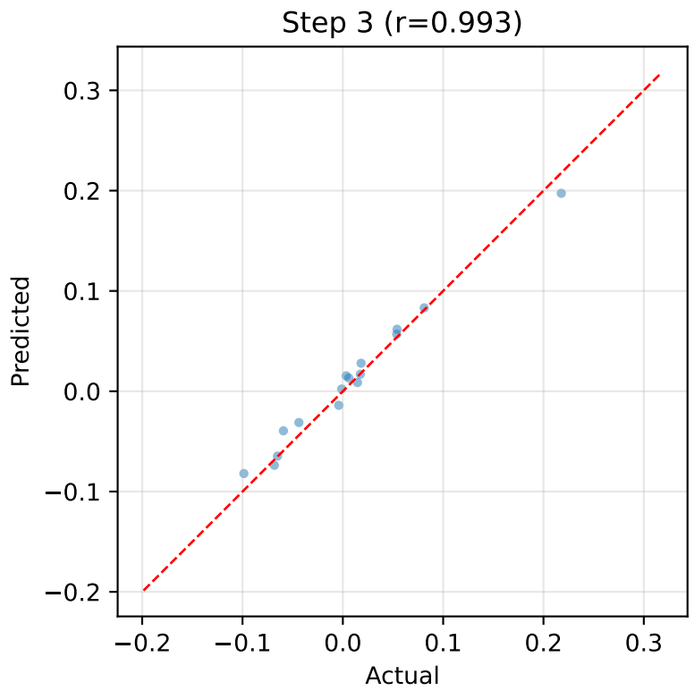


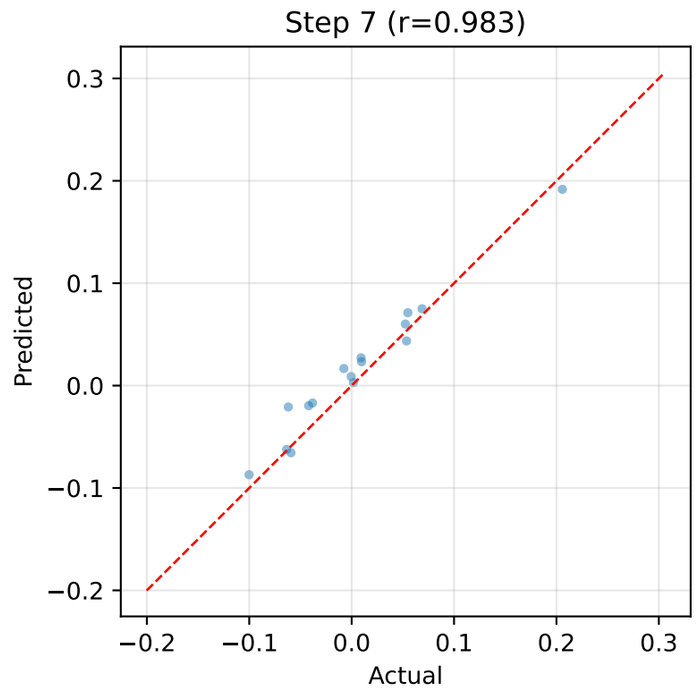


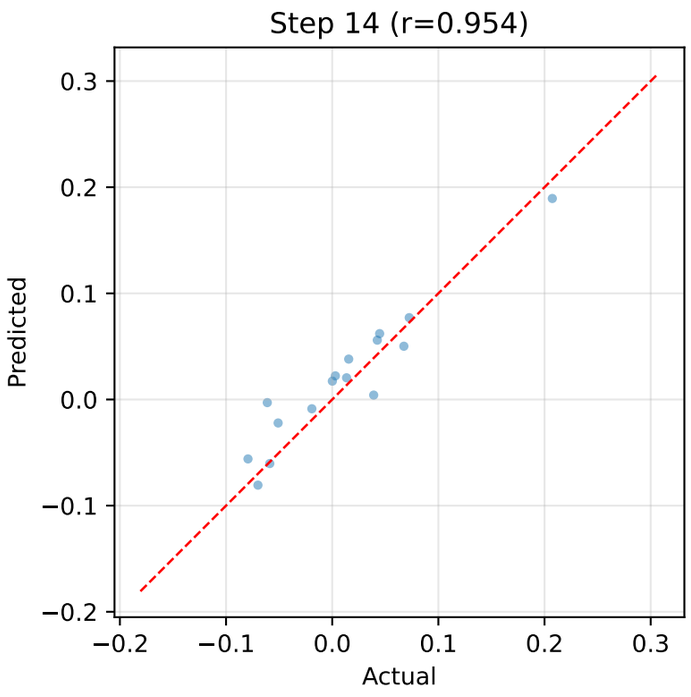


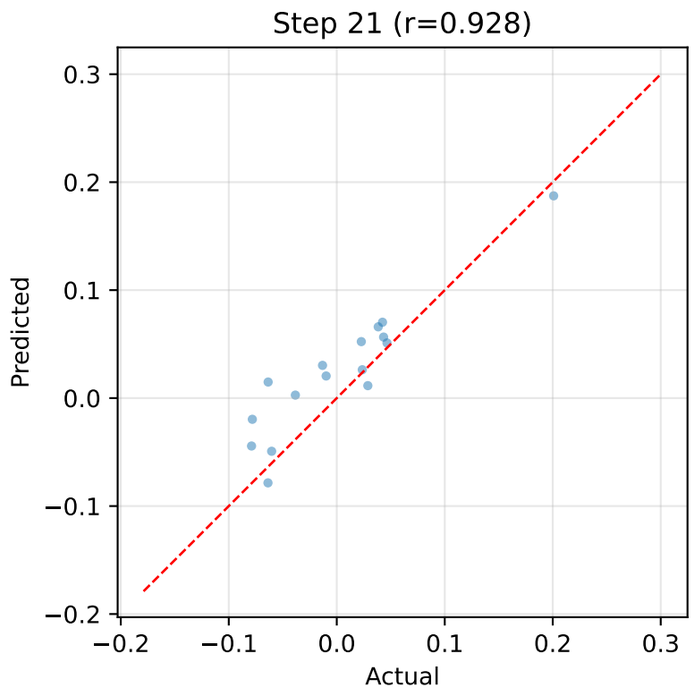
ENSO 3.4 Scatter — Validation

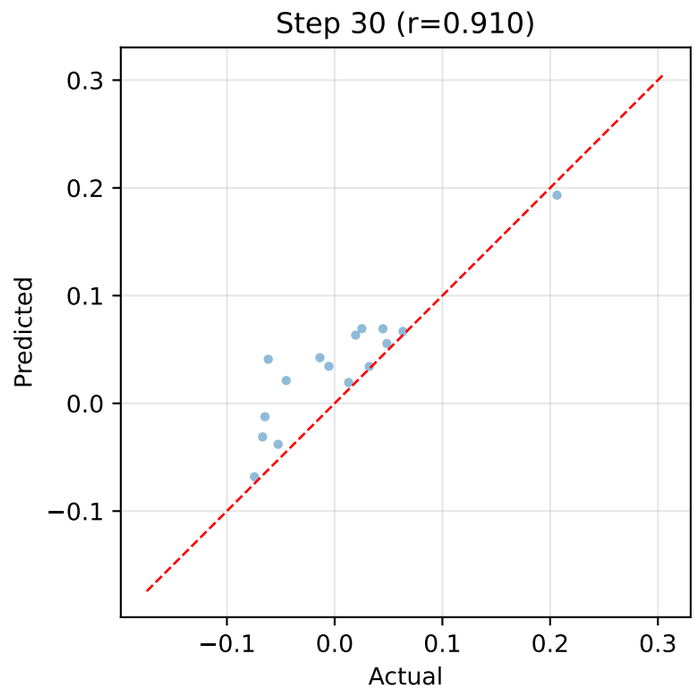


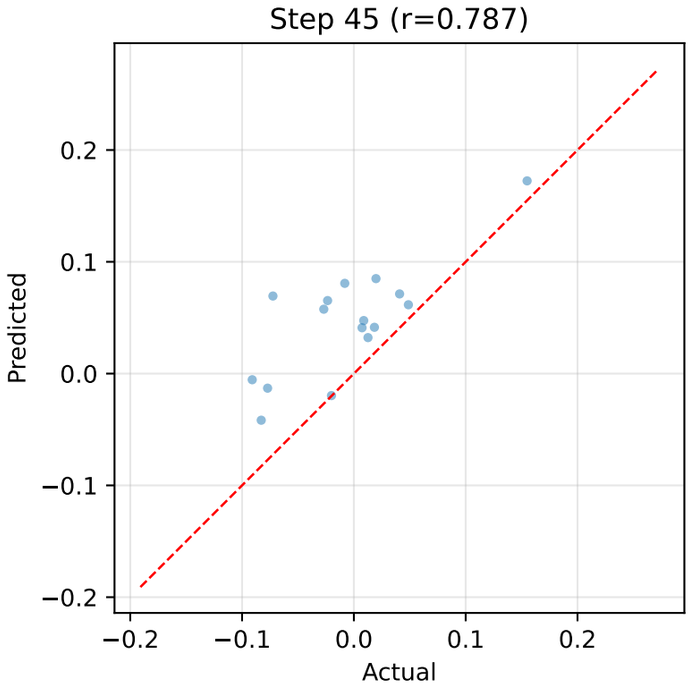


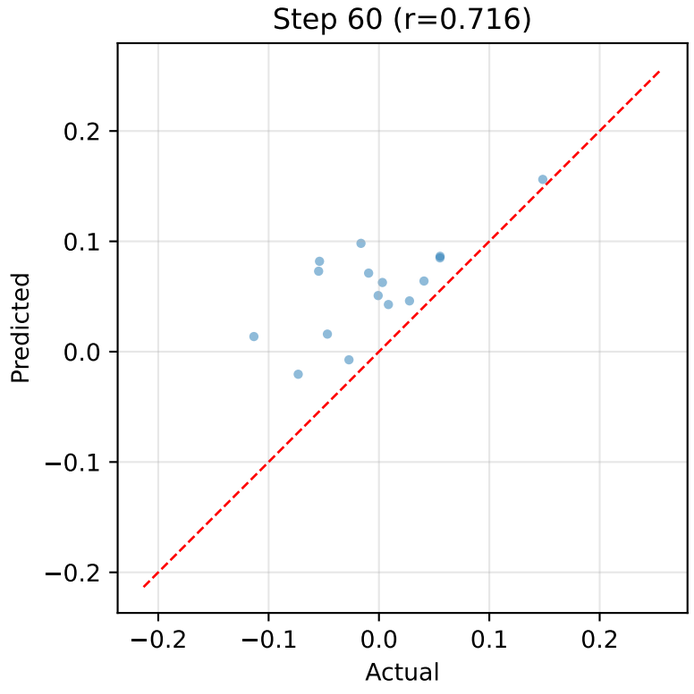


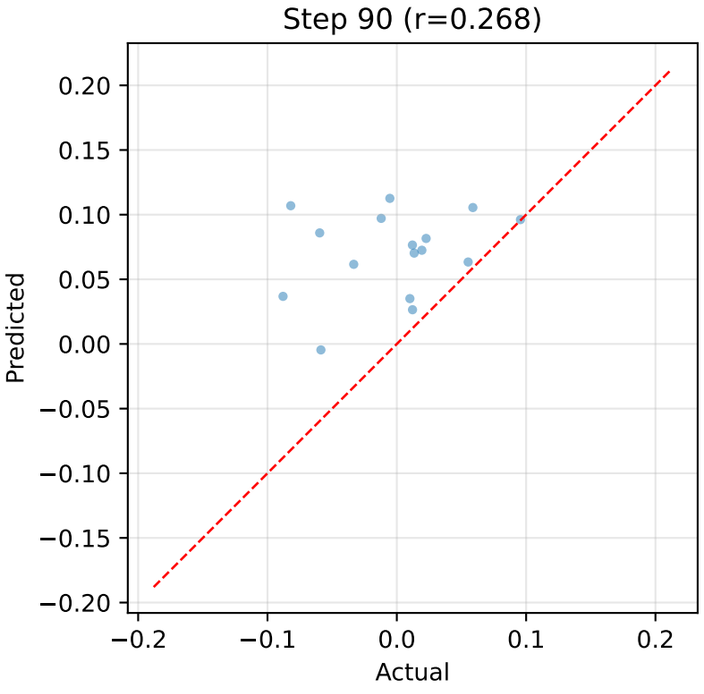








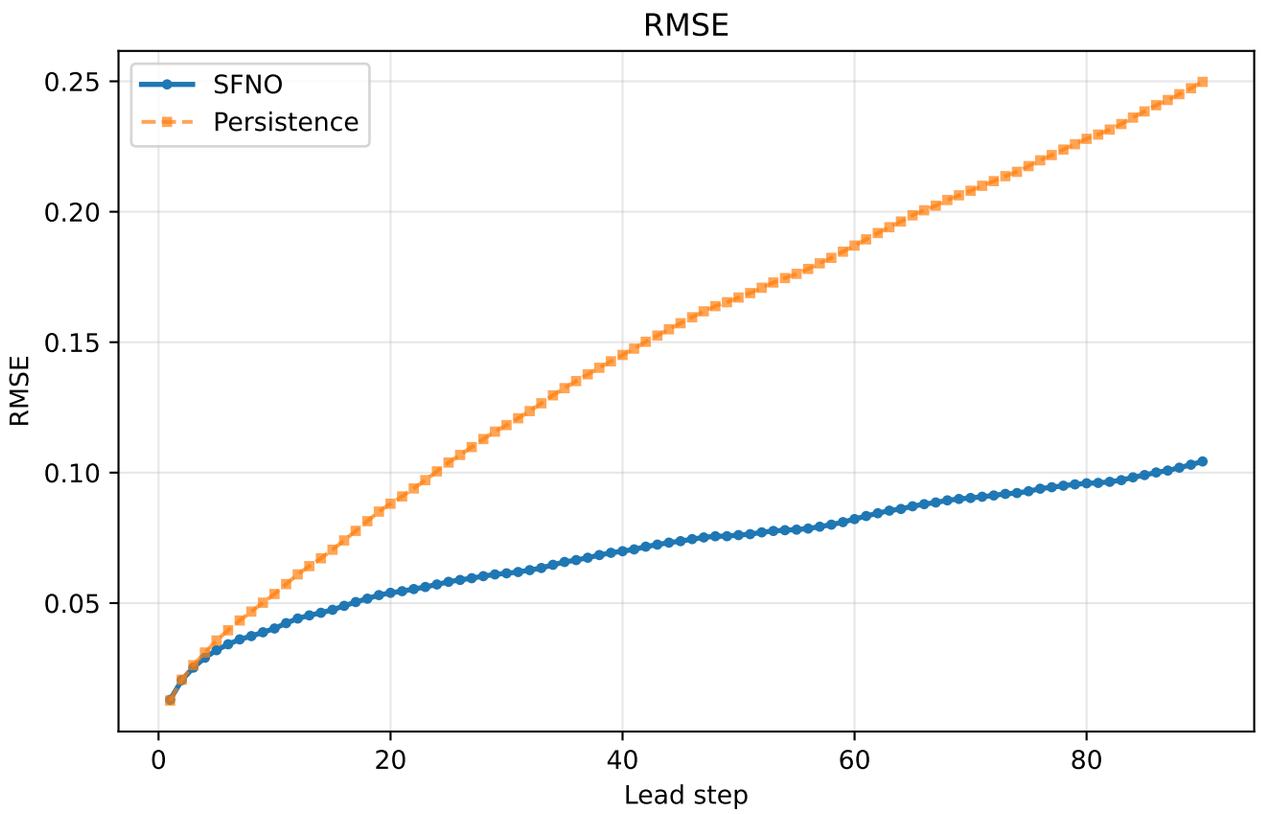


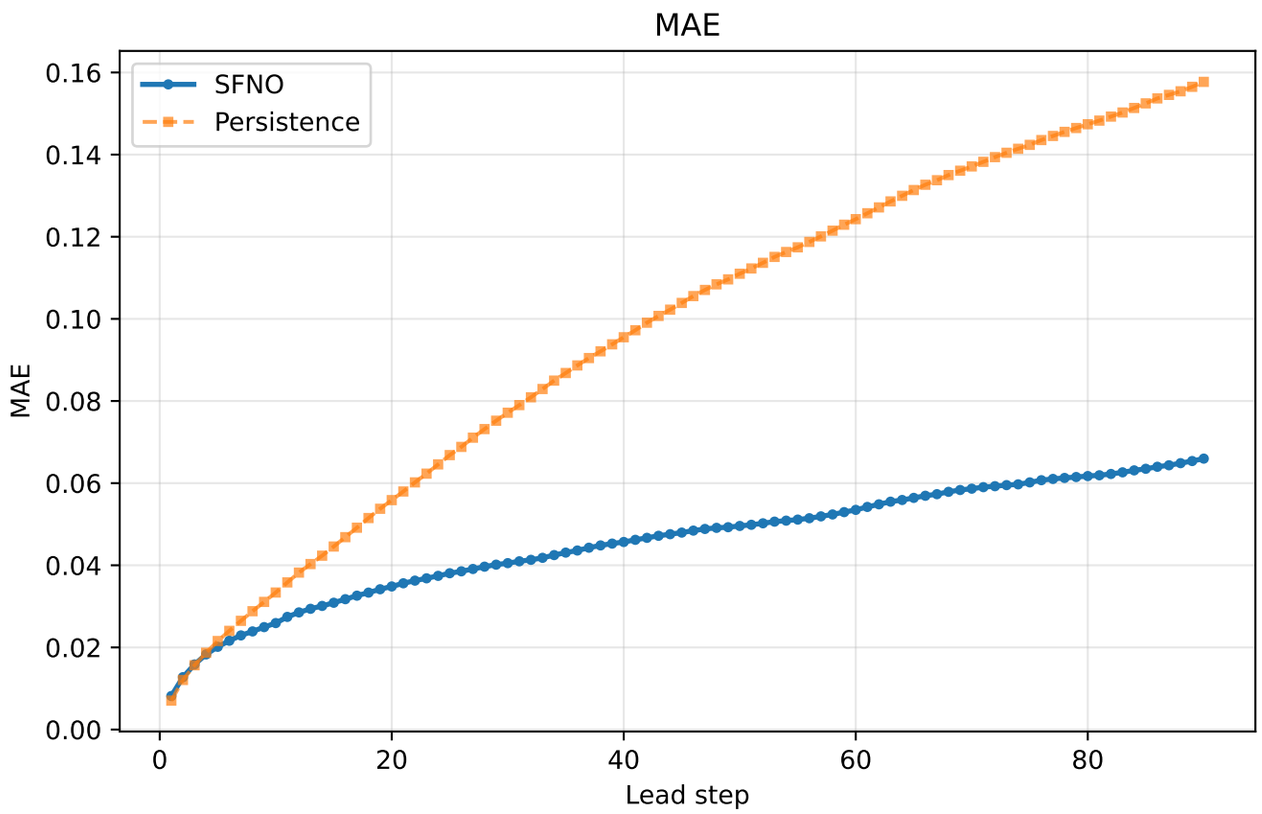


Metrics by Lead Step — Test

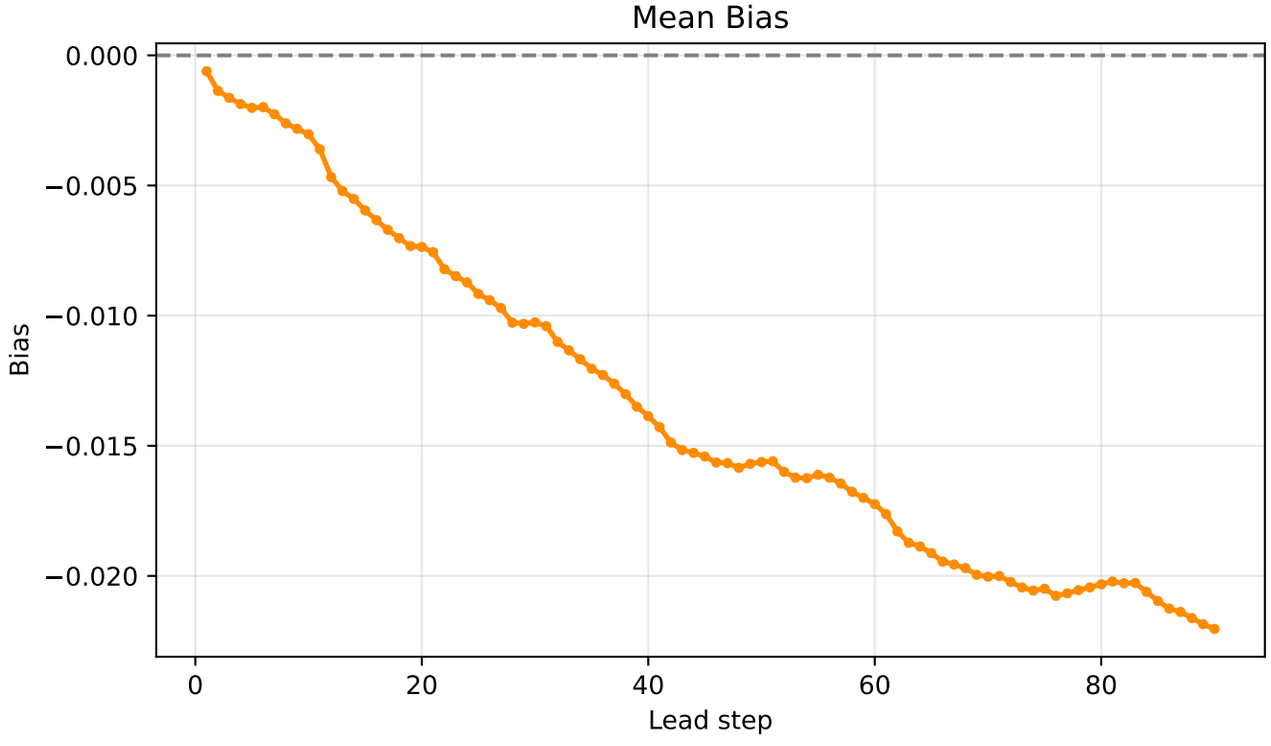
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0129	0.0126	-2.3%	0.0082	-0.0006	0.9805	0.9796	0.9998
2	0.0205	0.0206	0.9%	0.0128	-0.0014	0.9500	0.9462	0.9996
3	0.0253	0.0262	3.6%	0.0158	-0.0016	0.9210	0.9132	0.9994
4	0.0291	0.0311	6.4%	0.0183	-0.0019	0.8923	0.8780	0.9992
5	0.0319	0.0357	10.4%	0.0202	-0.0020	0.8667	0.8411	0.9990
6	0.0342	0.0396	13.6%	0.0216	-0.0020	0.8433	0.8072	0.9989
7	0.0361	0.0433	16.7%	0.0229	-0.0023	0.8221	0.7728	0.9987
8	0.0373	0.0467	20.1%	0.0239	-0.0026	0.8042	0.7389	0.9986
9	0.0388	0.0501	22.6%	0.0249	-0.0028	0.7868	0.7063	0.9985
10	0.0403	0.0535	24.7%	0.0259	-0.0030	0.7694	0.6743	0.9984
11	0.0423	0.0573	26.1%	0.0274	-0.0036	0.7484	0.6410	0.9983
12	0.0441	0.0610	27.7%	0.0285	-0.0047	0.7262	0.6068	0.9981
13	0.0453	0.0642	29.4%	0.0294	-0.0052	0.7086	0.5769	0.9980
14	0.0463	0.0672	31.1%	0.0301	-0.0055	0.6931	0.5500	0.9979
15	0.0474	0.0705	32.7%	0.0309	-0.0060	0.6770	0.5219	0.9978
16	0.0489	0.0740	33.9%	0.0317	-0.0063	0.6581	0.4938	0.9977
17	0.0504	0.0777	35.1%	0.0326	-0.0067	0.6391	0.4629	0.9975
18	0.0517	0.0814	36.5%	0.0333	-0.0070	0.6211	0.4320	0.9974
19	0.0531	0.0851	37.6%	0.0342	-0.0073	0.6019	0.4030	0.9973
20	0.0539	0.0881	38.8%	0.0348	-0.0074	0.5854	0.3801	0.9972
21	0.0546	0.0909	40.0%	0.0356	-0.0076	0.5693	0.3592	0.9971
22	0.0554	0.0939	41.1%	0.0363	-0.0082	0.5533	0.3356	0.9970
23	0.0562	0.0971	42.2%	0.0368	-0.0085	0.5405	0.3131	0.9969
24	0.0571	0.1005	43.1%	0.0374	-0.0087	0.5259	0.2889	0.9968
25	0.0581	0.1039	44.0%	0.0381	-0.0092	0.5096	0.2670	0.9967
26	0.0589	0.1068	44.9%	0.0385	-0.0094	0.4921	0.2487	0.9967
27	0.0595	0.1098	45.8%	0.0391	-0.0097	0.4746	0.2287	0.9966
28	0.0603	0.1129	46.6%	0.0397	-0.0103	0.4584	0.2138	0.9965
29	0.0610	0.1157	47.3%	0.0401	-0.0103	0.4420	0.2032	0.9964
30	0.0614	0.1182	48.1%	0.0405	-0.0103	0.4262	0.1947	0.9964
31	0.0619	0.1208	48.8%	0.0410	-0.0104	0.4094	0.1854	0.9963
32	0.0626	0.1236	49.3%	0.0413	-0.0110	0.3927	0.1758	0.9962
33	0.0634	0.1266	49.9%	0.0418	-0.0113	0.3755	0.1635	0.9961
34	0.0647	0.1296	50.1%	0.0425	-0.0117	0.3555	0.1532	0.9960
35	0.0658	0.1324	50.3%	0.0431	-0.0120	0.3352	0.1445	0.9958
36	0.0665	0.1351	50.8%	0.0436	-0.0123	0.3158	0.1347	0.9957
37	0.0674	0.1377	51.1%	0.0443	-0.0126	0.2965	0.1282	0.9956
38	0.0684	0.1402	51.2%	0.0448	-0.0130	0.2781	0.1243	0.9955
39	0.0693	0.1427	51.5%	0.0453	-0.0135	0.2635	0.1209	0.9954
40	0.0699	0.1451	51.8%	0.0457	-0.0139	0.2508	0.1145	0.9953
41	0.0706	0.1475	52.2%	0.0462	-0.0143	0.2336	0.1073	0.9952
42	0.0716	0.1502	52.3%	0.0467	-0.0149	0.2188	0.1002	0.9950
43	0.0724	0.1526	52.5%	0.0472	-0.0152	0.2059	0.0960	0.9949
44	0.0732	0.1549	52.8%	0.0476	-0.0153	0.1907	0.0912	0.9948
45	0.0737	0.1573	53.1%	0.0480	-0.0154	0.1754	0.0831	0.9947
46	0.0745	0.1596	53.3%	0.0484	-0.0156	0.1611	0.0757	0.9946
47	0.0752	0.1618	53.6%	0.0488	-0.0157	0.1528	0.0700	0.9945
48	0.0756	0.1638	53.8%	0.0491	-0.0158	0.1460	0.0685	0.9945
49	0.0756	0.1653	54.3%	0.0493	-0.0157	0.1427	0.0718	0.9945
50	0.0761	0.1671	54.5%	0.0496	-0.0156	0.1383	0.0747	0.9944
51	0.0764	0.1688	54.7%	0.0499	-0.0156	0.1314	0.0768	0.9944
52	0.0771	0.1709	54.9%	0.0502	-0.0160	0.1219	0.0753	0.9943
53	0.0776	0.1729	55.1%	0.0506	-0.0162	0.1139	0.0730	0.9942
54	0.0779	0.1746	55.4%	0.0509	-0.0162	0.1085	0.0736	0.9941
55	0.0782	0.1762	55.6%	0.0511	-0.0161	0.1043	0.0758	0.9941
56	0.0786	0.1781	55.9%	0.0515	-0.0162	0.0972	0.0736	0.9940
57	0.0793	0.1802	56.0%	0.0519	-0.0165	0.0889	0.0694	0.9939
58	0.0801	0.1823	56.1%	0.0524	-0.0168	0.0816	0.0654	0.9938
59	0.0810	0.1847	56.1%	0.0529	-0.0170	0.0710	0.0584	0.9937
60	0.0822	0.1870	56.1%	0.0535	-0.0172	0.0618	0.0535	0.9935
61	0.0834	0.1894	56.0%	0.0542	-0.0176	0.0512	0.0470	0.9933
62	0.0844	0.1918	56.0%	0.0548	-0.0183	0.0388	0.0385	0.9931
63	0.0854	0.1941	56.0%	0.0555	-0.0187	0.0259	0.0316	0.9930
64	0.0861	0.1963	56.1%	0.0559	-0.0189	0.0193	0.0248	0.9928
65	0.0871	0.1986	56.1%	0.0564	-0.0191	0.0132	0.0192	0.9927
66	0.0879	0.2006	56.2%	0.0569	-0.0195	0.0062	0.0171	0.9925
67	0.0886	0.2023	56.2%	0.0573	-0.0196	0.0029	0.0183	0.9924
68	0.0894	0.2045	56.3%	0.0579	-0.0197	-0.0020	0.0135	0.9923
69	0.0899	0.2063	56.4%	0.0583	-0.0200	-0.0038	0.0125	0.9922
70	0.0903	0.2080	56.6%	0.0587	-0.0200	-0.0044	0.0122	0.9921
71	0.0908	0.2099	56.8%	0.0590	-0.0200	-0.0067	0.0110	0.9920
72	0.0913	0.2117	56.9%	0.0593	-0.0202	-0.0063	0.0120	0.9920
73	0.0919	0.2136	57.0%	0.0595	-0.0204	-0.0076	0.0109	0.9918
74	0.0922	0.2153	57.2%	0.0597	-0.0206	-0.0058	0.0133	0.9918
75	0.0929	0.2175	57.3%	0.0602	-0.0205	-0.0051	0.0086	0.9917
76	0.0938	0.2197	57.3%	0.0607	-0.0208	-0.0096	0.0050	0.9915
77	0.0944	0.2217	57.4%	0.0610	-0.0207	-0.0098	0.0049	0.9914
78	0.0950	0.2238	57.6%	0.0613	-0.0205	-0.0083	0.0037	0.9913
79	0.0955	0.2259	57.7%	0.0615	-0.0204	-0.0072	0.0019	0.9912
80	0.0959	0.2279	57.9%	0.0617	-0.0203	-0.0081	-0.0015	0.9911
81	0.0961	0.2296	58.2%	0.0619	-0.0202	-0.0066	-0.0020	0.9911
82	0.0965	0.2315	58.3%	0.0622	-0.0203	-0.0085	-0.0056	0.9910
83	0.0971	0.2336	58.4%	0.0626	-0.0203	-0.0097	-0.0076	0.9909
84	0.0982	0.2361	58.4%	0.0631	-0.0206	-0.0097	-0.0090	0.9907
85	0.0991	0.2385	58.5%	0.0635	-0.0210	-0.0103	-0.0119	0.9905
86	0.1001	0.2408	58.4%	0.0640	-0.0213	-0.0124	-0.0147	0.9903
87	0.1008	0.2428	58.5%	0.0644	-0.0214	-0.0129	-0.0142	0.9902
88	0.1019	0.2451	58.4%	0.0649	-0.0216	-0.0156	-0.0133	0.9900
89	0.1030	0.2474	58.3%	0.0654	-0.0219	-0.0181	-0.0151	0.9897
90	0.1043	0.2498	58.2%	0.0660	-0.0220	-0.0217	-0.0180	0.9895

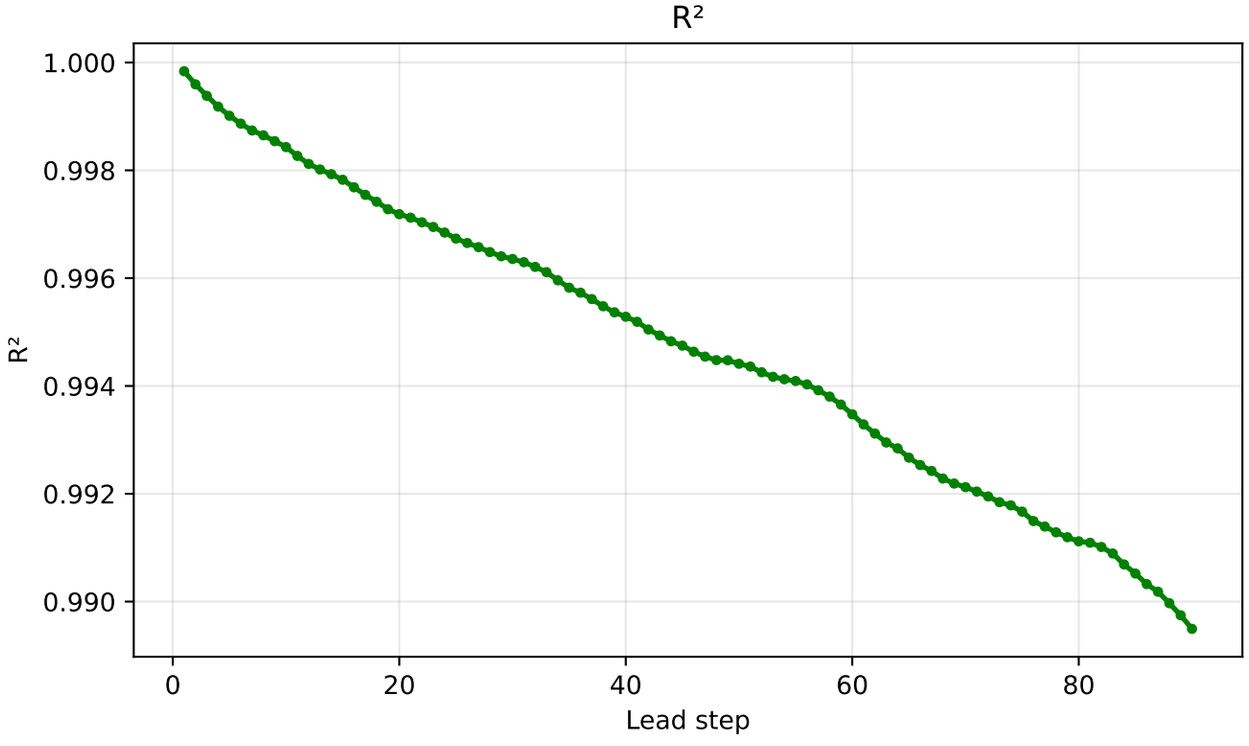
Forecast Skill vs Lead Step — Test





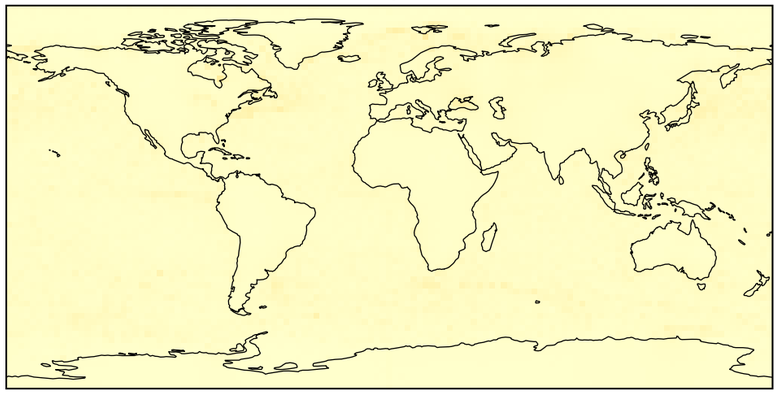
Bias and R² — Test



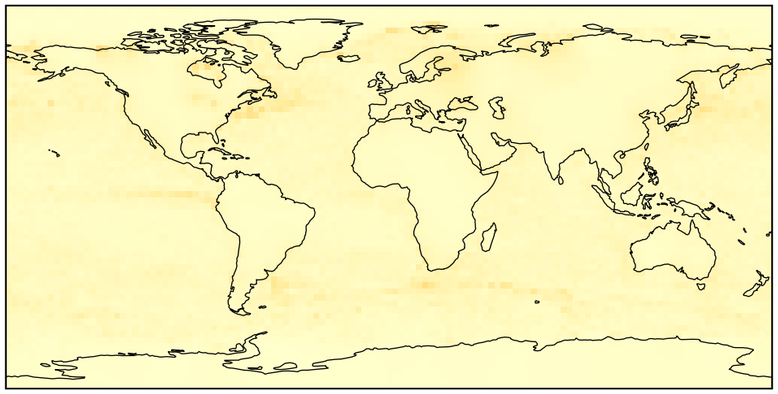


Spatial RMSE (all grid) — Test

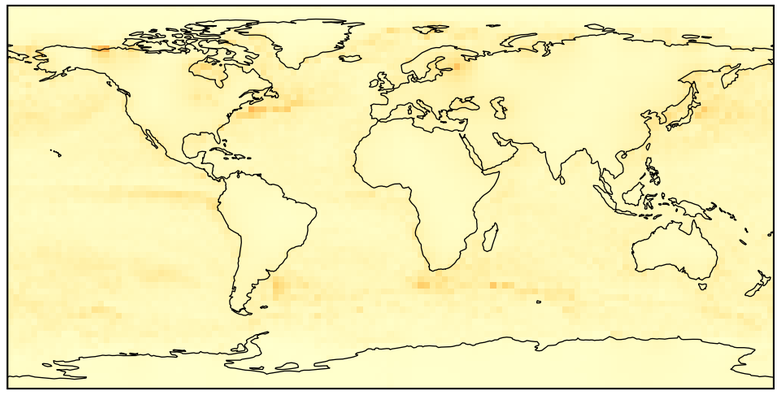
Step 1



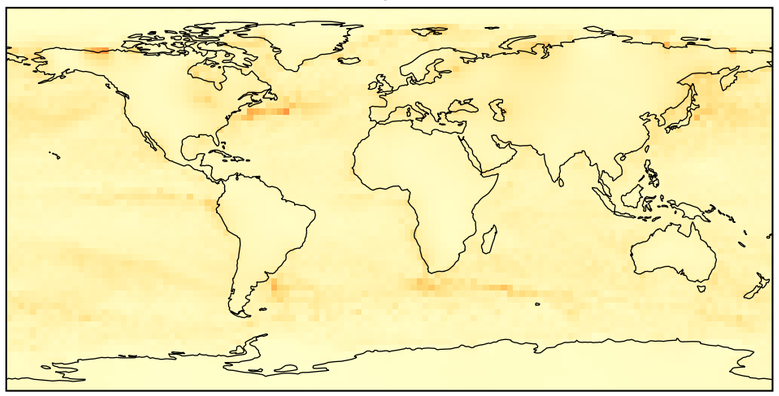
Step 3



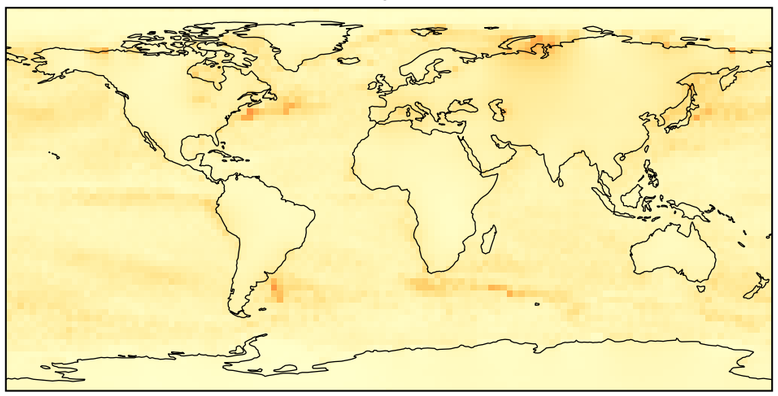
Step 7



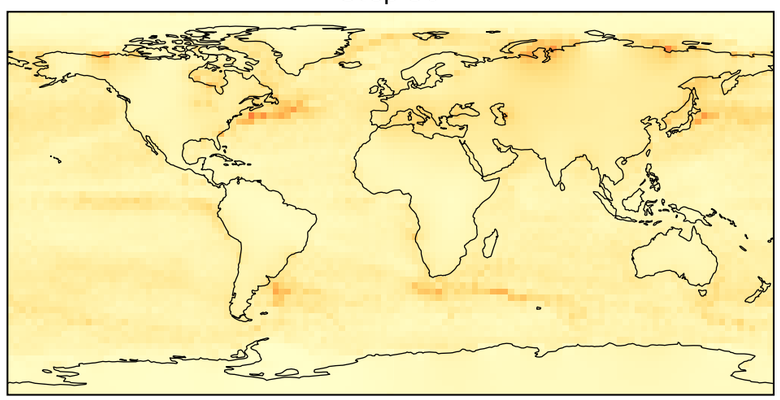
Step 14



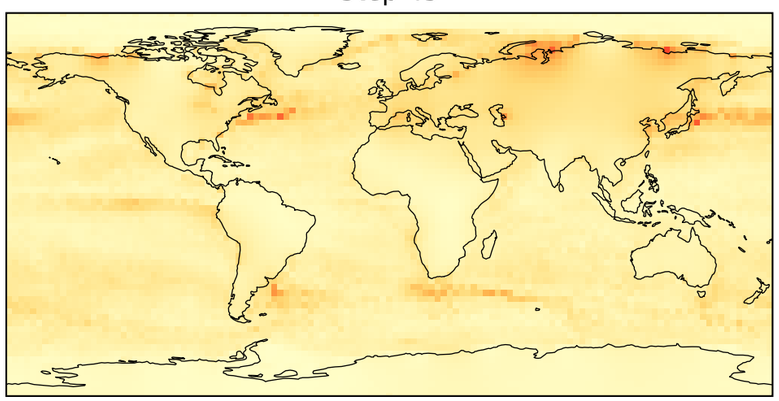
Step 21



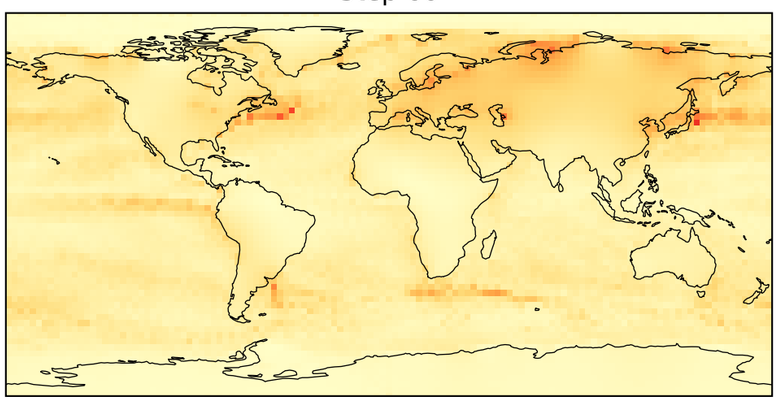
Step 30



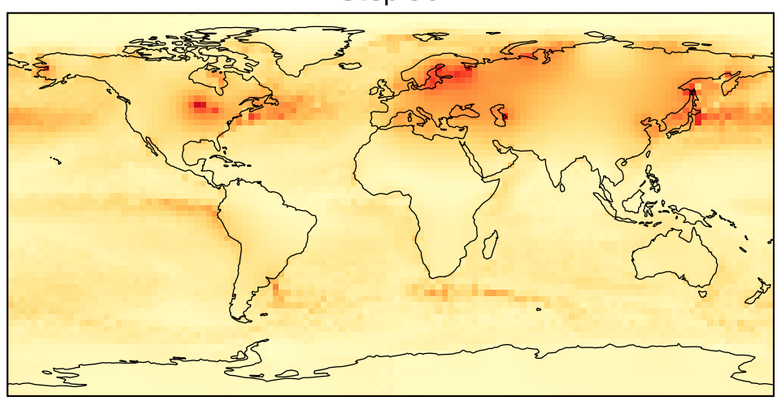
Step 45



Step 60

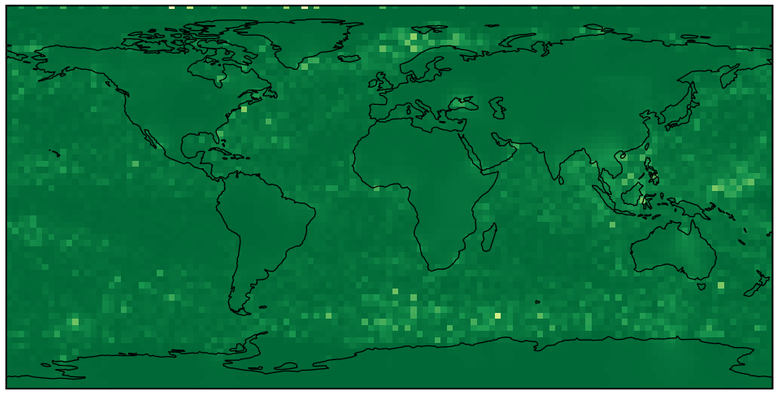


Step 90

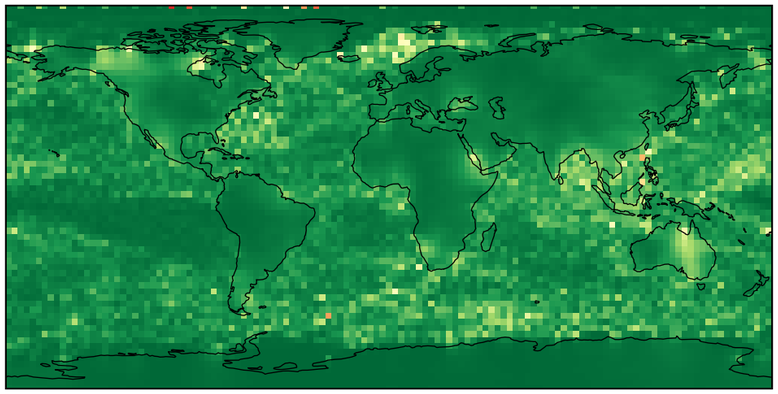


Spatial ACC (all grid) — Test

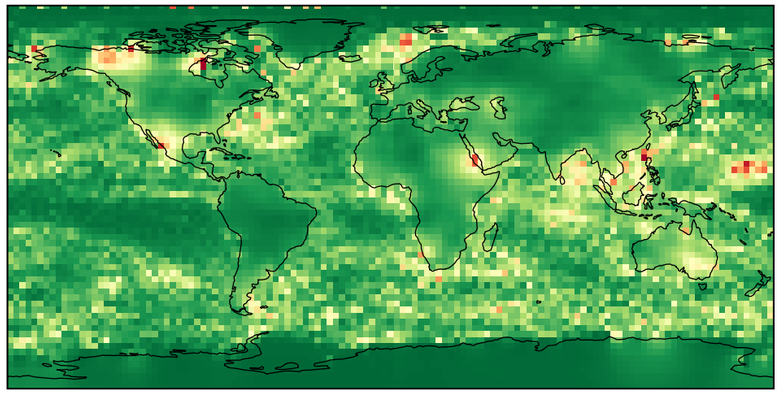
Step 1



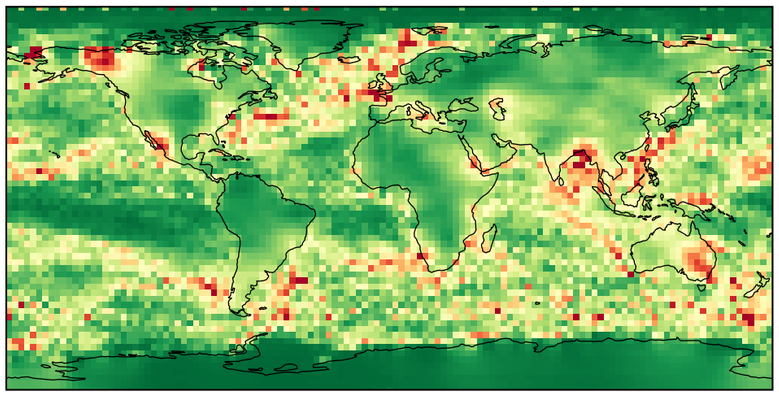
Step 3



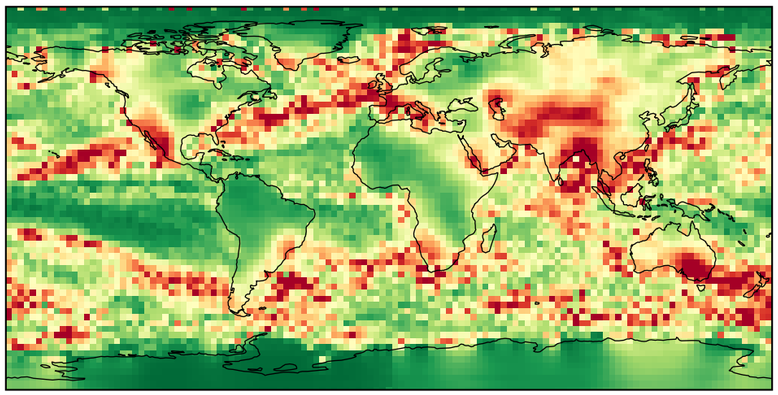
Step 7



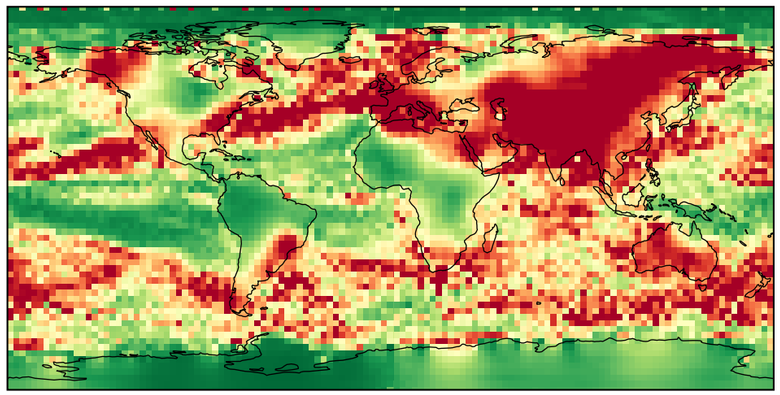
Step 14



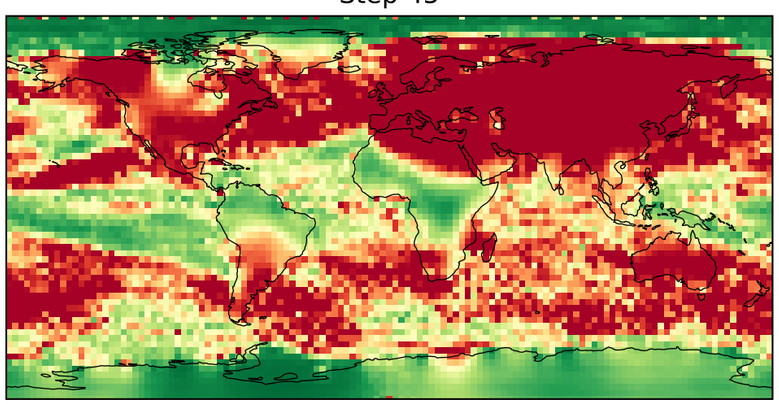
Step 21



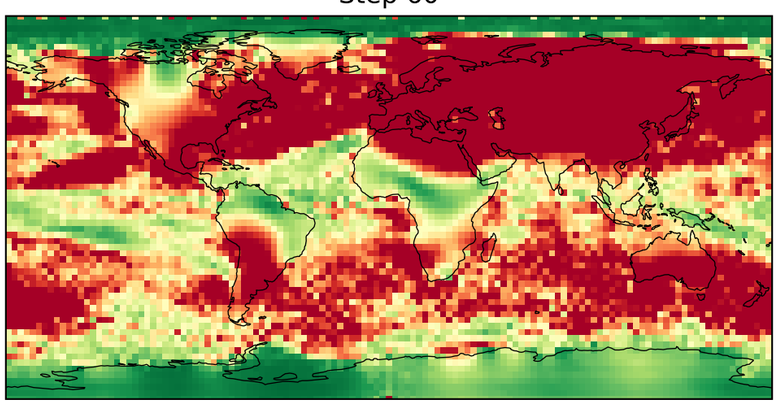
Step 30



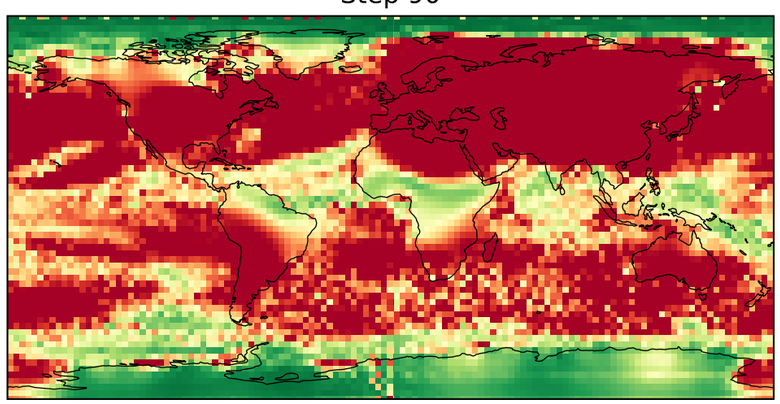
Step 45



Step 60

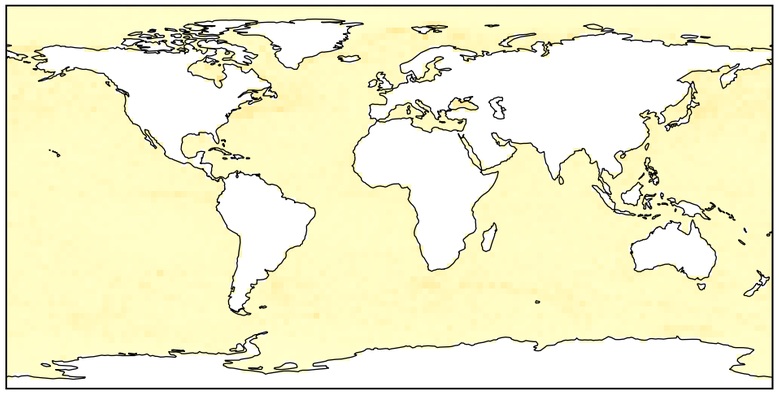


Step 90

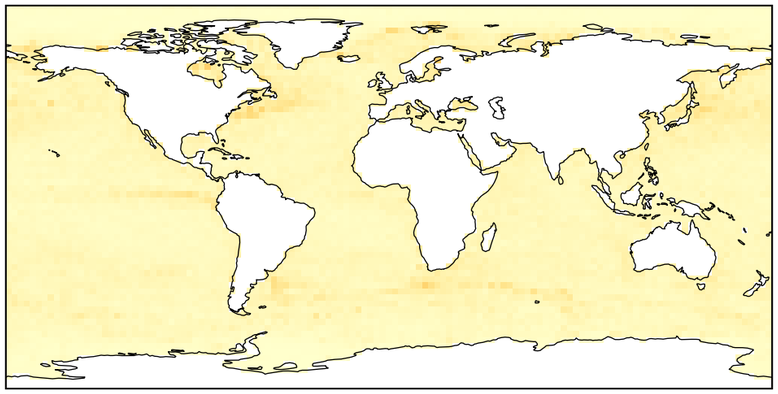


Spatial RMSE (ocean only) — Test

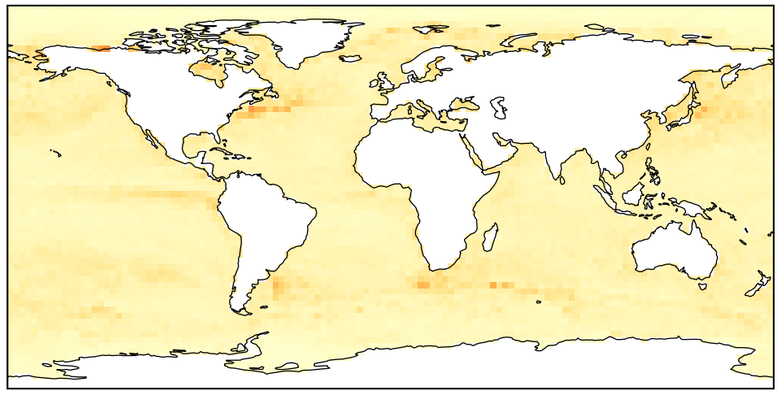
Step 1



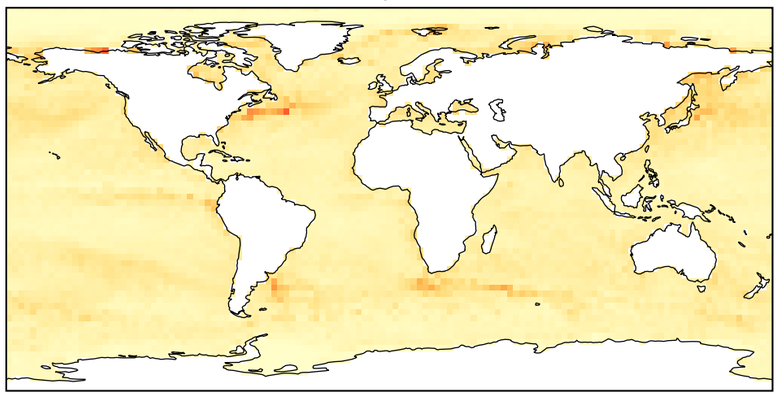
Step 3



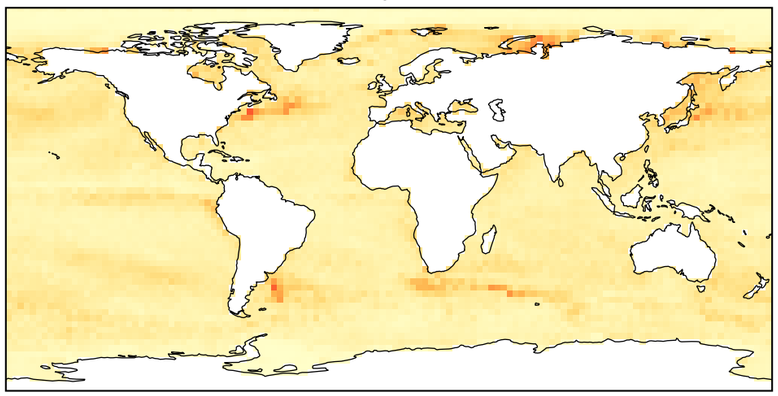
Step 7



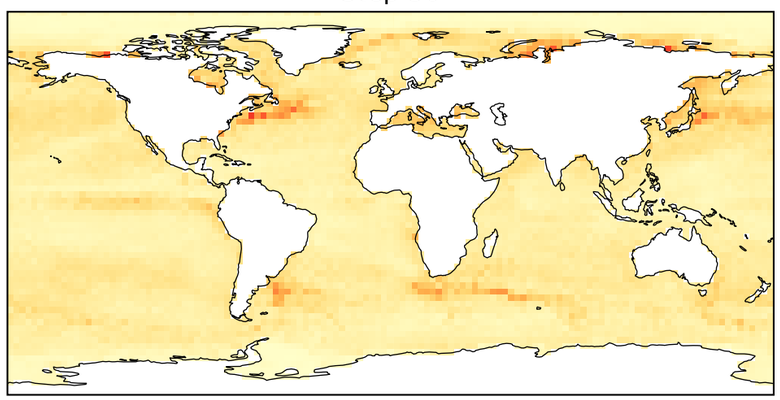
Step 14



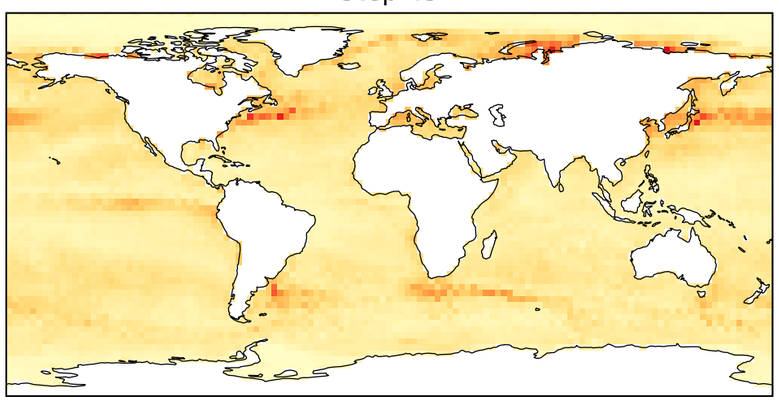
Step 21



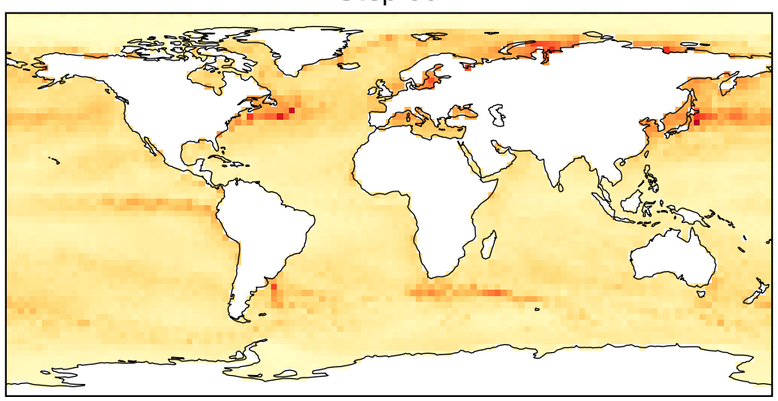
Step 30



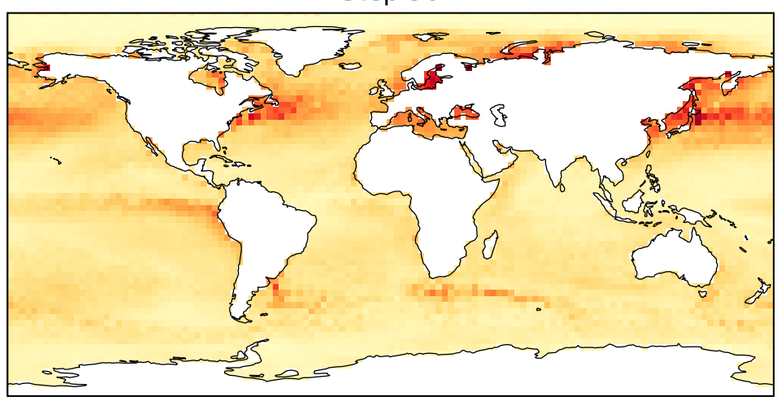
Step 45



Step 60

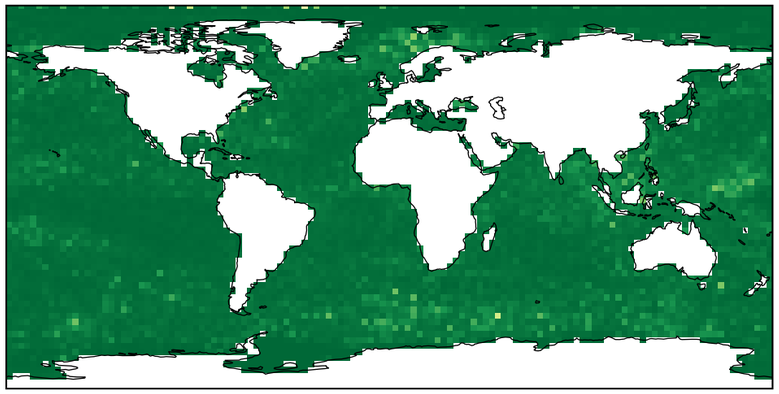


Step 90

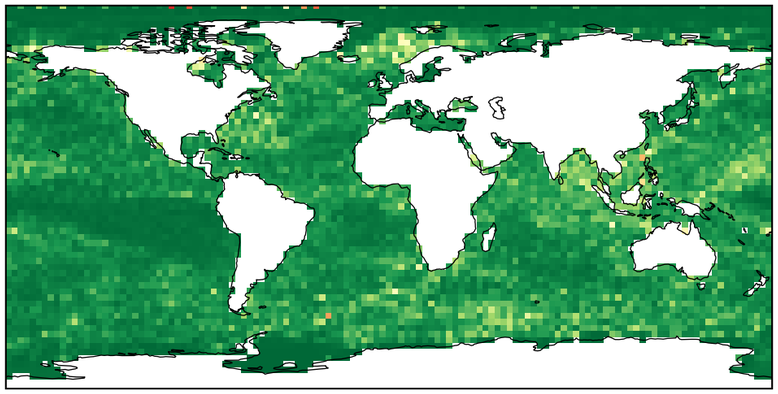


Spatial ACC (ocean only) — Test

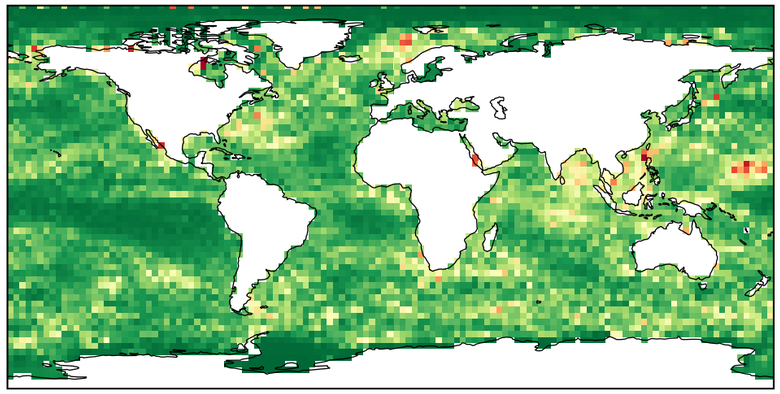
Step 1



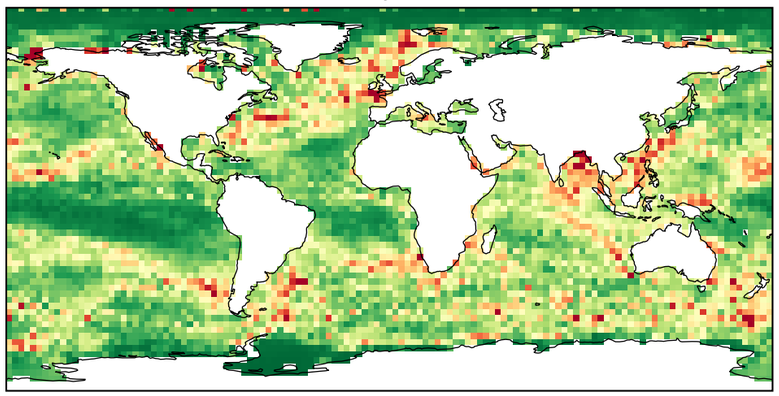
Step 3



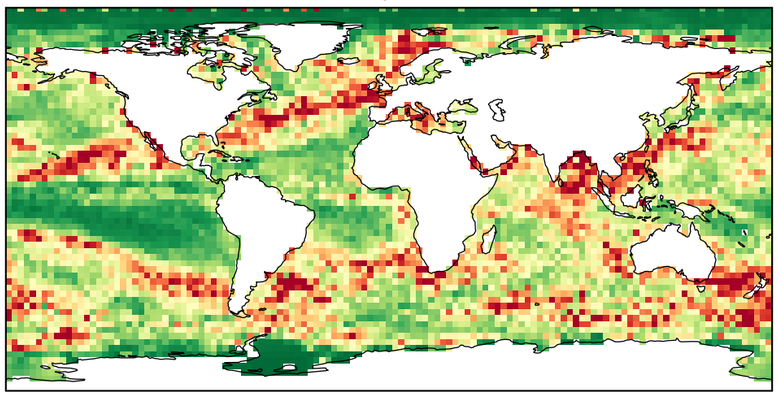
Step 7



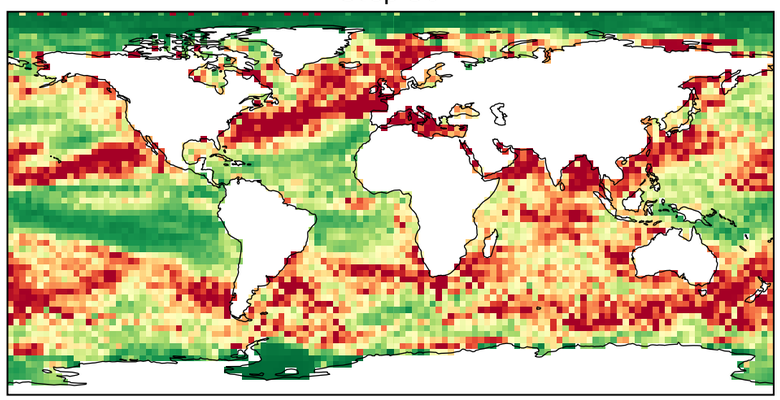
Step 14



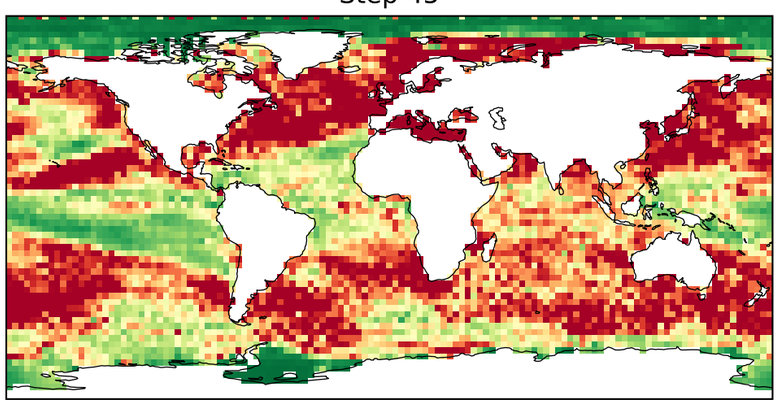
Step 21



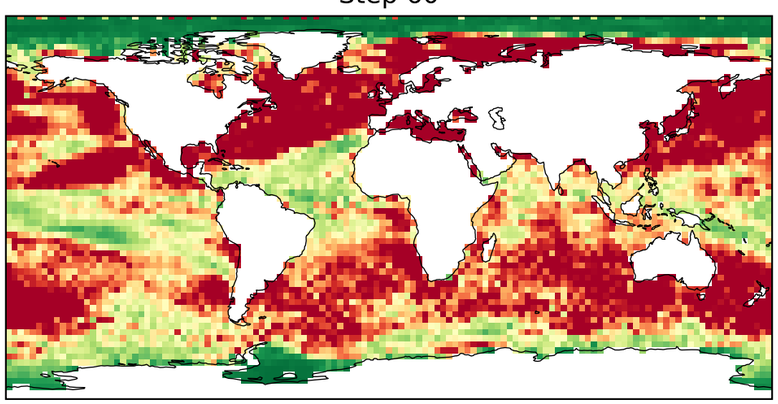
Step 30



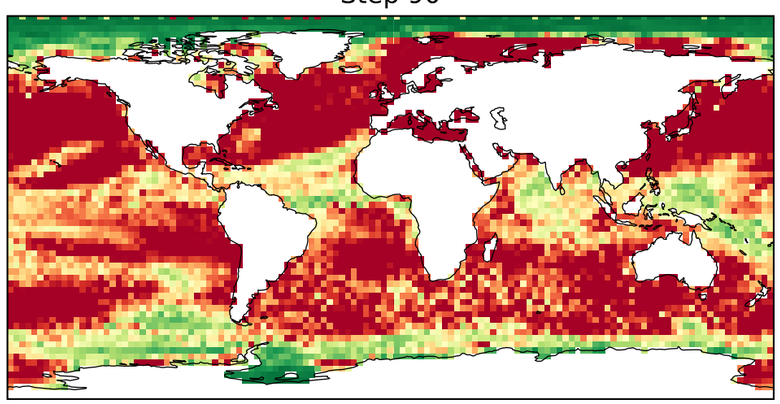
Step 45



Step 60

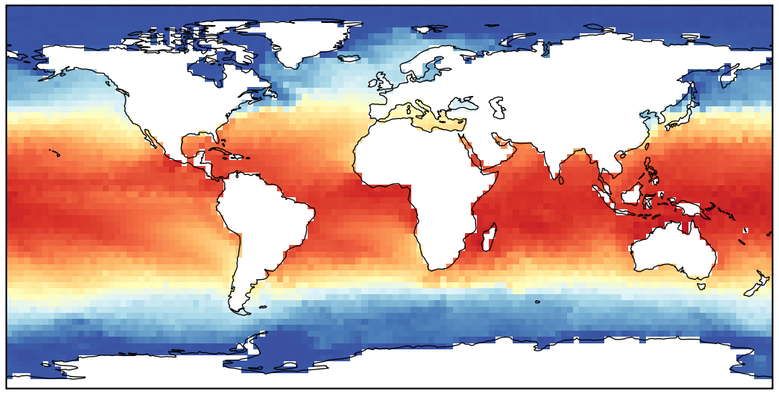


Step 90

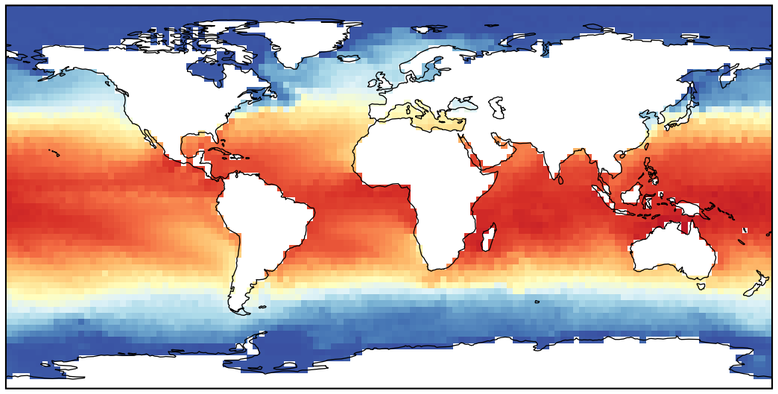


Chain init: 2020-01-11 — Test (°C)

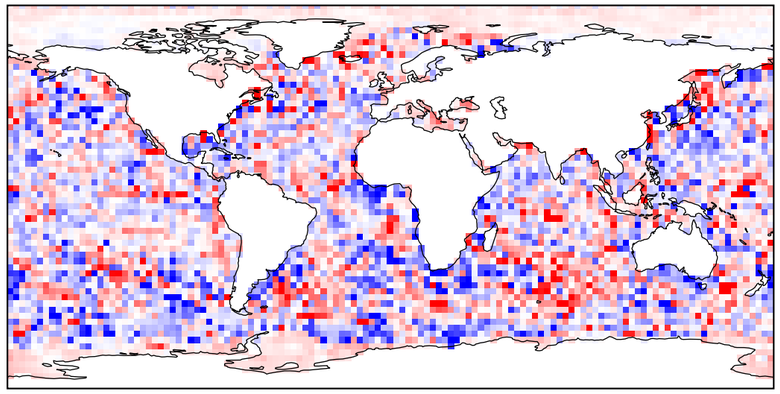
Actual — Step 1 (°C)



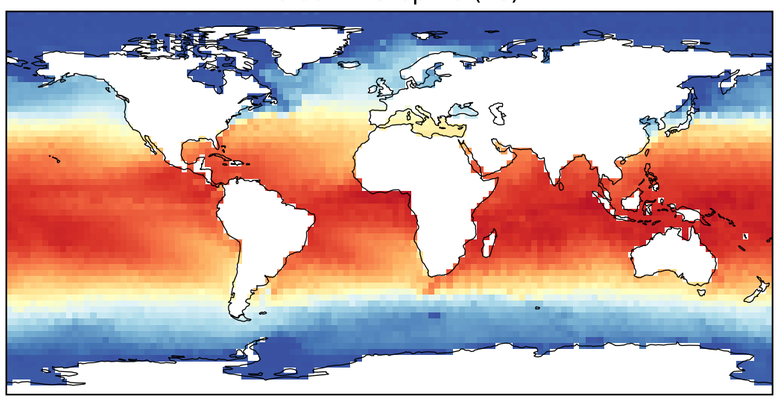
SFNO — Step 1 (°C)



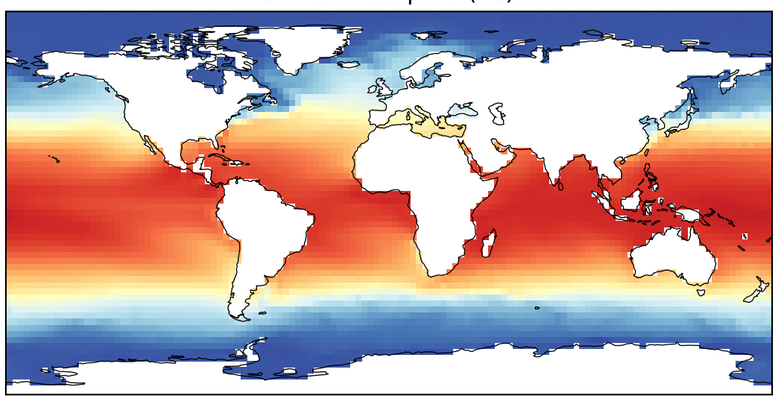
Error — Step 1 (°C)



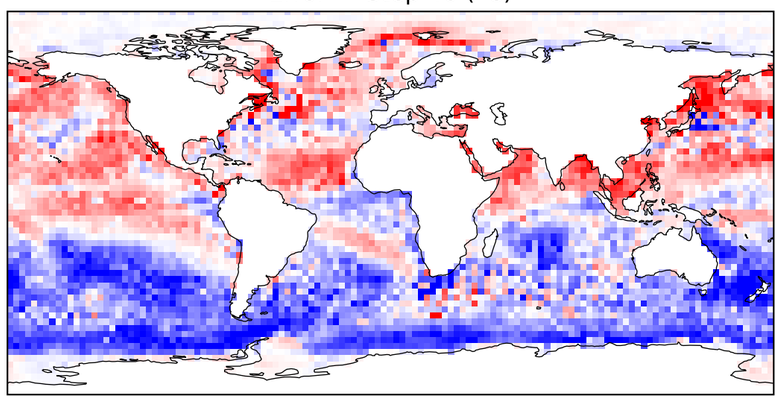
Actual — Step 46 (°C)



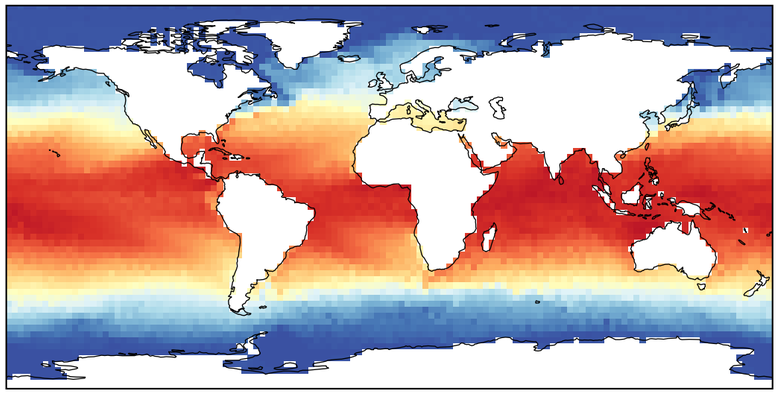
SFNO — Step 46 (°C)



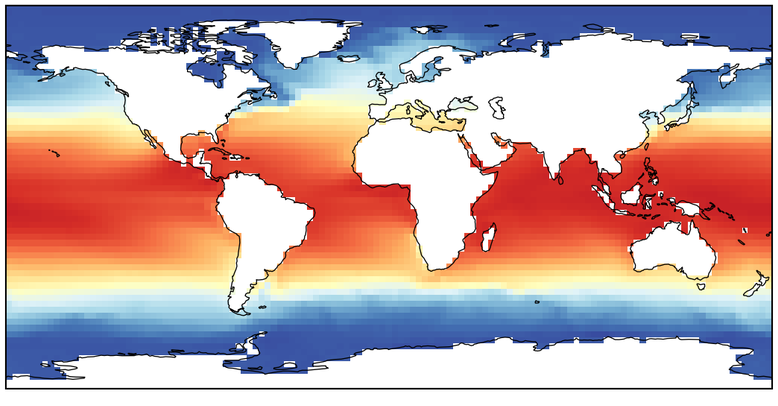
Error — Step 46 (°C)



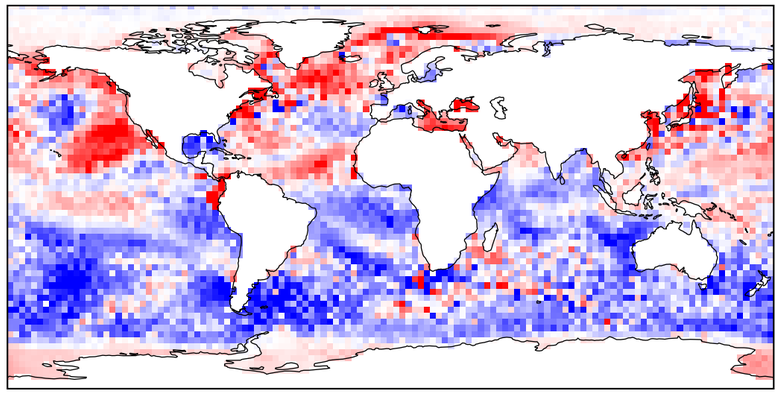
Actual — Step 90 (°C)

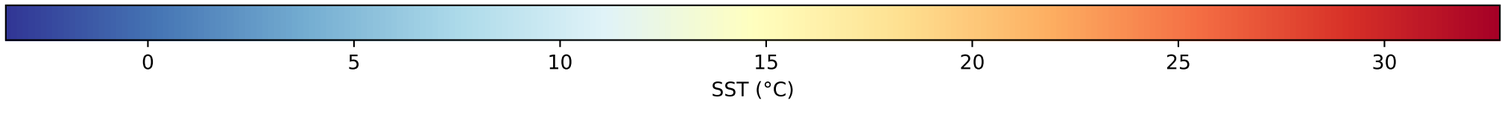


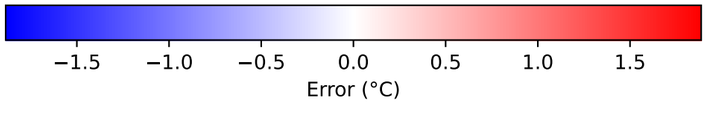
SFNO — Step 90 (°C)



Error — Step 90 (°C)

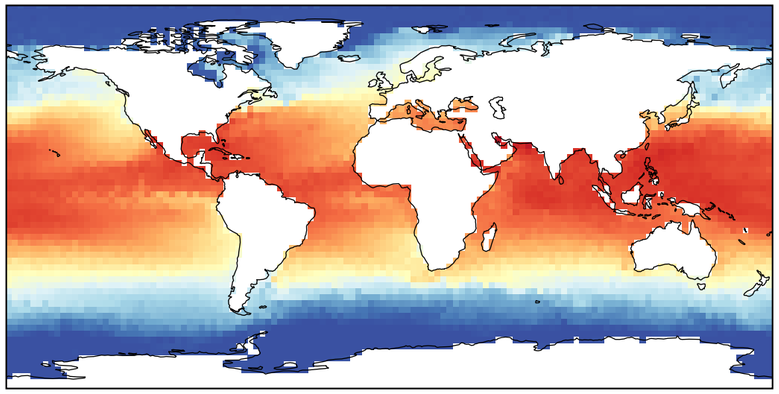




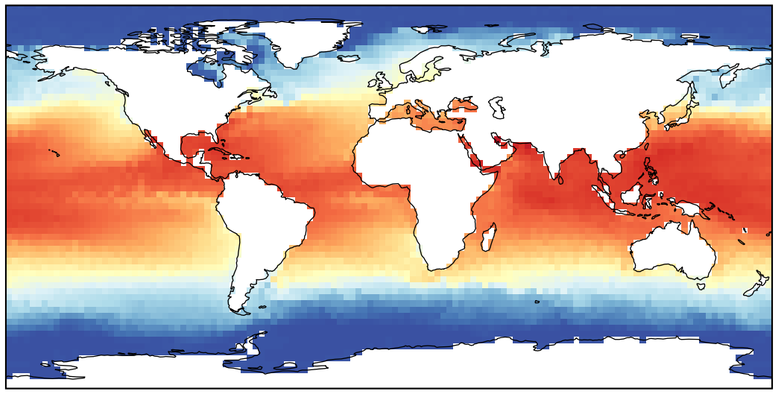


Chain init: 2020-07-09 — Test (°C)

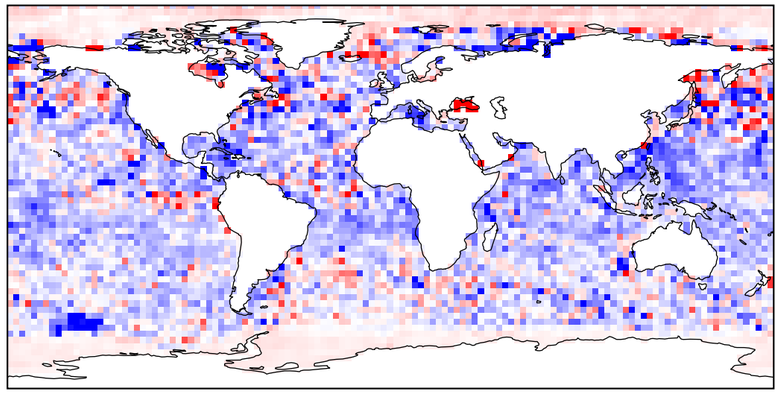
Actual — Step 1 (°C)



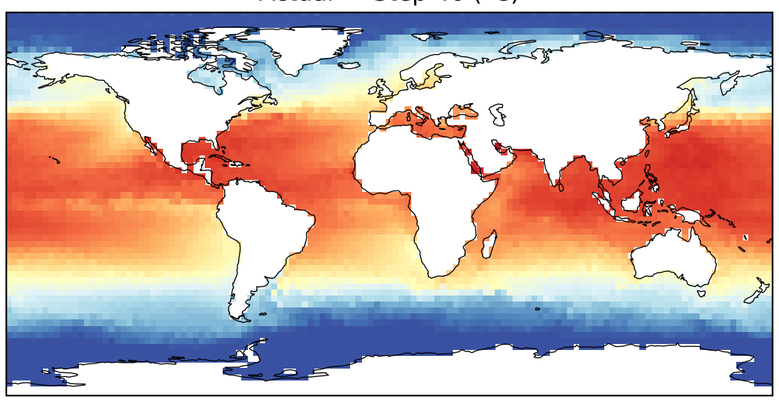
SFNO — Step 1 (°C)



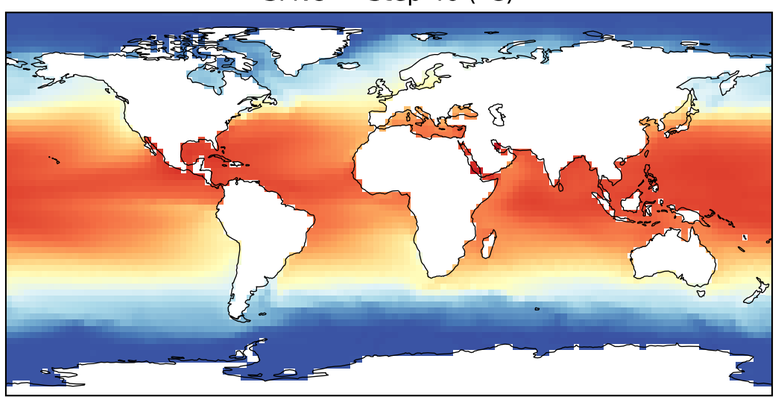
Error — Step 1 (°C)



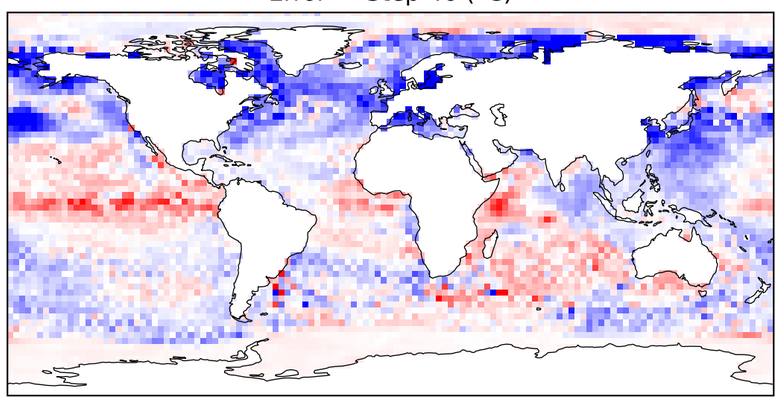
Actual — Step 46 (°C)



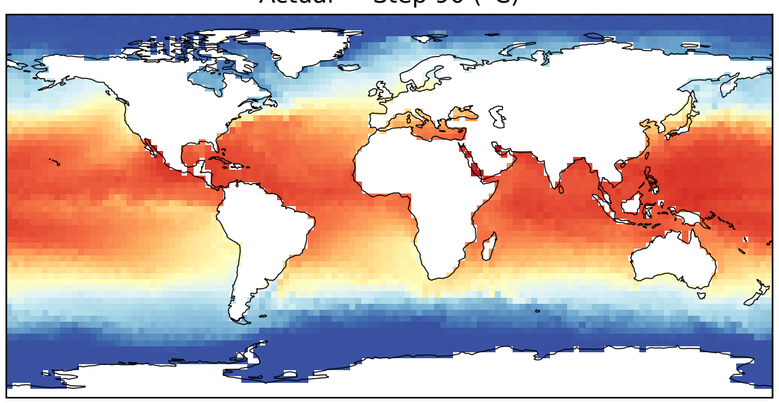
SFNO — Step 46 (°C)



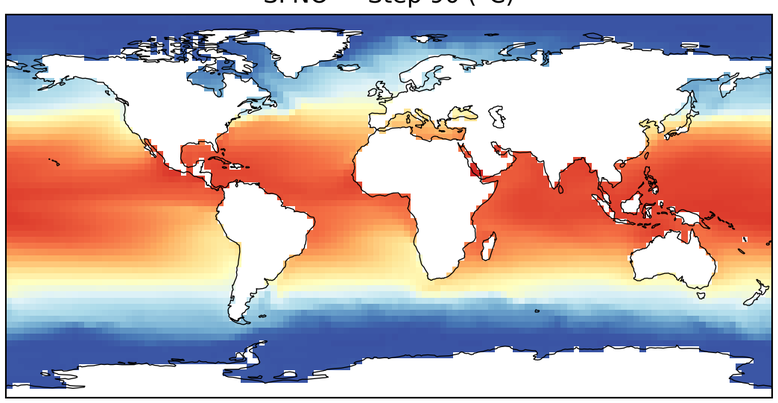
Error — Step 46 (°C)



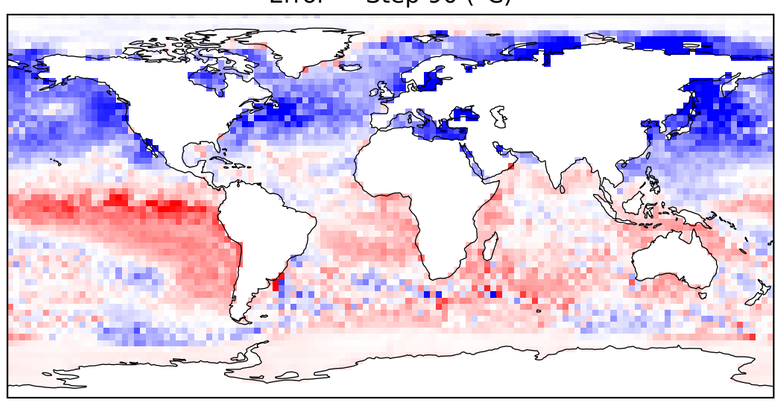
Actual — Step 90 (°C)

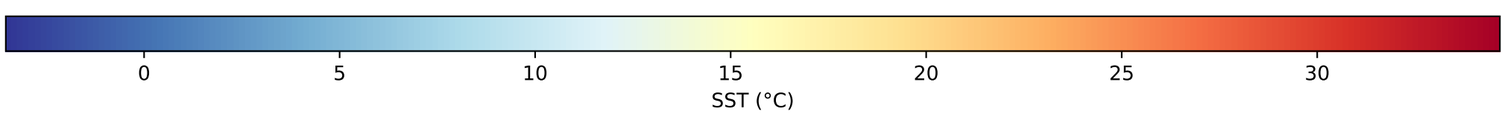


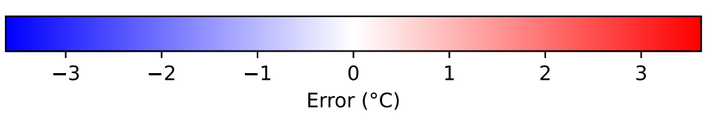
SFNO — Step 90 (°C)



Error — Step 90 (°C)

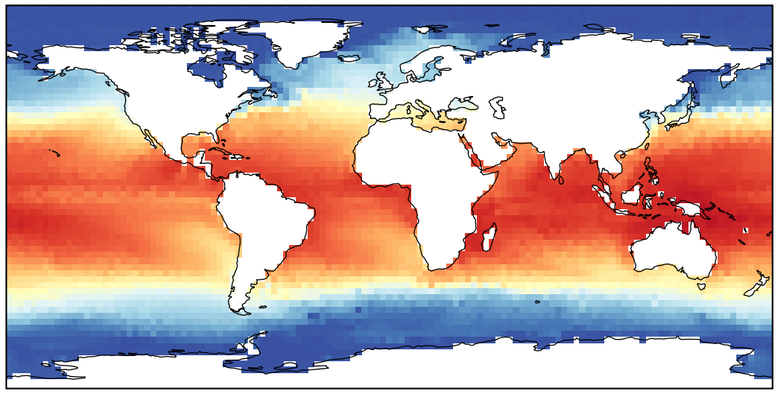




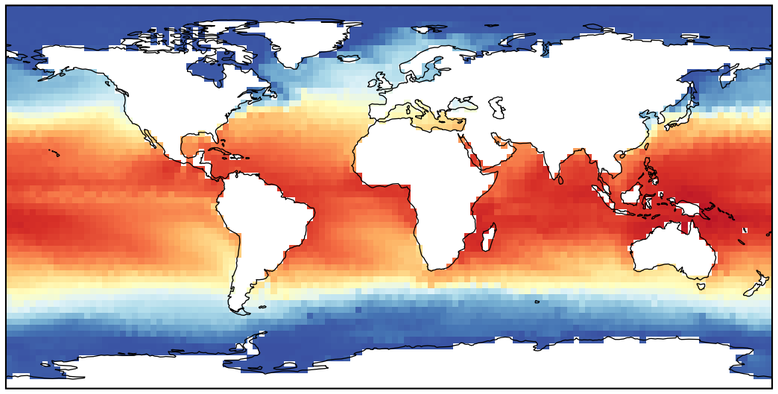


Chain init: 2021-01-06 — Test (°C)

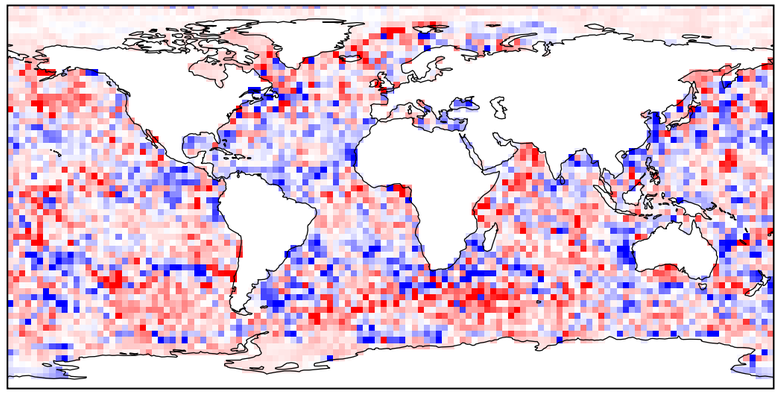
Actual — Step 1 (°C)



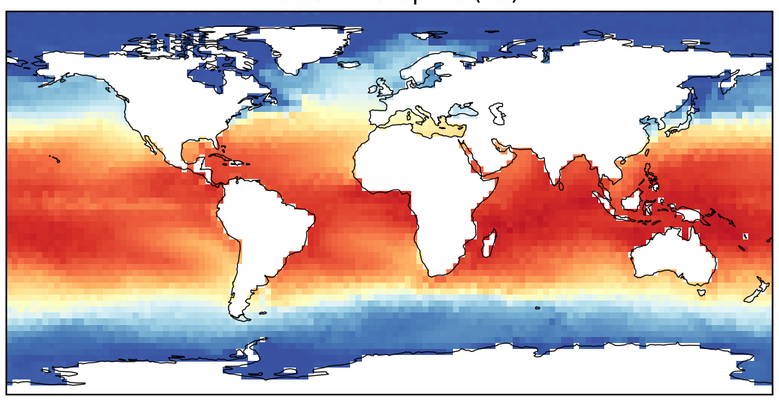
SFNO — Step 1 (°C)



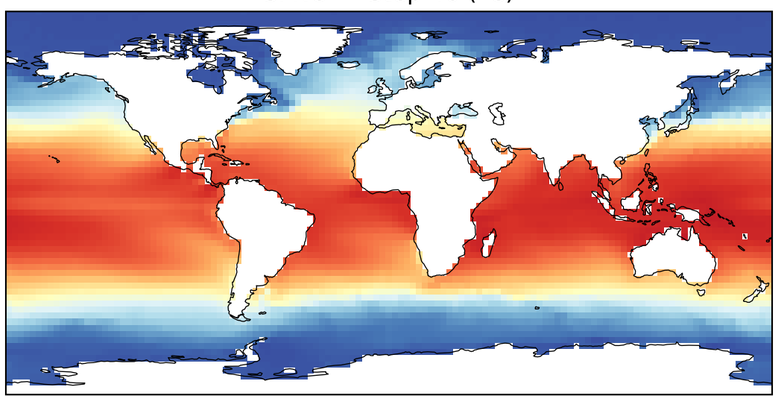
Error — Step 1 (°C)



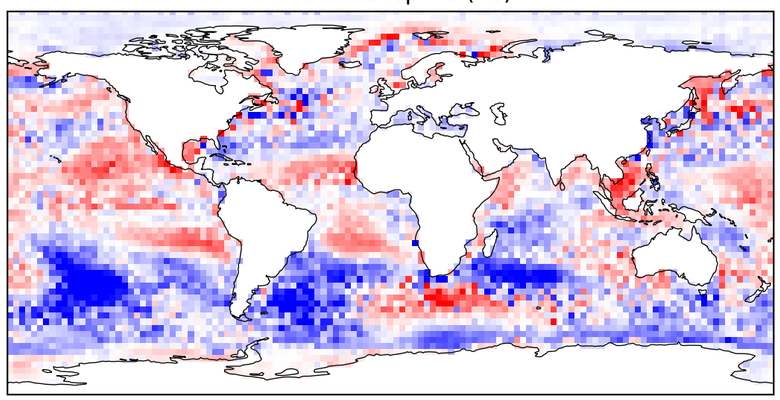
Actual — Step 46 (°C)



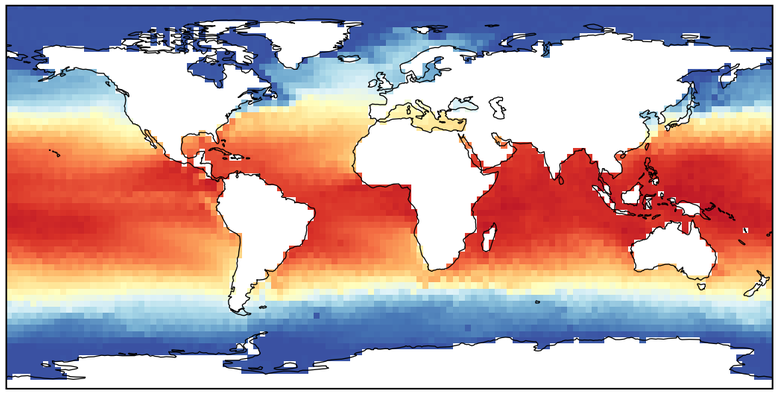
SFNO — Step 46 (°C)



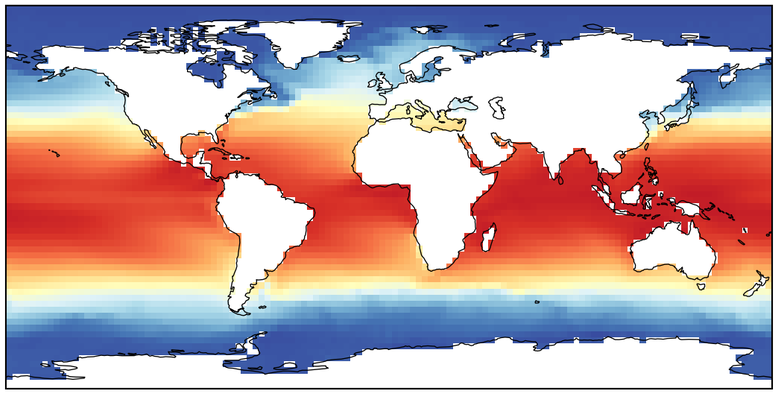
Error — Step 46 (°C)



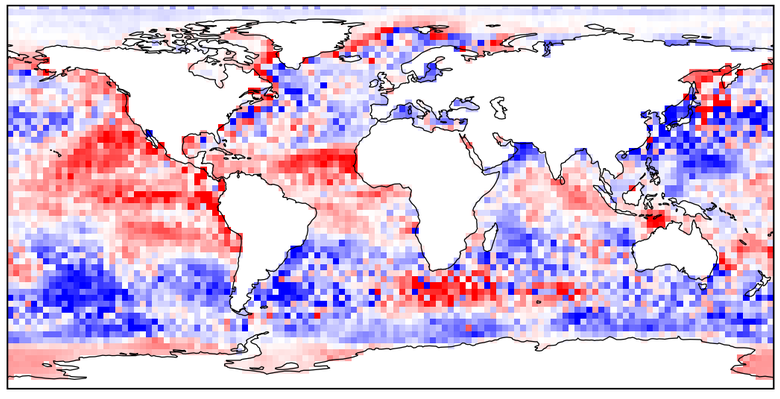
Actual — Step 90 (°C)

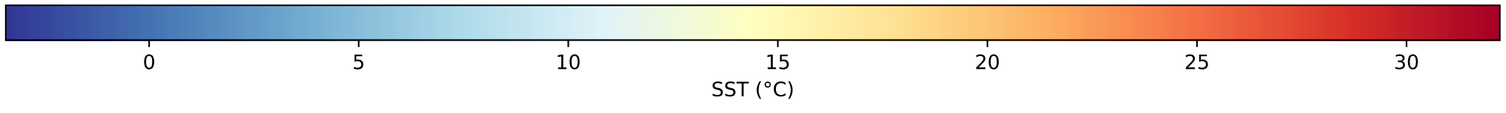


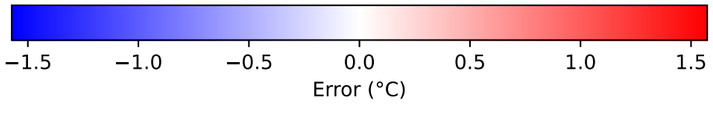
SFNO — Step 90 (°C)



Error — Step 90 (°C)

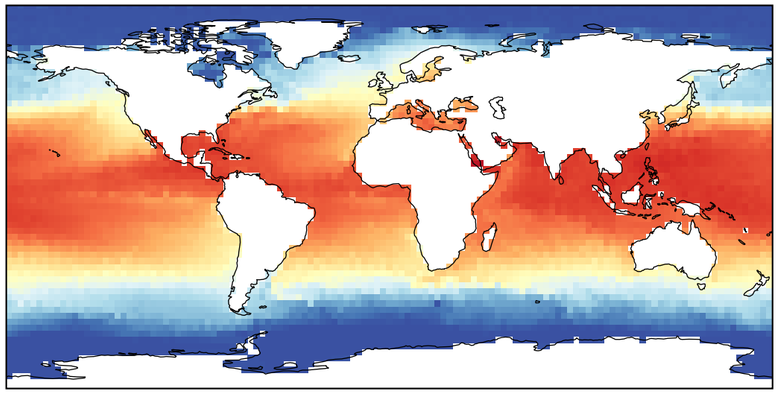




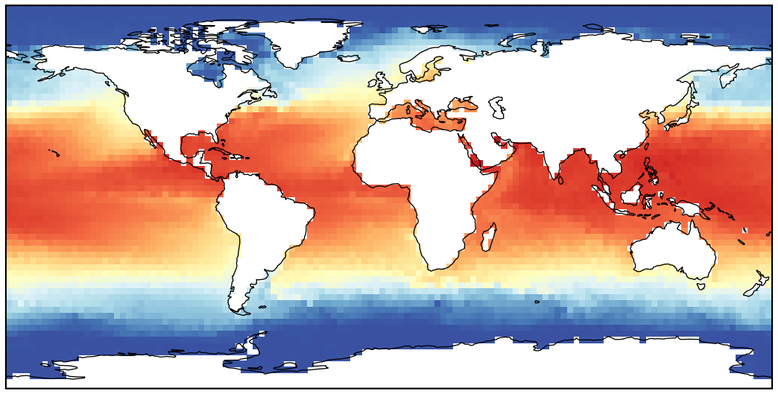


Chain init: 2021-07-05 — Test (°C)

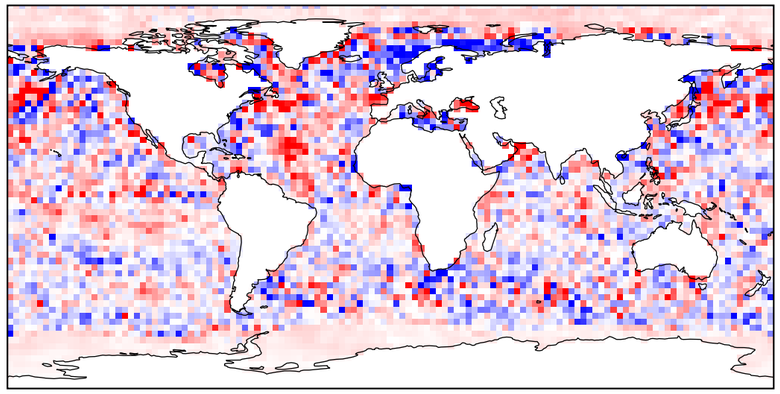
Actual — Step 1 (°C)



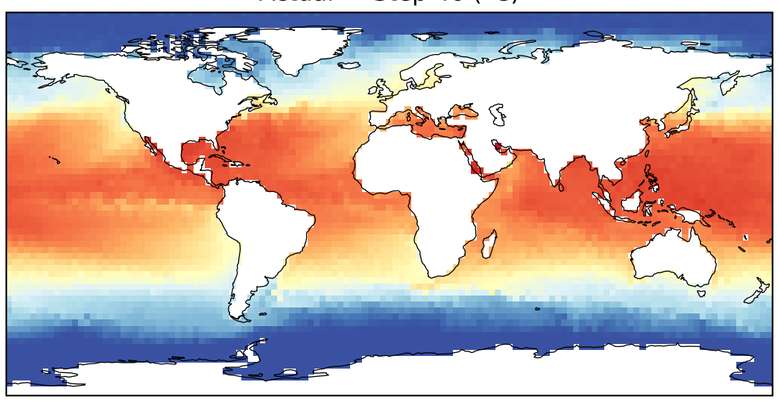
SFNO — Step 1 (°C)



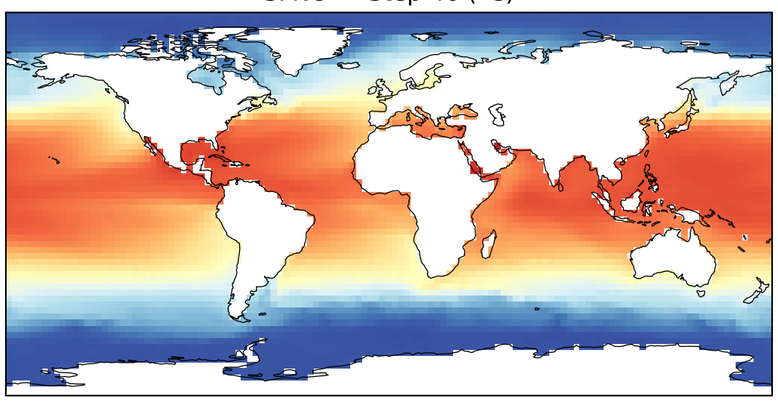
Error — Step 1 (°C)



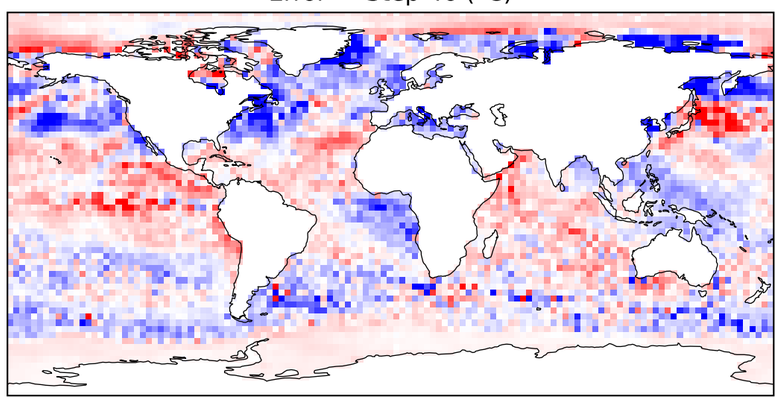
Actual — Step 46 (°C)



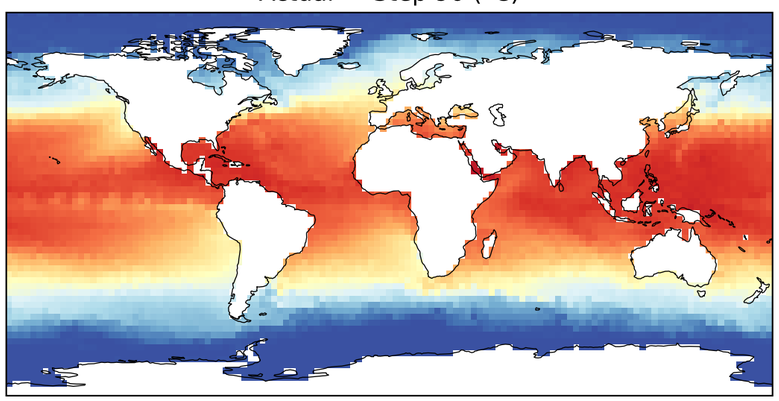
SFNO — Step 46 (°C)



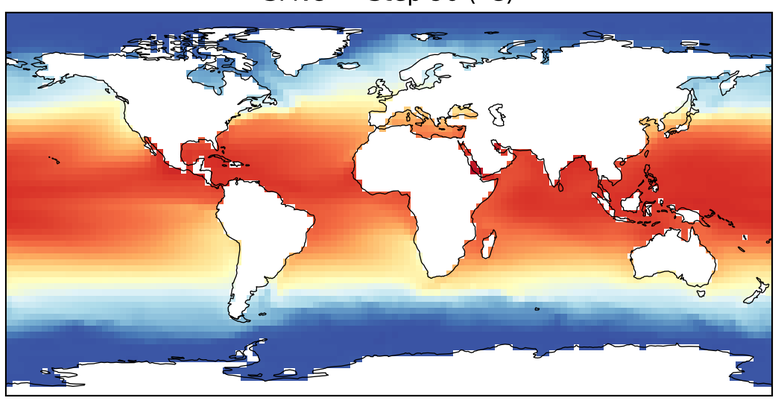
Error — Step 46 (°C)



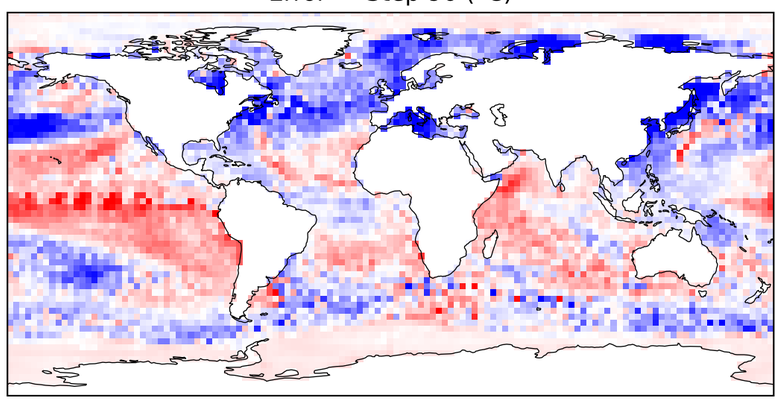
Actual — Step 90 (°C)

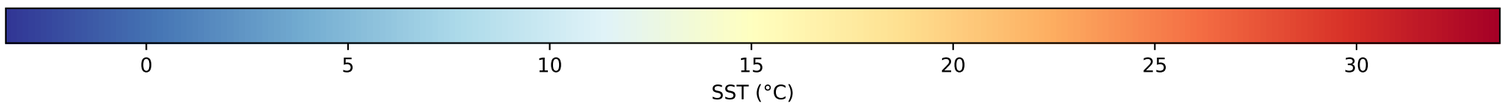


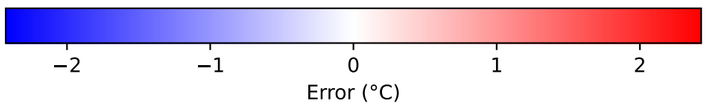
SFNO — Step 90 (°C)



Error — Step 90 (°C)

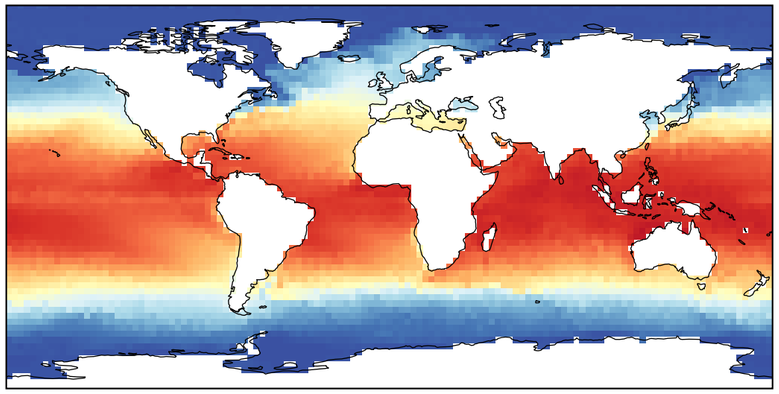




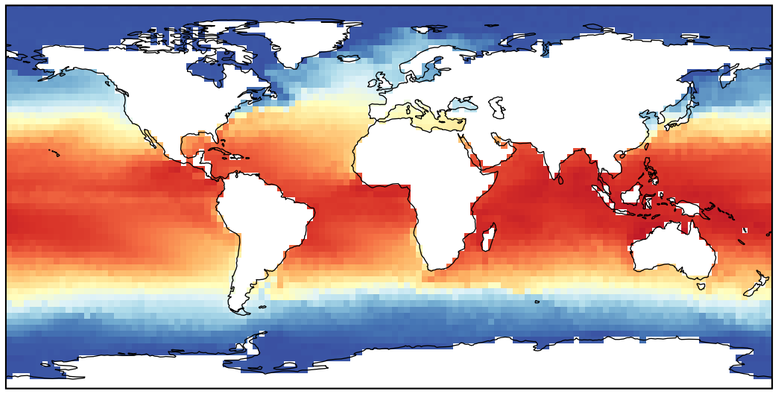


Chain init: 2022-04-01 — Test (°C)

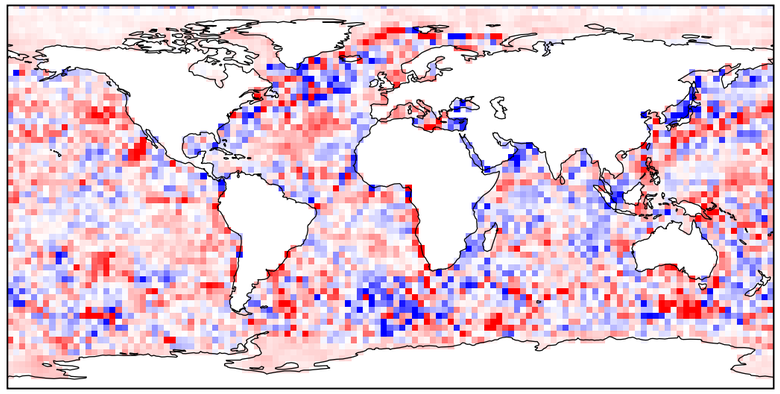
Actual — Step 1 (°C)



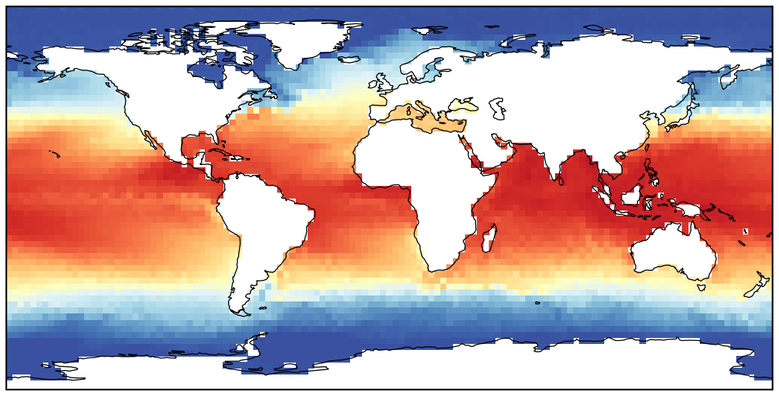
SFNO — Step 1 (°C)



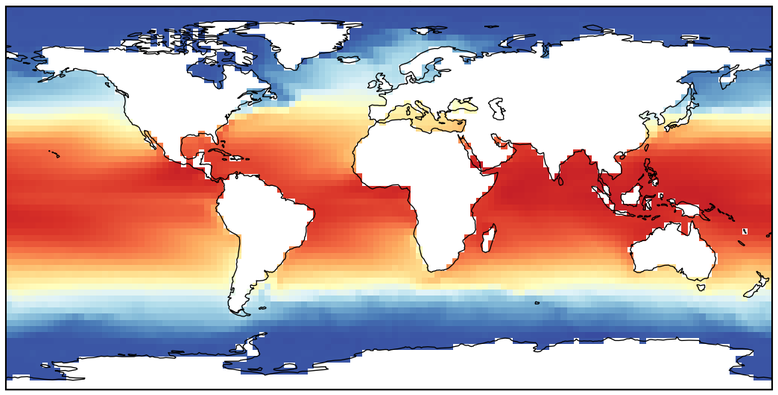
Error — Step 1 (°C)



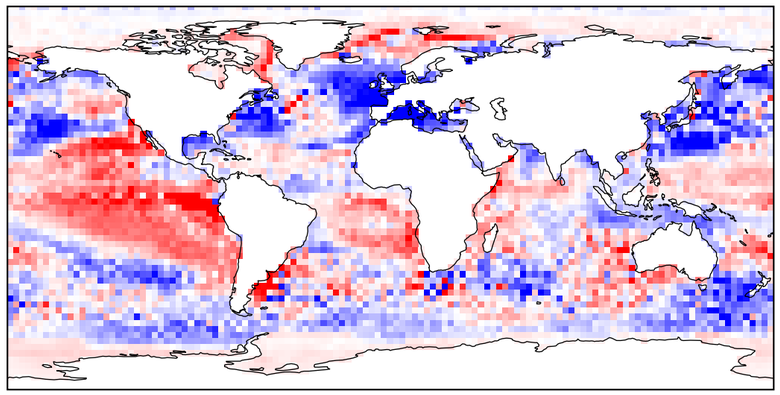
Actual — Step 46 (°C)



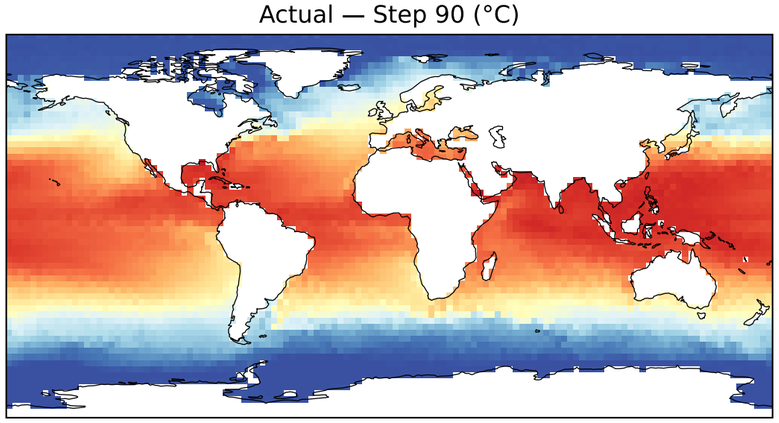
SFNO — Step 46 (°C)



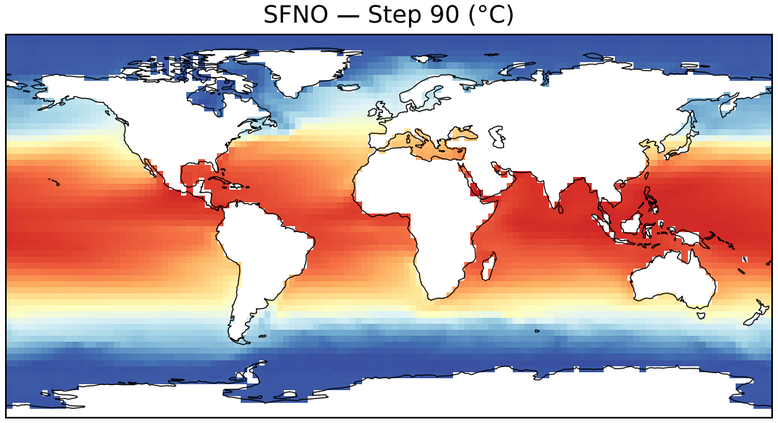
Error — Step 46 (°C)



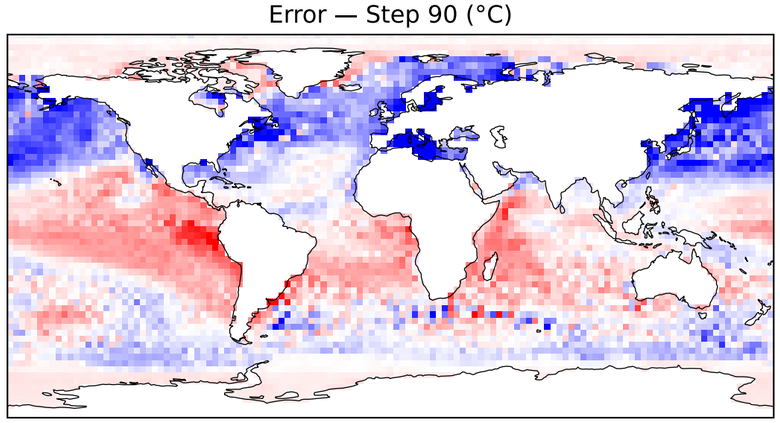
Actual — Step 90 (°C)

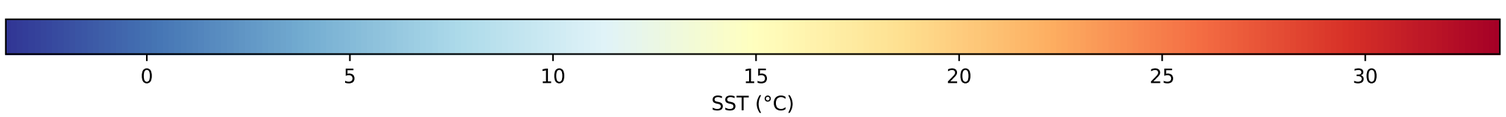


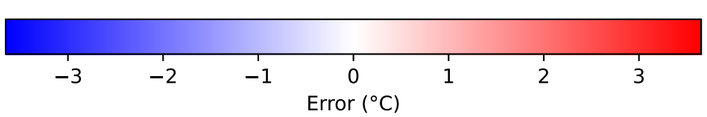
SFNO — Step 90 (°C)



Error — Step 90 (°C)

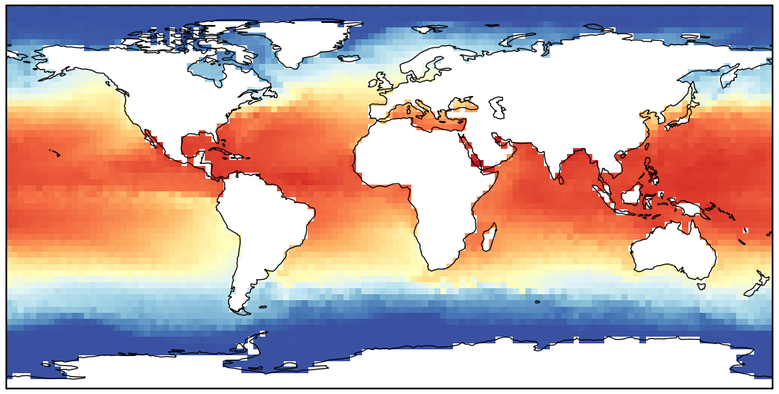




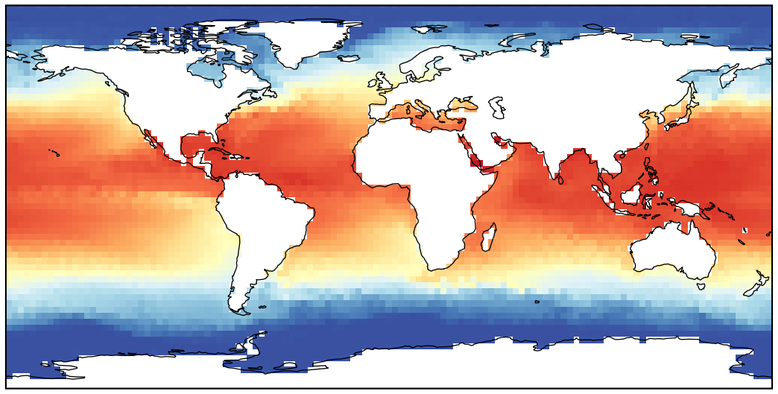


Chain init: 2022-09-28 — Test (°C)

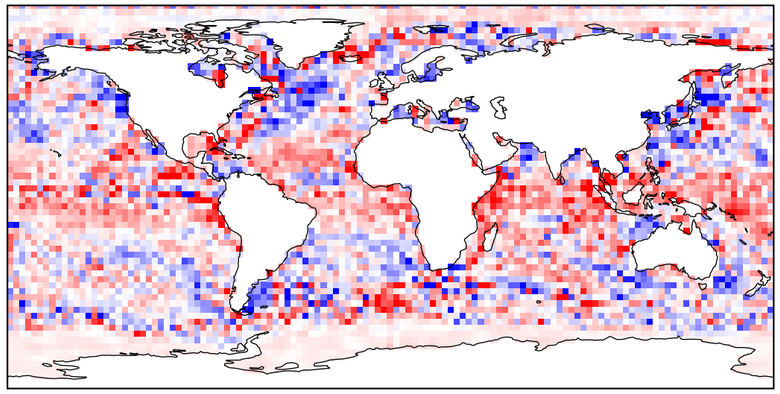
Actual — Step 1 (°C)



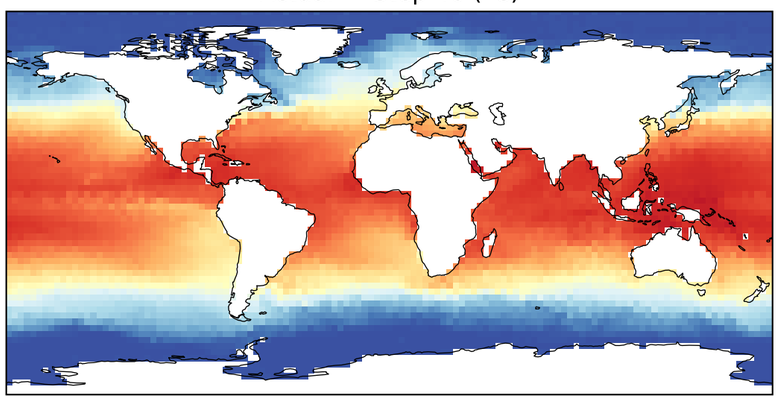
SFNO — Step 1 (°C)



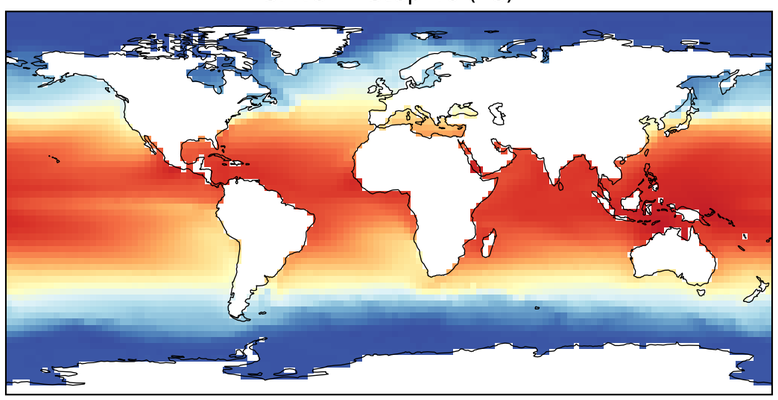
Error — Step 1 (°C)



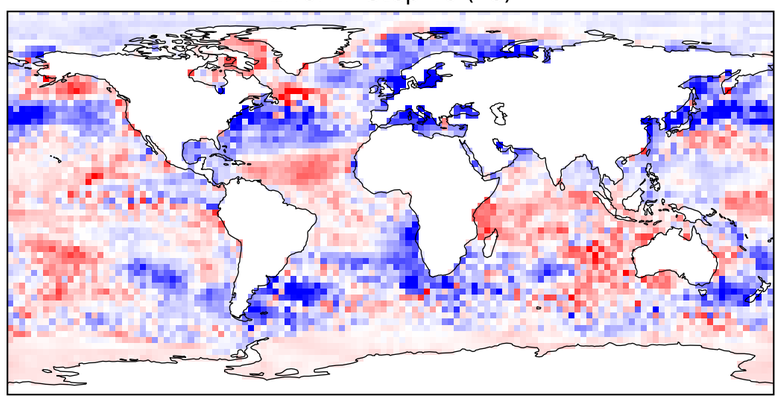
Actual — Step 46 (°C)



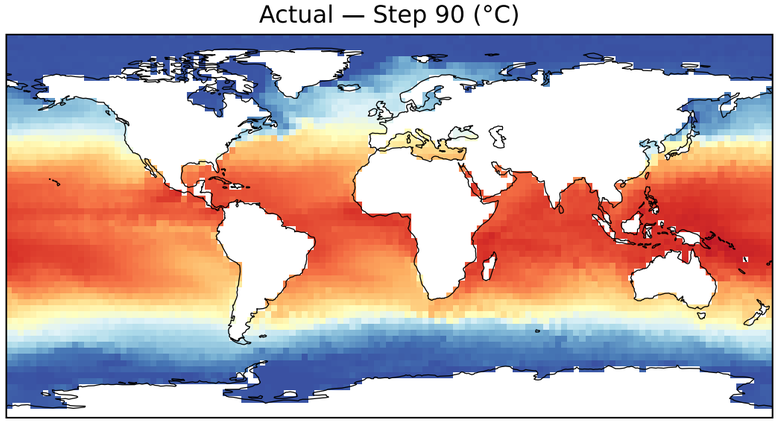
SFNO — Step 46 (°C)



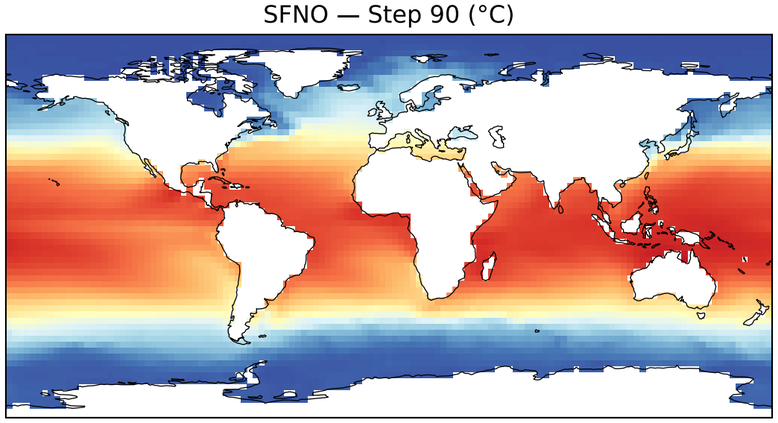
Error — Step 46 (°C)



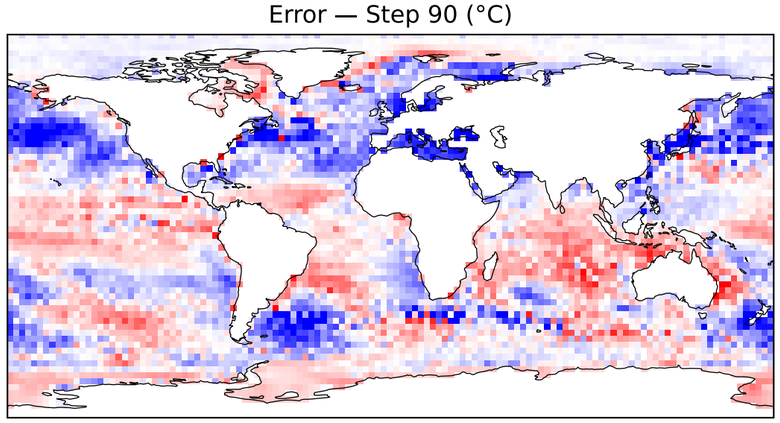
Actual — Step 90 (°C)

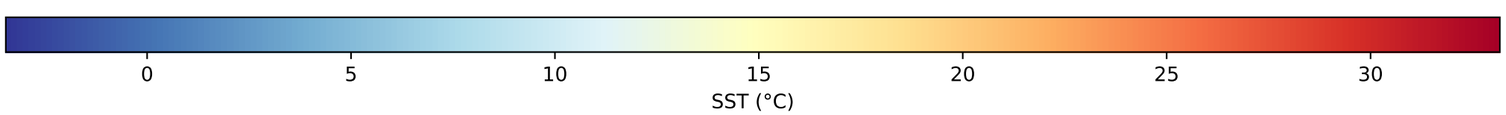


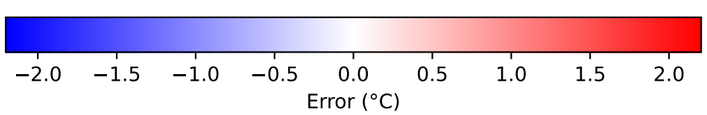
SFNO — Step 90 (°C)



Error — Step 90 (°C)

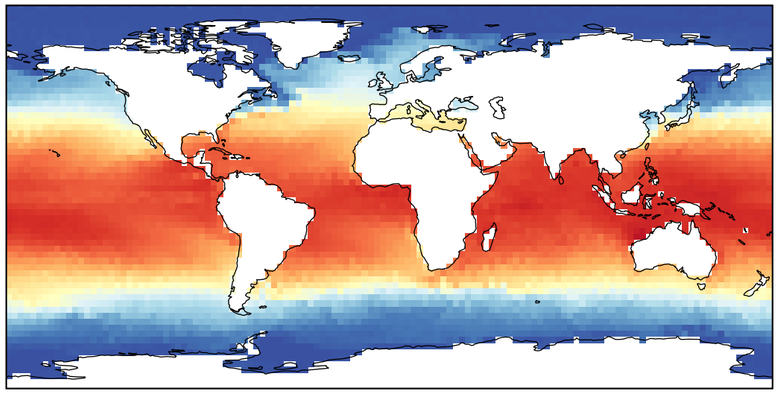




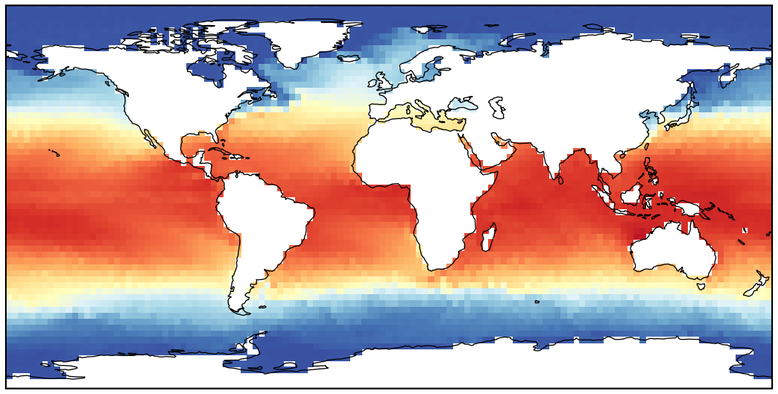


Chain init: 2023-03-27 — Test (°C)

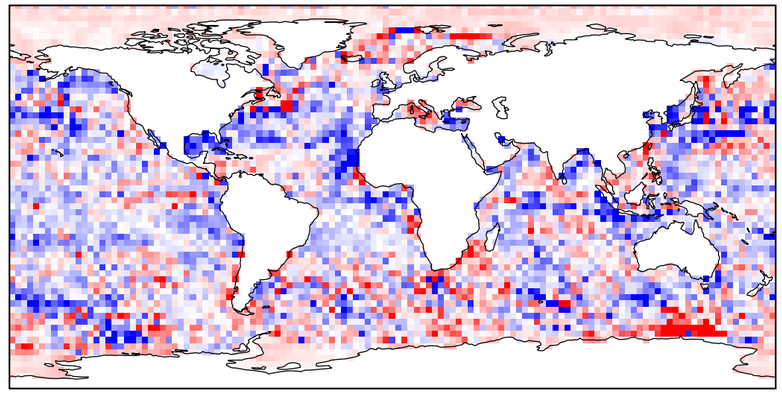
Actual — Step 1 (°C)



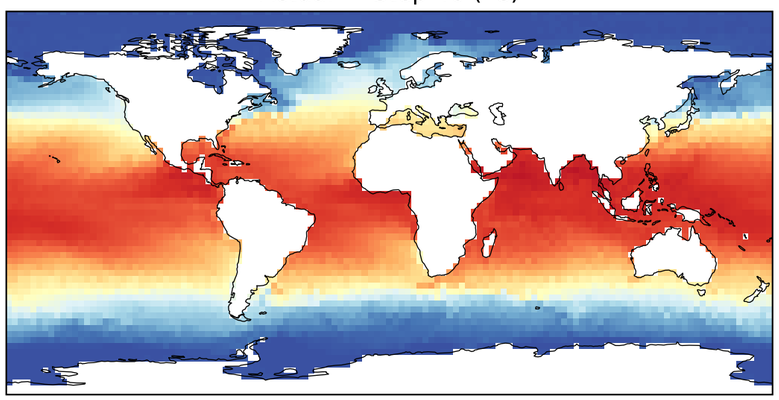
SFNO — Step 1 (°C)



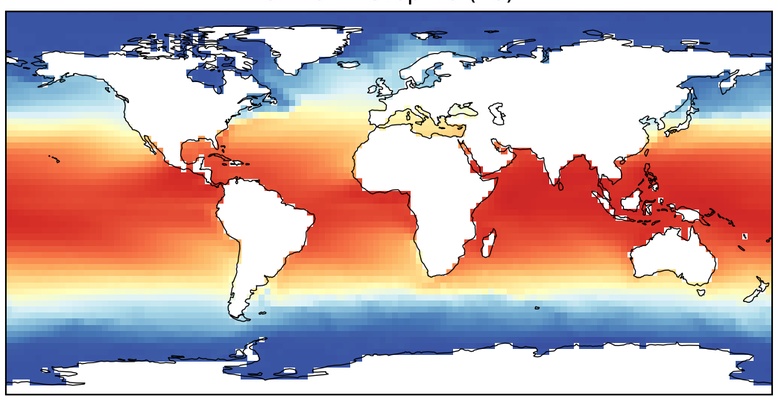
Error — Step 1 (°C)



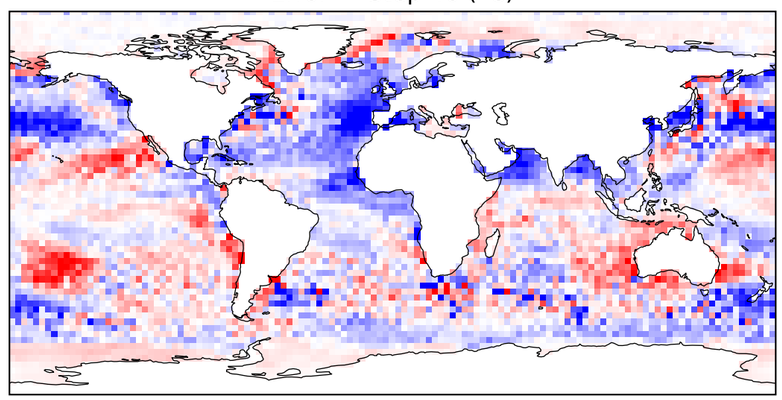
Actual — Step 46 (°C)



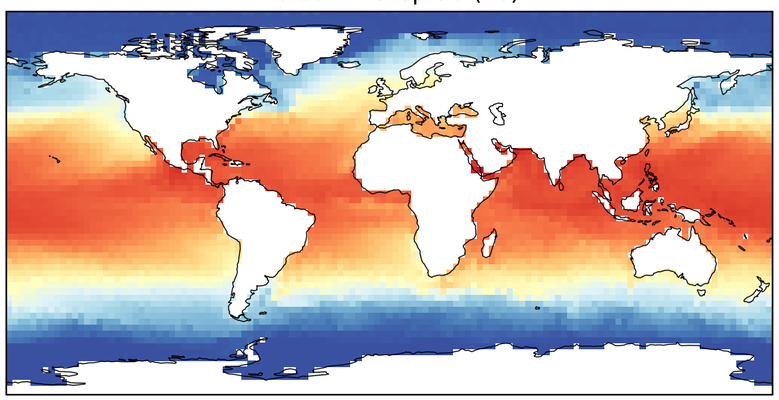
SFNO — Step 46 (°C)



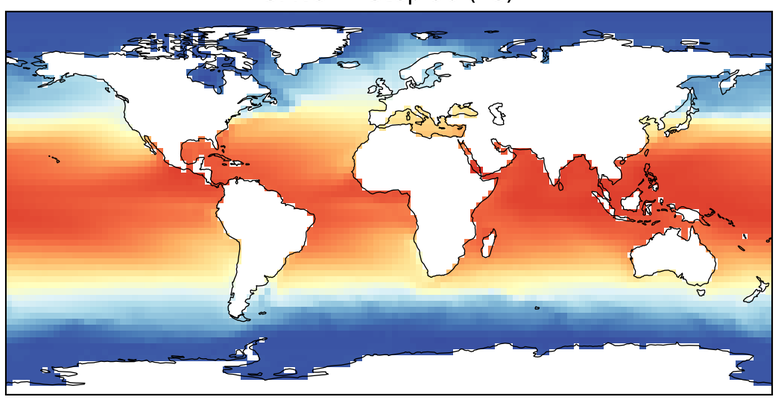
Error — Step 46 (°C)



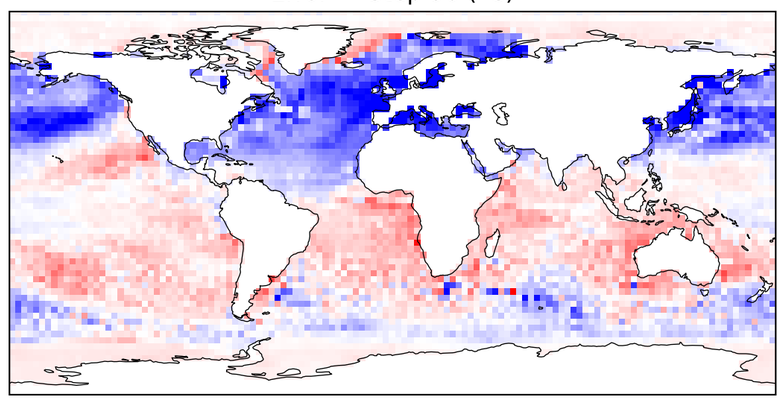
Actual — Step 90 (°C)

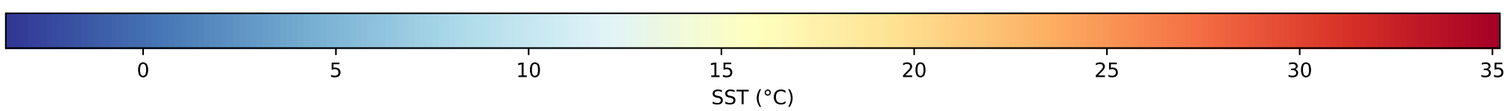


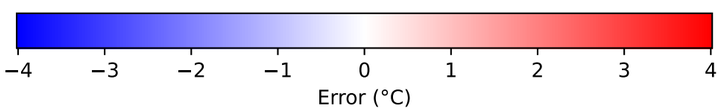
SFNO — Step 90 (°C)



Error — Step 90 (°C)

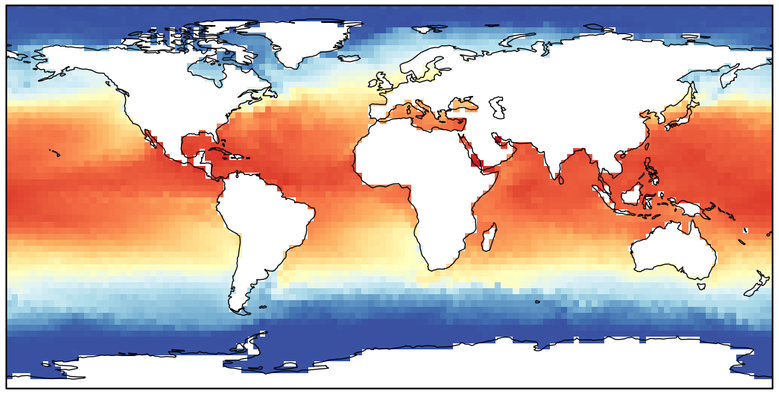




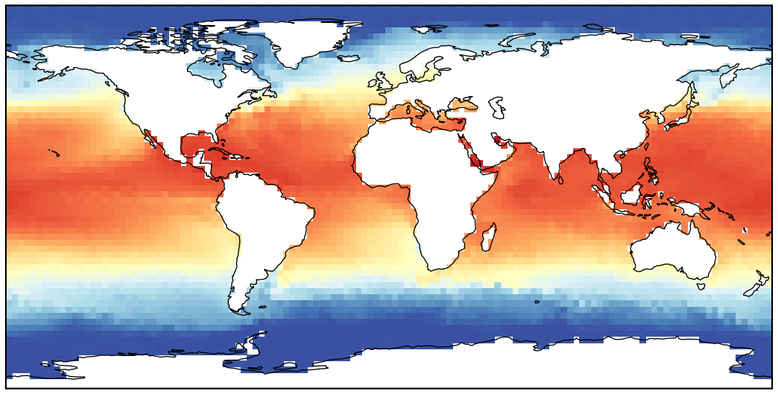


Chain init: 2023-09-23 — Test (°C)

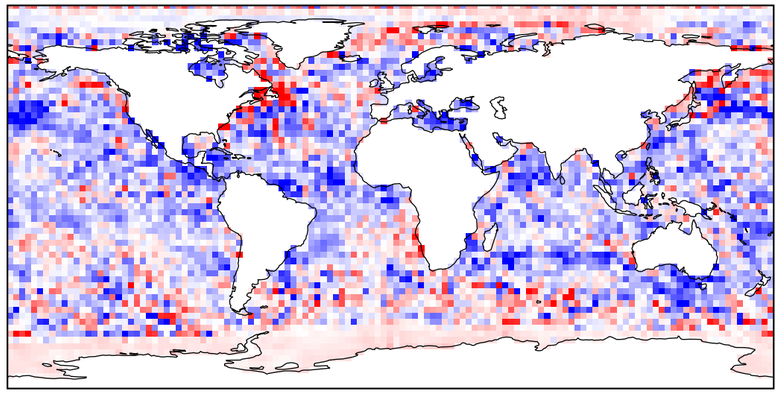
Actual — Step 1 (°C)



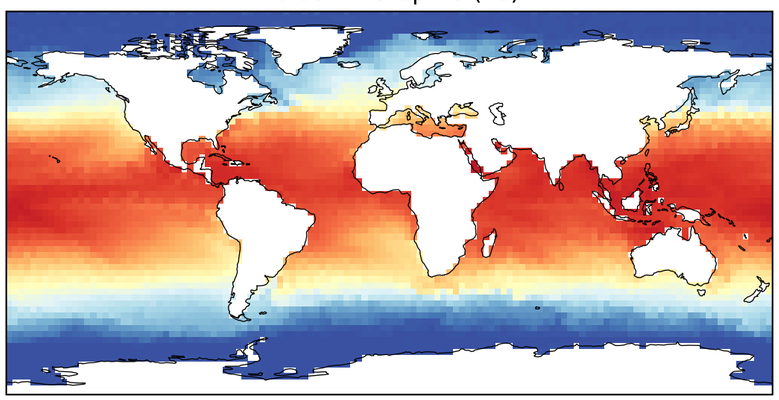
SFNO — Step 1 (°C)



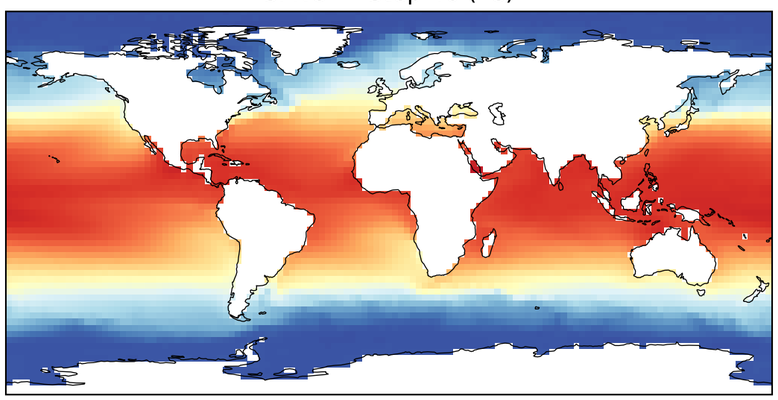
Error — Step 1 (°C)



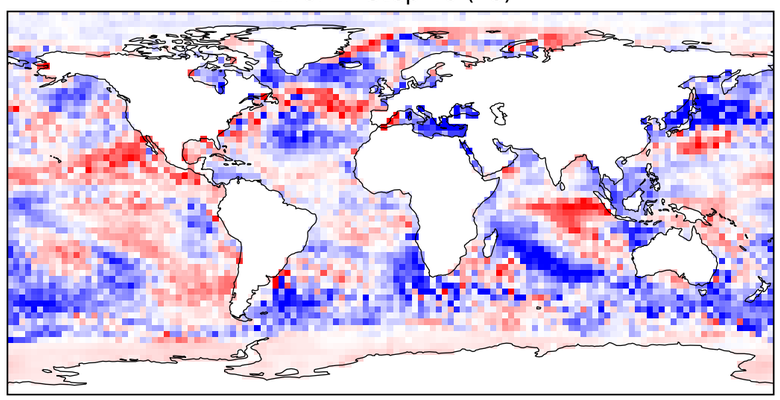
Actual — Step 46 (°C)



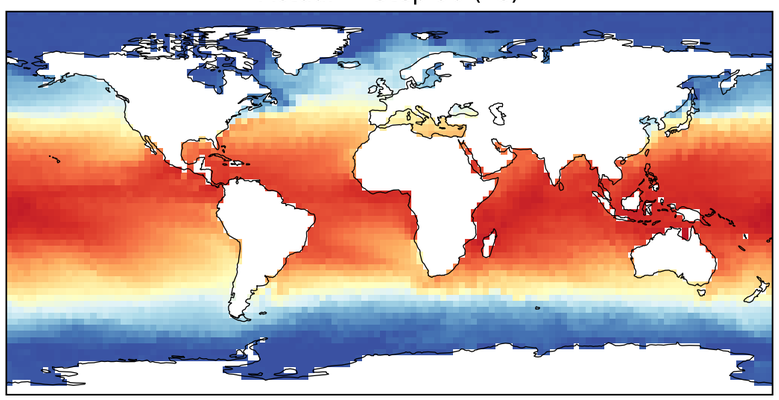
SFNO — Step 46 (°C)



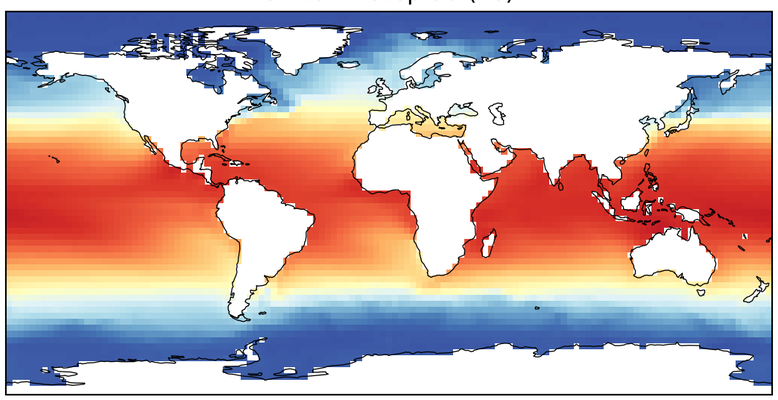
Error — Step 46 (°C)



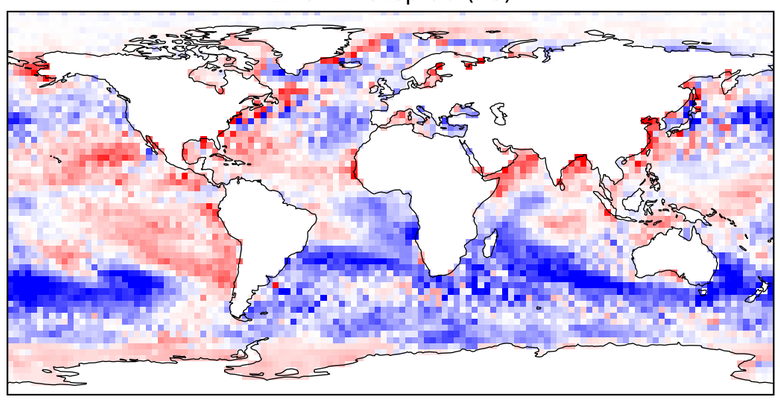
Actual — Step 90 (°C)

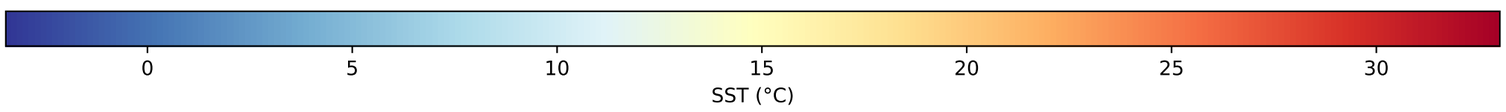


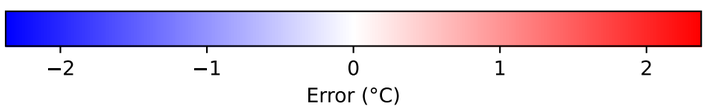
SFNO — Step 90 (°C)



Error — Step 90 (°C)

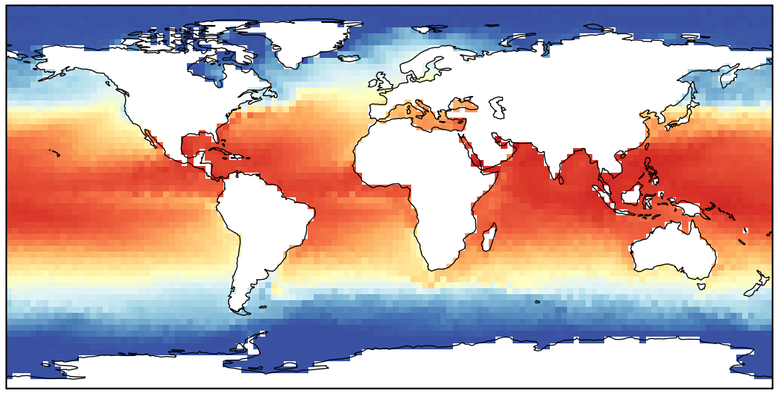




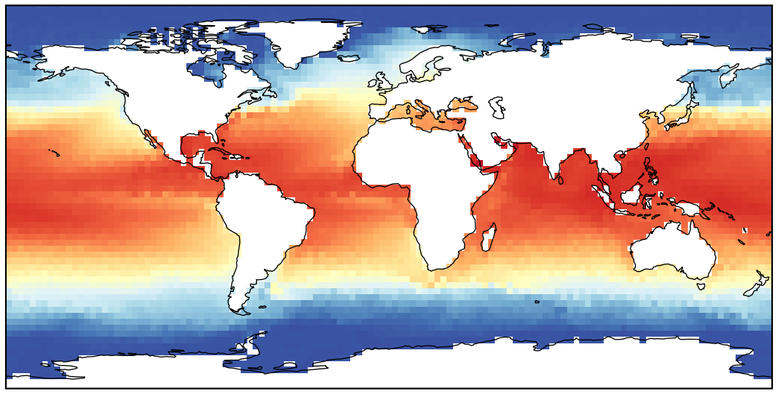


Chain init: 2024-06-19 — Test (°C)

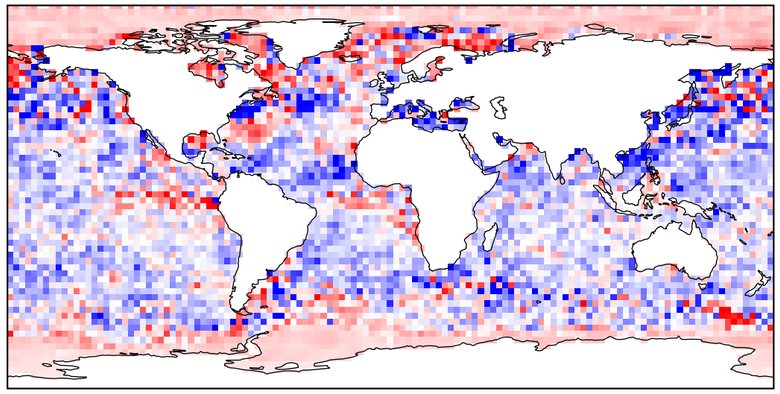
Actual — Step 1 (°C)



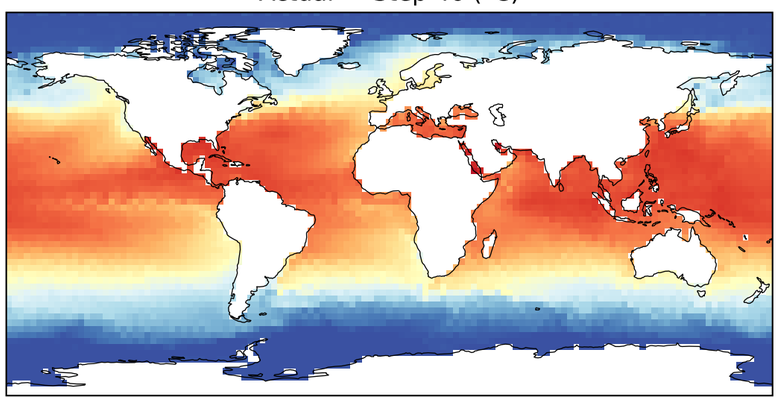
SFNO — Step 1 (°C)



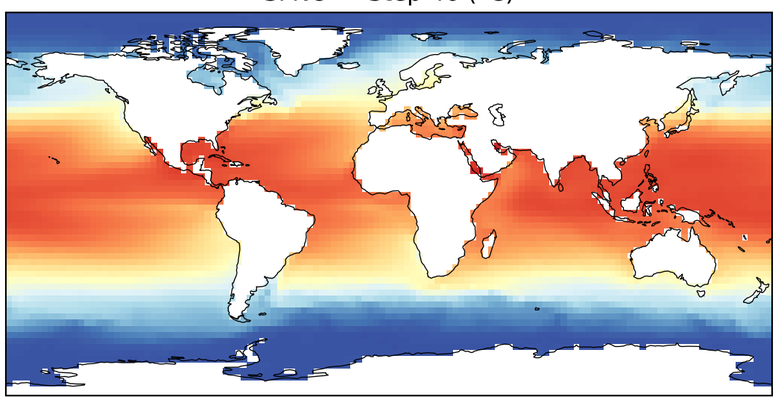
Error — Step 1 (°C)



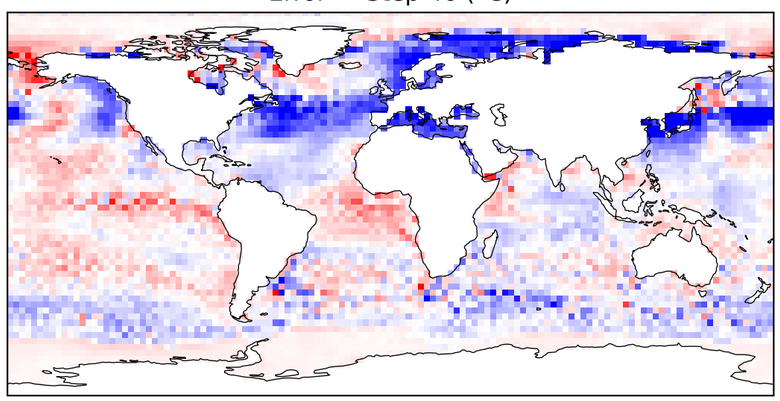
Actual — Step 46 (°C)



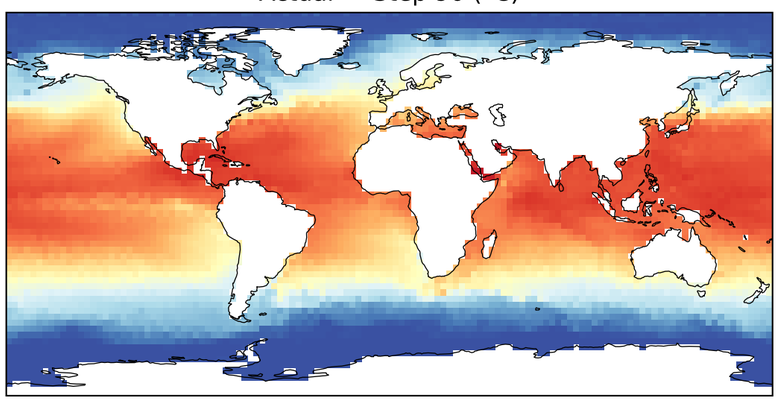
SFNO — Step 46 (°C)



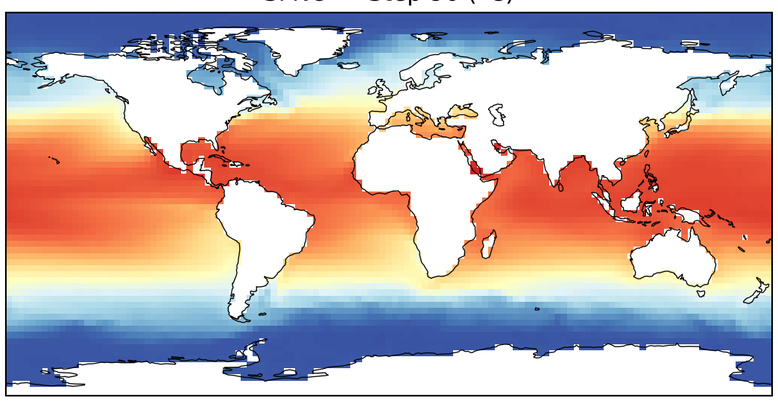
Error — Step 46 (°C)



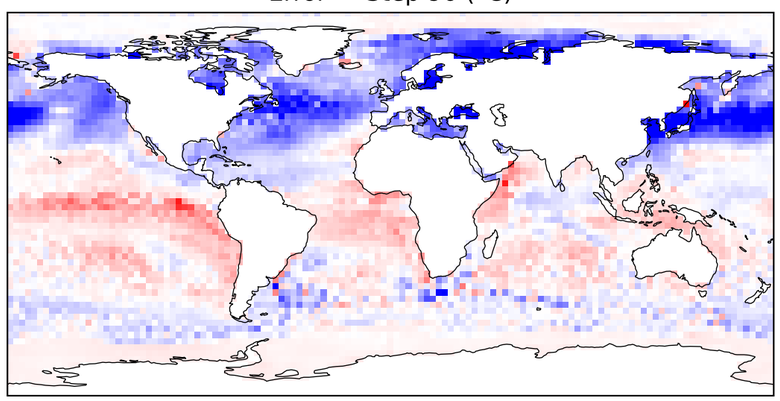
Actual — Step 90 (°C)

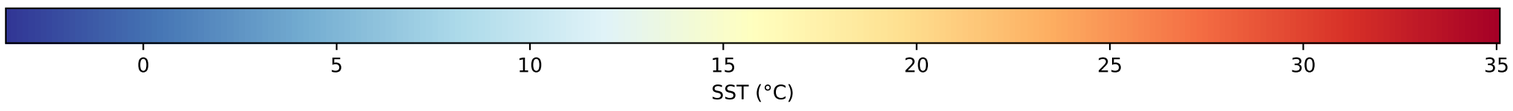


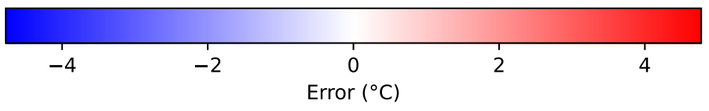
SFNO — Step 90 (°C)



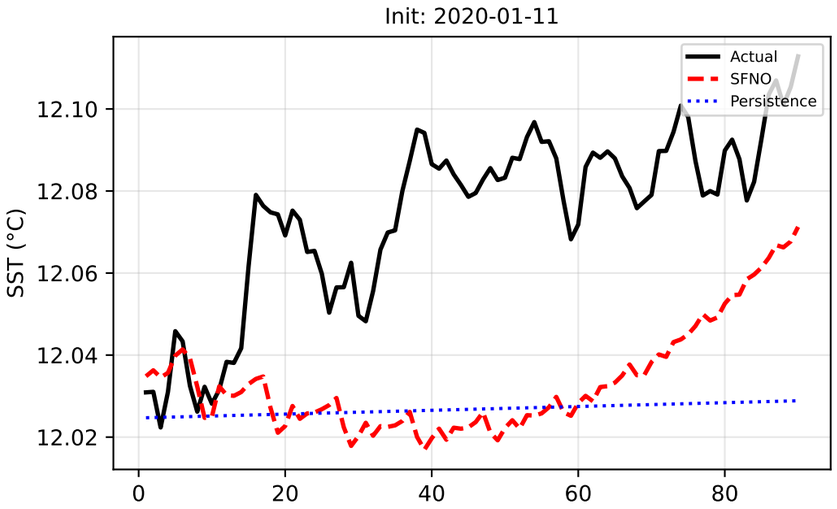
Error — Step 90 (°C)

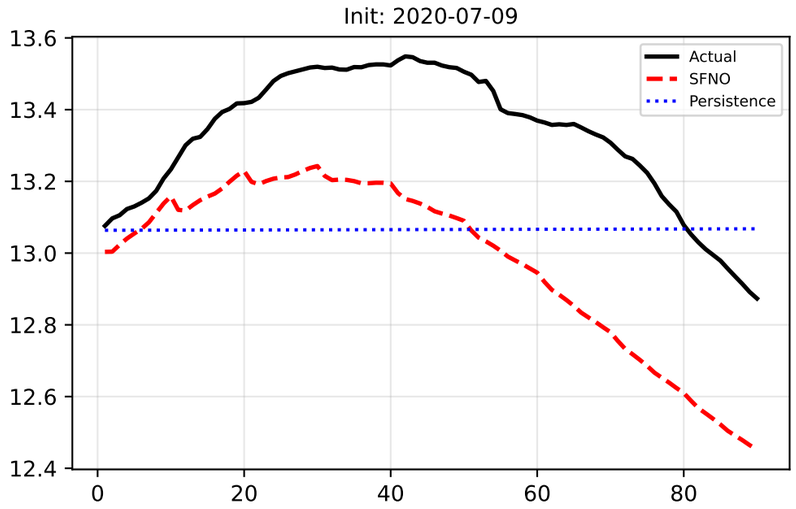


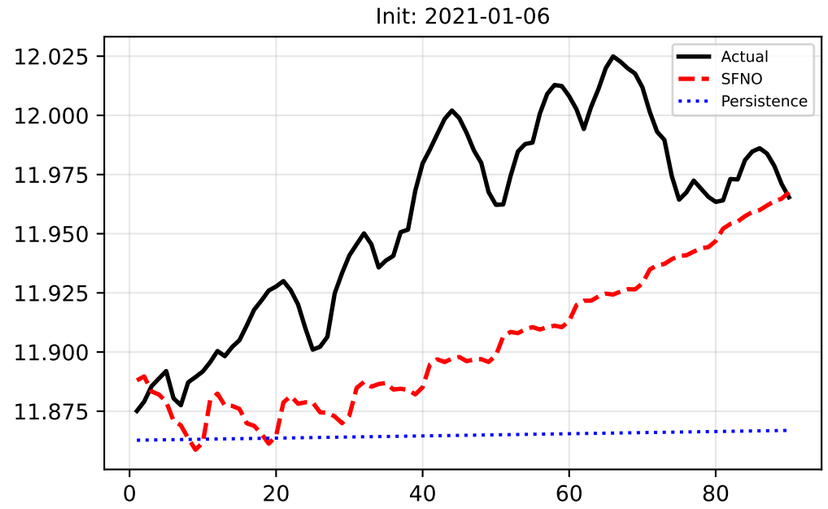


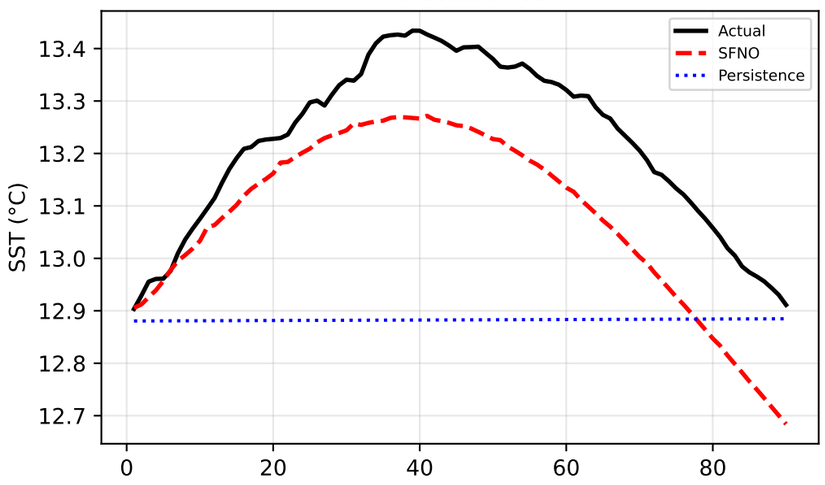


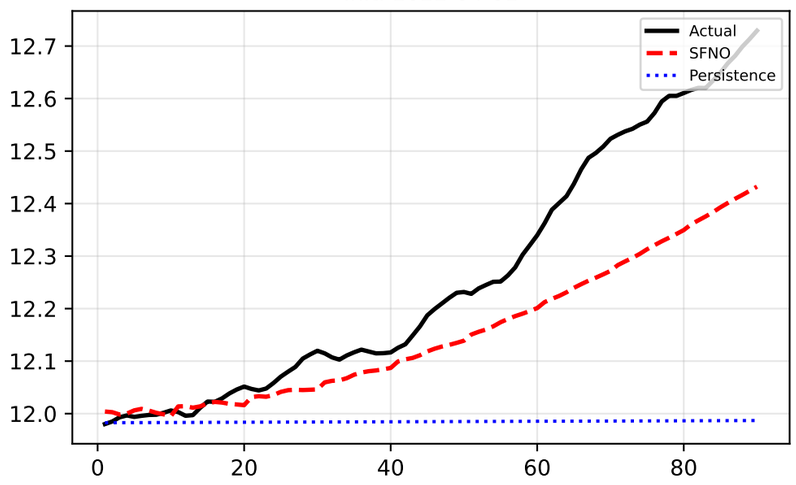
Global-Mean SST — Test (°C)

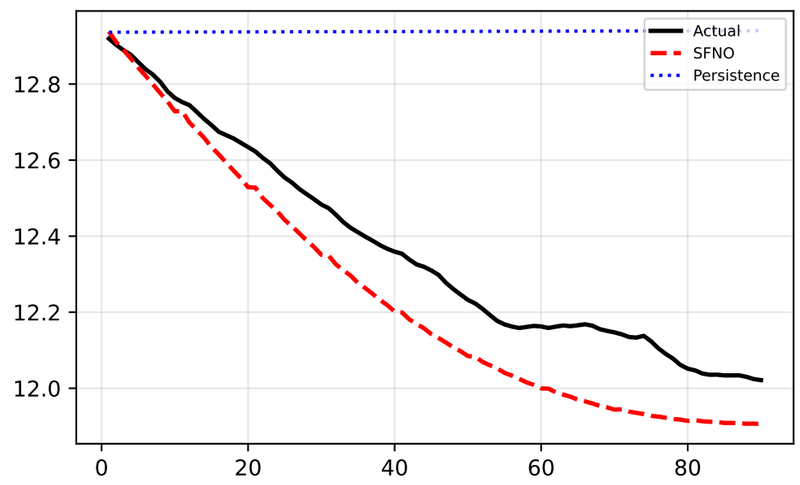


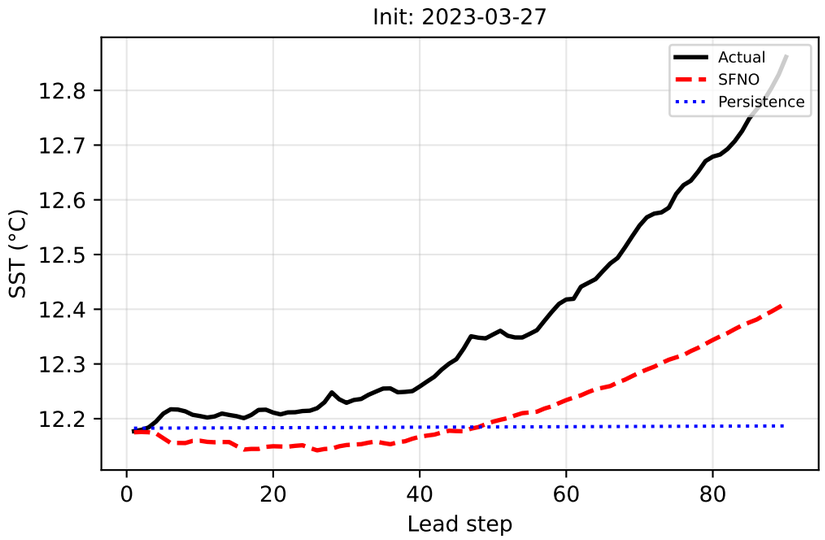


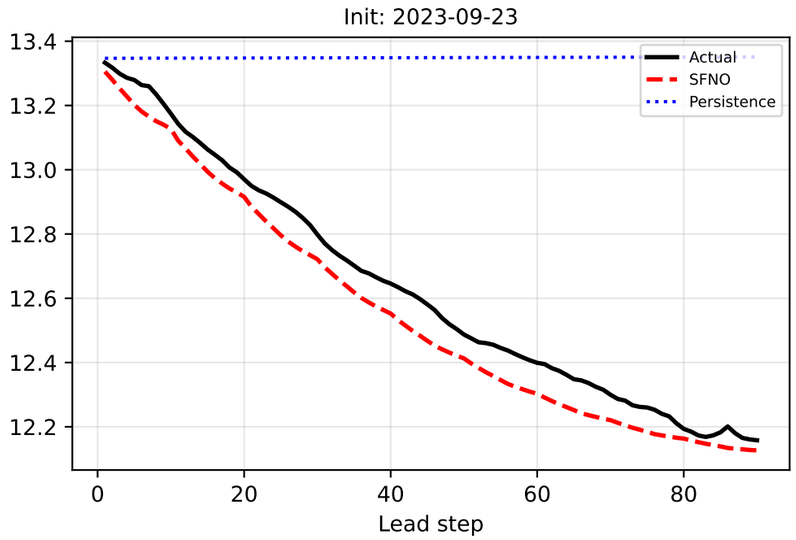


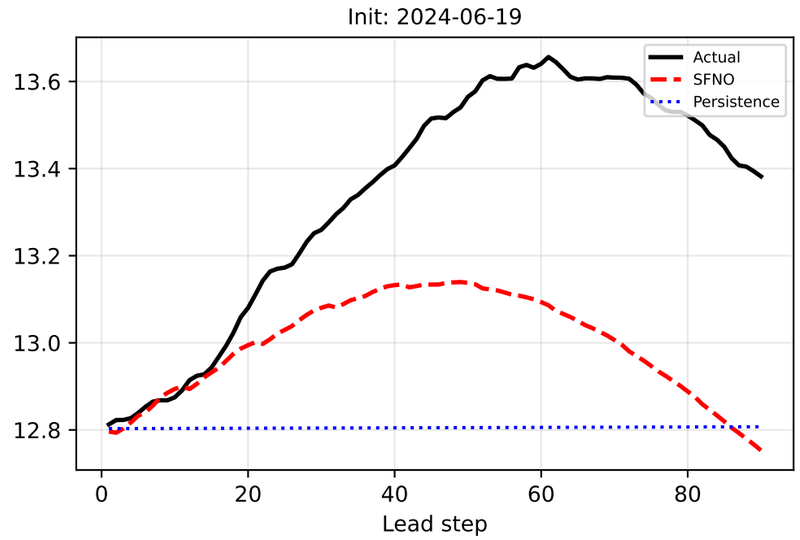






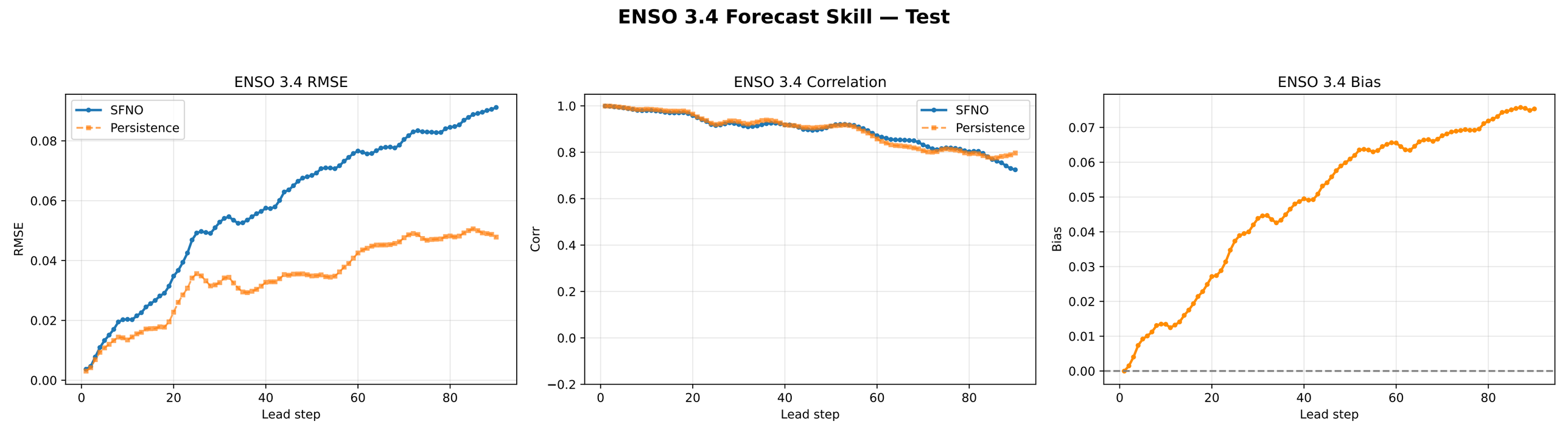




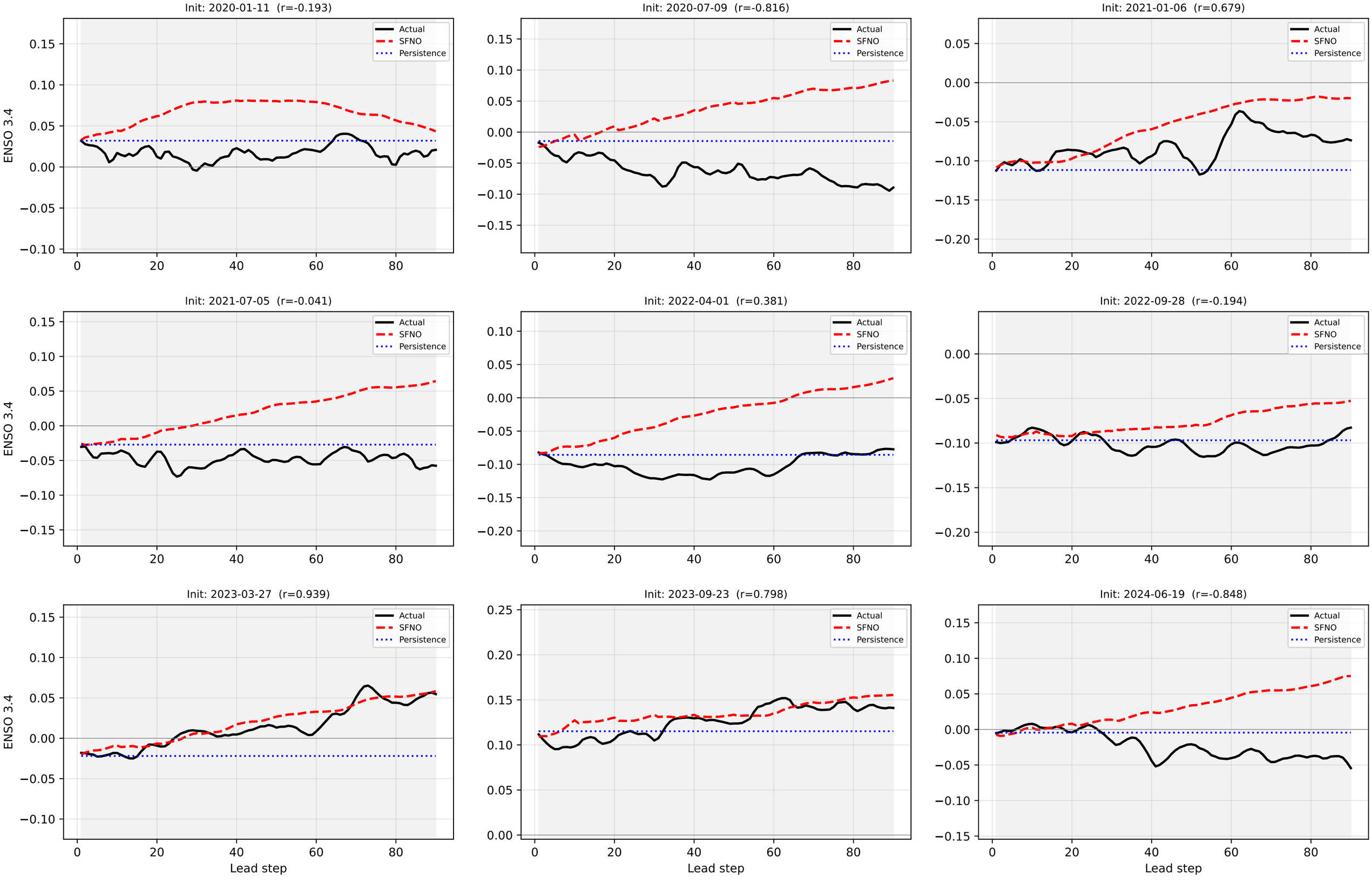


ENSO 3.4 Metrics — Test

Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0037	0.0031	0.9990	0.9992	-0.0000
2	0.0046	0.0042	0.9982	0.9989	0.0015
3	0.0078	0.0069	0.9957	0.9972	0.0040
4	0.0109	0.0093	0.9936	0.9950	0.0073
5	0.0133	0.0108	0.9912	0.9927	0.0092
6	0.0151	0.0120	0.9880	0.9896	0.0101
7	0.0170	0.0133	0.9846	0.9867	0.0112
8	0.0195	0.0144	0.9806	0.9842	0.0131
9	0.0202	0.0142	0.9794	0.9843	0.0135
10	0.0203	0.0135	0.9798	0.9857	0.0134
11	0.0202	0.0145	0.9802	0.9850	0.0124
12	0.0215	0.0155	0.9780	0.9833	0.0132
13	0.0226	0.0160	0.9760	0.9817	0.0141
14	0.0245	0.0171	0.9722	0.9784	0.0160
15	0.0256	0.0172	0.9701	0.9772	0.0175
16	0.0267	0.0173	0.9696	0.9774	0.0193
17	0.0282	0.0178	0.9699	0.9770	0.0214
18	0.0291	0.0177	0.9707	0.9777	0.0228
19	0.0314	0.0195	0.9669	0.9736	0.0249
20	0.0348	0.0227	0.9583	0.9644	0.0272
21	0.0367	0.0260	0.9491	0.9526	0.0274
22	0.0394	0.0285	0.9402	0.9440	0.0288
23	0.0425	0.0308	0.9325	0.9369	0.0314
24	0.0469	0.0341	0.9196	0.9251	0.0347
25	0.0492	0.0356	0.9152	0.9197	0.0373
26	0.0497	0.0349	0.9177	0.9226	0.0389
27	0.0494	0.0332	0.9223	0.9287	0.0395
28	0.0491	0.0315	0.9269	0.9351	0.0400
29	0.0510	0.0319	0.9240	0.9350	0.0420
30	0.0528	0.0326	0.9198	0.9322	0.0439
31	0.0541	0.0342	0.9136	0.9251	0.0446
32	0.0546	0.0344	0.9096	0.9217	0.0447
33	0.0535	0.0325	0.9110	0.9261	0.0435
34	0.0524	0.0308	0.9133	0.9305	0.0426
35	0.0526	0.0295	0.9185	0.9367	0.0433
36	0.0535	0.0293	0.9232	0.9386	0.0449
37	0.0546	0.0297	0.9250	0.9370	0.0465
38	0.0557	0.0304	0.9252	0.9336	0.0480
39	0.0564	0.0314	0.9225	0.9265	0.0487
40	0.0575	0.0327	0.9177	0.9182	0.0495
41	0.0574	0.0329	0.9180	0.9161	0.0491
42	0.0579	0.0329	0.9144	0.9151	0.0492
43	0.0601	0.0339	0.9077	0.9113	0.0509
44	0.0629	0.0353	0.8986	0.9066	0.0532
45	0.0636	0.0351	0.8968	0.9069	0.0541
46	0.0650	0.0355	0.8945	0.9049	0.0558
47	0.0665	0.0355	0.8962	0.9067	0.0576
48	0.0676	0.0355	0.8996	0.9080	0.0589
49	0.0680	0.0352	0.9054	0.9099	0.0599
50	0.0684	0.0349	0.9130	0.9127	0.0609
51	0.0692	0.0349	0.9188	0.9148	0.0620
52	0.0707	0.0352	0.9198	0.9157	0.0635
53	0.0710	0.0346	0.9201	0.9176	0.0637
54	0.0709	0.0345	0.9179	0.9155	0.0635
55	0.0707	0.0348	0.9158	0.9104	0.0630
56	0.0717	0.0362	0.9092	0.9010	0.0634
57	0.0732	0.0378	0.9023	0.8912	0.0645
58	0.0745	0.0390	0.8933	0.8831	0.0651
59	0.0758	0.0408	0.8820	0.8706	0.0656
60	0.0766	0.0425	0.8713	0.8575	0.0655
61	0.0762	0.0435	0.8653	0.8473	0.0645
62	0.0756	0.0441	0.8605	0.8398	0.0636
63	0.0758	0.0448	0.8551	0.8324	0.0634
64	0.0767	0.0452	0.8537	0.8290	0.0646
65	0.0776	0.0452	0.8533	0.8279	0.0659
66	0.0778	0.0452	0.8522	0.8256	0.0664
67	0.0779	0.0453	0.8509	0.8230	0.0665
68	0.0776	0.0457	0.8499	0.8187	0.0660
69	0.0786	0.0462	0.8434	0.8146	0.0666
70	0.0805	0.0476	0.8316	0.8065	0.0676
71	0.0817	0.0486	0.8240	0.8014	0.0681
72	0.0830	0.0490	0.8152	0.8005	0.0687
73	0.0834	0.0487	0.8111	0.8030	0.0689
74	0.0831	0.0474	0.8156	0.8113	0.0691
75	0.0830	0.0468	0.8189	0.8146	0.0694
76	0.0829	0.0470	0.8186	0.8124	0.0692
77	0.0828	0.0471	0.8167	0.8097	0.0692
78	0.0828	0.0472	0.8135	0.8062	0.0695
79	0.0840	0.0480	0.8066	0.7976	0.0711
80	0.0845	0.0482	0.8022	0.7935	0.0719
81	0.0848	0.0479	0.8040	0.7956	0.0724
82	0.0854	0.0482	0.8040	0.7931	0.0731
83	0.0869	0.0492	0.7953	0.7855	0.0743
84	0.0878	0.0500	0.7809	0.7786	0.0746
85	0.0888	0.0506	0.7689	0.7738	0.0751
86	0.0892	0.0500	0.7616	0.7761	0.0755
87	0.0896	0.0492	0.7547	0.7818	0.0757
88	0.0901	0.0490	0.7412	0.7847	0.0755
89	0.0905	0.0487	0.7302	0.7888	0.0749
90	0.0912	0.0478	0.7245	0.7967	0.0753


ENSO 3.4 Forecast Skill — Test
ENSO 3.4 RMSE
SFNO
Persistence
RMSE
Lead step
ENSO 3.4 Correlation
Corr
Lead step
ENSO 3.4 Bias
Bias
Lead step

ENSO 3.4 Index — Test



ENSO 3.4 Scatter — Test